

Post-Quantum Persistent Coercion-Resistant Voting from Deniable Encryption

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Abstract. E-voting systems must balance the voters’ need for privacy with the election’s goal of fairness and transparency. There are numerous post-quantum voting systems, but none of them are able to provide stronger security guarantees than ballot privacy (no entity can learn the content of ballots). Even for a ballot privacy property, the existing post-quantum schemes provide informal or weak guarantees. On the classical side, stronger properties like Coercion Resistance (coercer cannot know whether voter followed his instruction or not) has been studied for two decades, but existing definition assumes the coercer departs once casting closes, which leaves the voter’s post-tally local view, credentials, and proof randomness unmodeled.

In this work, we close both gaps. First, we introduce *Persistent Coercion Resistance* (PCR), a privacy notion stronger than coercion resistance that models a coercer who, even after the tally is published, may demand a voter’s complete local state across every cast ballot. We show $\text{PCR} \Rightarrow \text{CR}$, develop a quantitative framework for it, and we show that any scheme lacking a randomness-faking suffers a coercer-distinguishing advantage of at least $1 - 1/n_C - O(1/\sqrt{n_V})$, where n_C is number of candidates and n_V is number of voters. Second, we design **Onyx**, the first post-quantum verifiable voting system that provides guarantees beyond ballot privacy, built from deniable fully homomorphic encryption. We formally prove **Onyx** satisfies PCR and verifiability under LWE, MLWE, and MSIS, then provide machine-check privacy proofs. Finally, we implement a complete election cycle in 0.18 s at 128-bit post-quantum security on different CPUs.

1 Introduction

The idea of electronic voting is to modernize democratic processes through more accessibility and efficiency [1]; however, apart from the fact that people trust more in simple election systems with easily understandable procedures than complicated ones [2], designing voting schemes that can meet a wide range of security and privacy requirements remains one of the challenging problems in cryptography [3]. The paradox lies between transparency toward public verifiability and a high level of privacy for individual votes [2], as the level of privacy decreases by increasing verifiability and vice versa. This challenge gets more complicated by the advent of quantum computers, which will be in a position to break today’s classically encrypted ballots [4]. Transition to post-quantum (PQ) e-voting needs to happen as soon as possible, because of the Store Now, Decrypt Later attack [5], but migration from classical cryptosystems to post-quantum ones is not trivial and all applications face many issues for this transition [6], from the substantial increase in key and ciphertext sizes imposed by post-quantum schemes [7] to interoperability and backward compatibility challenges arising during the coexistence of classical and quantum-resistant [8] algorithms.

However, in voting systems, apart from these issues, this migration has even extra challenges, as mentioned earlier on balancing privacy and verifiability. It is expected a voting system to have privacy [9] (no entity can learn the content of ballots) and verifiability [10–13] (voters can verify their ballots were recorded and anyone can verify correct tallying). As summarized in Table 1, our study shows a consistent pattern: no existing scheme simultaneously achieves strong ballot privacy, formal verifiability, and practical efficiency under post-quantum assumptions, so the current PQ e-voting schemes need to go beyond innovations in building block and consider bigger picture by taking account formal cryptographic privacy and verifiability properties in voting, because elections are more than just counting ballots [1]. Revoting support [14], important for usability and coercion mitigation, is similarly absent from all but one [15] scheme.

The most fundamental privacy property that has been well-studied is BPRIV [9] (no entity can learn the content of ballots), and an stronger notion is **mb**-BPRIV [16] (ballot privacy against an adversary who controls the ballot box). **du**-**mb**-BPRIV [17] is an equivalent of **mb**-BPRIV, where verification is delayed until after tallying. There are stronger privacy properties such as *receipt-freeness* (voters cannot construct proofs of their choices to third parties) [18], and currently, the strongest privacy property in voting systems is the coercer cannot know whether voter followed his instruction or not [19], which is called *Coercion Resistance* (CR). So, CR implies all aforementioned properties.

Coercion resistance has been formalized in several ways, none of which agree on a single canonical experiment. The early definition of Juels, Catalano, and Jakobsson [20] is simulation-based: it compares the real protocol against an idealized one that leaks only the election result, and demands that no adversary can tell whether a coerced voter obeyed or evaded. Haines and Smyth [21] surveyed the many variants and observed that the original CR game is technically unrealizable, since the real world inevitably leaks the public board size while the ideal world does not. Kuesters et al. [13] proposed a quantitative δ -CR framework measuring the explicit gap between honest-evasion and full-cooperation worlds. Most recently, Cortier et al. [22] generalized the JCJ definition to quantify over all voter-behavior distributions and to model revoting and authority-injected fake ballots, which yields the strongest CR definition.

Problem 1. Every existing CR definition [13, 20–22] shares a single decisive modelling assumption: once the casting window closes, the coerced voter is left alone and unobserved. So, none of these definitions accounts for an adversary who returns *after* the tally is published and demands the voter’s complete local view, including the credential, the casting coins ρ consumed at every ballot, and the proof randomness r_π . As a direct result, a CR-secure scheme is permitted to embed a verbatim record of the voter’s genuine choice inside the local randomness of the ballot, because that record is never inspected by the model. Two further gaps follow from the same choice: simultaneous coercion of several voters is not first-class, since the experiment hands the adversary a single target; and persistent revoting confers no extra protection, because the revote chain is never disclosed. Figure 1 maps these unmodeled leakages onto the election timeline, which isolates the post-tally window left entirely uncaptured by prior CR and BPRIV definitions.

Problem 2. None of the existing PQ voting systems can provide stronger guarantees than BPRIV, even most of the existing ones do not provide any proof or formal guarantee for their privacy. As Table 1 shows, the existing post-quantum voting progress stops at BPRIV, and most schemes fall short even of that. Among twelve surveyed constructions [4,

15, 23–32], only a handful [15, 26, 28] prove standard BPRIV, while others rely on weaker notions [23, 33], offer only proof sketches [15], or provide no formal privacy guarantee at all [24, 25, 29–32]. No post-quantum scheme is coercion-resistant, let alone resistant to a persistent coercer, and none combines formal privacy, verifiability, revoting, and concrete parameters in one place. Coercion resistance is precisely the high-coercion setting where post-quantum migration matters most, yet it is the topic the literature has left untouched.

1.1 Contributions

We address both problems by introducing a privacy notion stronger than coercion resistance and by realizing it in the first post-quantum voting system that provably attains it, which is built around deniable encryption that lets a voter re-explain every coin ever consumed. The Figure 1 and Figure 2 demonstrate our motivation for this work.

Persistent Coercion Resistance. To address Problem 1, we introduce *Persistent Coercion Resistance* (PCR), a privacy notion stronger than CR that models a coercer who remains active beyond casting and may, even after the tally is published, demand a coerced voter’s entire local view: the credential, every casting coin, and the full sequence of claimed votes. The voter answers with a single Fake algorithm, and PCR requires that no coercer can distinguish a genuine view from a fabricated one. We prove $\text{PCR} \Rightarrow \text{CR}$ (Theorem 1), which we close exactly the post-tally window of Figure 1. Beyond the qualitative definition, we develop a quantitative framework that decomposes the persistent leakage into four channels, which derives closed-form and asymptotic ($1/\sqrt{n_V}$) advantage bounds, and shows that any scheme lacking a Fake algorithm suffers advantage $\geq 1 - 1/n_C - O(1/\sqrt{n_V})$, assuming n_V and n_C as numbers of voters and candidates. We validate these observations across twelve experiments (Figure 18).

Onyx Electronic Voting System. To address Problem 2, we design Onyx, the first post-quantum verifiable voting system that has stronger privacy than ballot privacy [9], and prove it satisfies PCR (Theorem 2) under LWE, MLWE, and MSIS, as well as universal verifiability (Theorem 3); these results place Onyx alone in the previously empty cell of Figure 2. We achieve persistent deniability through deniable encryption: deniable encryption allows a user to decrypt the same ciphertext to different plausible plaintexts depending on which secret the user uses, so the user can reveal a “decoy” message under coercion while hiding the real one [34]. By propagating randomness deniability through every voter-side layer, Onyx lets a coerced voter re-explain every coin behind every cast ballot. We further provide machine-checked proofs of Onyx’s PCR in EasyCrypt. Also, we implement and benchmark Onyx across four platforms which completes a full election cycle in 0.18 s, and concrete 128-bit post-quantum parameters.

Non-Interactive Prover-Deniable Zero-Knowledge Proof. As part of realizing Onyx, we construct the first non-interactive prover-deniable zero-knowledge proof system in the post-quantum setting (Section 5.2). Existing deniable proof systems, such as designated-verifier proofs [35, 36], online/offline deniable proofs [37], and transferable zero-knowledge [38], do not simultaneously achieve non-interactive prover-side randomness deniability under standard post-quantum assumptions. Our lattice-based NIZK, built on LNP [39] with a Micciancio–Peikert equivocation trapdoor [40], equips the prover with a Fake algorithm that produces alternative randomness consistent with any

published proof under any other valid witness. It is worth noting that our construction does not contradict the impossibility result of Pass [41]: that result rules out fully *adversary*-deniable non-interactive zero-knowledge in the standard random-oracle model, whereas our notion is *prover*-deniable, which is a weaker and achievable guarantee. This building block in Onyx is what allows the ballot’s proof randomness to be denied alongside its plaintext and credential coins.

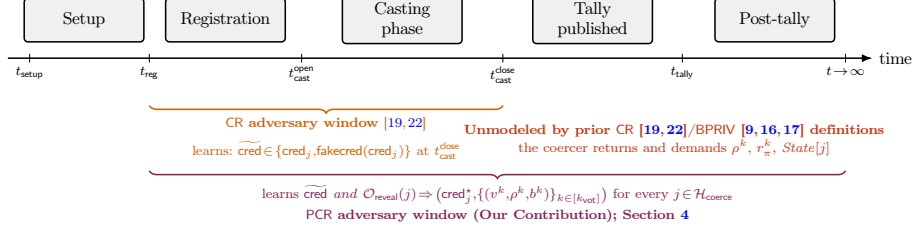


Fig. 1: Persistent-coercion in Problem 1: PCR extends the adversary’s window past $t_{\text{cast}}^{\text{close}}$.

Figure 1 maps the election phases on a time axis ($t_{\text{setup}}, t_{\text{reg}}, t_{\text{cast}}^{\text{open}}, t_{\text{cast}}^{\text{close}}, t_{\text{tally}}, t \rightarrow \infty$): the orange brace bounds coercion-resistance (CR) adversary, who is learning at $t_{\text{cast}}^{\text{close}}$ only the disclosed credential $\widetilde{\text{cred}} \in \{\text{cred}_j, \text{fakecred}(\text{cred}_j)\}$ (target voter j ’s genuine credential cred_j , or a fabrication via algorithm fakecred); the magenta brace bounds our persistent-coercion-resistance (PCR) adversary, additionally querying the post-tally oracle $\mathcal{O}_{\text{reveal}}(j) = (\text{cred}_j^*, \{(v^k, \rho^k, b^k)\}_{k \in [k_{\text{vot}]}})$ for every j in the coerced honest set $\mathcal{H}_{\text{coerce}}$, which exposes the (possibly faked) credential cred_j^* , plaintext v^k , casting coins ρ^k , ballot b^k , and (via the new algorithm Fake) the NIZK proof randomness r_π^k .

	BPRIV [9]	CR [19, 22]	PCR (Definition 2)
Classical	Helios [42], Belenios [43], ElectionGuard [44–46]	JCJ [20], Civitas [47], CHide [22]	None None of the existing scheme are PCR-secure
Post-Quantum	Epoque [23, 48], EVOLVE [26, 49], Homo-LWE [15], BPRIV-BC [28], LWE-Voting [32]	None None of the existing PQ scheme are CR-secure	Onyx (Section 5) is PCR-secure (Theorem 2) under LWE, MLWE, MSIS

Fig. 2: The gap in Problem 2 that we address. $\text{PCR} \Rightarrow \text{CR} \Rightarrow \text{BPRIV}$ (Theorem 1); schemes with no Fake function have $\text{Adv}^{\text{PCR}} \geq 1 - 1/n_C - O(1/\sqrt{n_V})$.

across all k_{vot} ballots, so the red region isolates leakage left unmodeled by every prior CR/ballot-privacy (BPRIV) definition, namely the full local state $State[j]$.

2 Related Works

In this work, it is expected from Onyx to compete with the voting systems in state-of-the-art of both classical coercion-resistant schemes and ballot-private post-quantum voting systems, which our evaluations show that Onyx provides stronger privacy guarantees. To check this, we first investigate both groups in the following.

2.1 Post-Quantum Voting Systems

The voting systems that are resistant to quantum attacks have diverse structures with minimal privacy and/or verifiability guarantees, where Table 1 summarizes our study.

As a note, there are two main methods of election tally [48, 50]: *Homomorphic* aggregates encrypted votes directly into an encrypted sum that is decrypted once, while *mix-net* anonymizes individual ciphertexts through a shuffle before decrypting each ballot.

Starting with Epoque [23], it splits each vote into secret shares, commits to each share, encrypts the shares under an IBE scheme (binding them to the voter’s identity) and sends them to multiple trustees, who tally via homomorphic addition of commitments verified by a ZKP of well-formedness; all trustees must be online and actively participate in both the ballot weeding and tallying phases. Epoque’s privacy proof uses a privacy definition called *Vote Privacy* [33] that Bernhard et al. [9] shows this is a weak definition of ballot privacy since it restricts the class of protocols, may yield inconsistent results under natural result functions (e.g., majority [51]), fails to apply to protocols outputting auxiliary data (e.g., tally proofs in ElectionGuard [44]), and misses privacy violations beyond same-vote detection [9]. Moreover, their Lemma 5 [48] omits the full hybrid argument for secret-share uniformity under commitment hiding. Also, Cortier et al. in [12] shows why the Epoque’s verifiability definition [13] is weak, since it does not distinguish ballot deletion from ballot modification [12]. Moreover, concrete post-quantum parameters for all building blocks are also absent in Epoque.

Hough et al. proposed NTRU Mix-Net [24], in which votes are NTRU-encrypted, anonymized through a verifiable mix-net, and recovered via threshold distributed decryption, with ZKPs ensuring honest execution at each stage. In this scheme, NTRU parameters must be carefully chosen to avoid the overstretched regime where $q > 0.0058 \cdot \sigma^2 \cdot d^{2.484}$ (with d the polynomial degree, q the modulus, and σ the noise distribution width) enables subfield lattice attacks, and the scheme’s privacy fundamentally relies on the assumption that at least one mix server remains honest without providing detection mechanisms for malicious mixing beyond proof verification. It is not straightforward how the privacy claims in [24] are achievable; although Hough et al. [24] does not propose a complete voting scheme, upon using their NTRU Mix-Net, they refer the privacy of resulting system to another mix-net by Aranha et al. [52], but even their theoretical system (full version’s Appendix H [50]) does not provide any proofs or formal guarantees.

In FOO-based Voting [4], a voter encrypts their vote, gets a threshold blind signature on it from multiple authorities, and anonymously posts it to a bulletin board, where authorities later reveal decryption shares so anyone can tally. It combines threshold lattice-based blind signatures [53–55] with Dual-Regev encryption [56, 57], but one-more

unforgeability of the underlying blind signature remains conjectural [58], and privacy breaks if the required perfectly anonymous submission channel is compromised.

Code-based Voting [25], the first code-based verifiable voting system, relies on the less-studied dual-LPN assumption whose cryptanalytic background is limited, concrete 128-bit parameters against ISD attacks remain uncertain [59], and ballot privacy claims lack formal proofs.

In EVOLVE [26], each vote is split into additive shares, committed to multiple authorities using a lattice-based homomorphic commitment, with an OR-proof ensuring vote validity and amortized ZKPs by authorities enabling homomorphic tallying; it is post-quantum secure but lacks individual verifiability, requires a large voter threshold for efficiency, and supports no revoting. EVOLVE modifies the BPRIV definition [9] but the adaptation sidesteps the simulator requirement by making the tally consistent with the board in both worlds; in standard BPRIV, $\beta=1$ requires a simulator Sim to produce a fake proof π' consistent with BB_1 , which EVOLVE never constructs, and whether their notion implies standard BPRIV is not formally analyzed. EVOLVE also proves only a weaker “realistic result” consistency rather than strong consistency [9]. EVOLVED [27] extends EVOLVE with Hamming Weight Proofs but inherits the same definitional gaps.

In LWE-Voting [32], each ballot is split into two Lindner–Peikert PKE [60] ciphertext halves routed to two physically separated verification servers, which jointly run an interactive zero-knowledge plaintext-content-validity check [32] before the tally server homomorphically aggregates and decrypts the combined ciphertext, with the result verified via a Sign-authenticated tally proof [32]. In this scheme, individual verifiability is explicitly left as an open problem, and universal verifiability is claimed via the tally-result verification game, but this holds only under the strong assumption that Verification Server A and Verification Server B are non-colluding—an assumption that the [32] motivates as “reasonable” but does not formalize in the KTV trust-model, and which, if violated, also falsifies the privacy argument via the LWE-ciphertext decomposition (Theorem 1 in [32]). In terms of PQ parameters, $n=512$ sits at the low end of LWE conservative recommendations; the $n=512$ plain-LWE parameter of their work was acceptable in 2018 but has degraded under the BKZ-2016 estimator [61] updates.

Chillotti et al. [15] present a homomorphic LWE-based voting scheme but provide only a proof sketch for ballot privacy. Abapour et al. [28] propose a blockchain-based voting system with or economic structures, which although lacks the flexibility for general use [62], but provide formal ballot privacy proof. In BPRIV-BC [28], votes are encoded as degree-1 polynomials committed via a lattice-based polynomial commitment scheme [63] and authorized through a posterior-secure [64] non-interactive blind signature [65] whose anonymity is retroactively strengthened after issuance.

In Kyber-BC [29], each voter’s ballot is encrypted as a polynomial under the voter’s own per-voter KEM [66] public key, hashed via [67], and stored on blockchain [68, 69], with the central voter-bank holding all secret keys in hardware security modules and decrypting every ciphertext directly at tally time.

In Hybrid-BC [30], votes are encrypted under a session key derived via ECDH [70] and gated by an ABE [30] access policy enforced by a smart contract on blockchain [71, 72], with ciphertexts stored off-chain on IPFS and content-identifier hashes pinned on-chain for retrieval at tally.

The same limitation of BPRIV-BC [28] applies to the blockchain-based schemes of Sekar et al. [29] and Mahanayak et al. [30] as well.

Gabriel et al. [31] presents a voting scheme based on steganography, which due to its informality, does not completely fit into conventional formal models in voting. Voter data is first encrypted using a multivariate cryptosystem [73, 74] and then covertly embedded into a cover image via LSB-steganography [31], transmitting the resulting stego-image over a public channel so that the ciphertext itself is hidden from any eavesdropper.

Scheme	Building Blocks	Privacy	Verifiability	Revoting	Parameters	Code
Epoque [23]	IBE [75], Com [76], NIZK [77]	Weak BPRIV [33]	indiv, univ	×	✓	p
NTRU-Mix [24]	PKE [78], Com [79], ZKP [80–82]	×	univ	×	✓	✓
FOO-based [4]	TPKE [56, 57], BSign [53–55]	×	partial univ	×	×	×
Code-based [25]	PKE [83, 84], Com [25], NIZK [25, 85]	×	indiv, univ	×	✓	×
EVOLVE [26]	PKE [26], ZKP [86], Sig [87], Com [88]	BPRIV [9]	partial univ	×	✓	p
EVOLVED [27]	PKE [27], Com [26], HW-Proof [89]	×	partial univ	×	✓	×
Homo-LWE [15]	FHE [90], Sign [91], NIZK [92]	BPRIV [9]	indiv, univ	✓	×	×
BPRIV-BC [28]	Com [63], PostSec [64], NIBSign [65]	BPRIV [9]	indiv, univ	×	×	×
Kyber-BC [29]	KEM [66], Block [68, 69], Hash [67]	×	partial indiv	×	✓	p
Hybrid-BC [30]	ABE [30], ECDH [70], Block [71, 72]	×	partial indiv	×	×	p
Steg-MQPC [31]	MQPC [73, 74], LSB-Steg [31]	×	×	×	✓	p
LWE-Voting [32]	PKE [60], ZKP [32], Sign [32]	×	univ	×	✓	p
Onyx (Section 5)	PKE [93], UT [94], Mix [52], ZKP [95]	PCR (Definition 2)	part indiv, univ	✓	✓	✓

Table 1: Comparison of post-quantum electronic voting schemes. **Parameters** column: ✓ if the paper provides a concrete numerical parameter set (security levels, modulus, dimensions, etc.), × if only asymptotic/abstract parameter formulas are given. **Code** column: ✓ if a publicly accessible implementation URL is given, p if the authors report having implemented the scheme but no public source code or repository is provided, × if no implementation exists. Here, consider zero-knowledge proof ZKP and non-interactive one as NIZK, hamming weight proof HW-Proof, multivariate quadratic polynomial cryptography MQPC, identity-based encryption IBE, commitment Com, public-key encryption PKE, blind signature BSign, threshold public-key encryption TPKE, digital signature Sig, fully homomorphic encryption FHE, polynomial commitment PCom, posterior security PostSec, non-interactive blind signature NIBSign, key encapsulation mechanism KEM, blockchain Block, hash function Hash, elliptic-curve Diffie–Hellman ECDH, attribute-based encryption ABE, least-significant-bit steganography LSB-Steg, and fully homomorphic signature FHS.

2.2 Coercion-resistant Voting Systems

Here, we go through the existing CR-secure schemes and the existing formal coercion resistance definitions. None of the CR-secure schemes has been publicly implemented and brachmarked yet, and all of the voting schemes that satisfy CR [20, 22, 47, 96–102] are vulnerable to quantum attacks.

JCJ. Juels, Catalano, and Jakobsson [20] introduced the seminal JCJ protocol that founded the fake-credential paradigm. Each voter receives a private credential from the registrar over an untappable channel; an encrypted public roster of all credentials is posted on the bulletin board. A voter casts a ballot by encrypting their vote together with their credential under the trustees’ threshold key, with a zero-knowledge proof of well-formedness. Under coercion, the voter hands over a freshly fabricated random credential of the same format which the coercer cannot distinguish from the genuine one. The tally uses plaintext-equivalence tests (PETs) to discard duplicate credentials and credentials not in the roster, followed by a verifiable mix-net and threshold decryption. JCJ’s main weakness is its cleansing step, which leaks the per-credential revote count and the total number of invalid ballots, information that Cortier et al. [22] show is exploitable by a coercer who knows the honest revoting distribution.

Civitas. Clarkson, Chong, and Myers [47] proposed Civitas as the first practical realisation of JCJ, which replaces the single registrar with multiple distributed registration tellers so that no single party learns a voter’s credential. The efficiency-relevant innovation is the *block-cleansing* optimisation: voters are publicly assigned to small blocks and credentials are checked against the roster block-by-block instead of against the full list, which reduces JCJ’s quadratic PET cost into something roughly linear. The casting channel, threshold tally, and evasion strategy are inherited from JCJ. Civitas’s limitation is that the public block structure makes each block’s ballot count observable, so a coercer who knows the targeted voter’s block sees a smaller and noisier crowd; Cortier et al. [22] show that Civitas leaks more during cleansing than JCJ does, and the block speed-up cannot be salvaged in any hiding pipeline.

Coercion-evidence schemes. Grewal et al. [96] proposed *Caveat Coercitor*, which abandons full prevention of coercion and instead aiming at *detection*. Caveat Coercitor truncates JCJ’s revote-PET phase as soon as one match per credential is found, and exposes a public coercion-evidence value that grows whenever a duplicate is detected. Voters submit ballots anonymously as in JCJ, but the evasion strategy is augmented with a procedure that lets a coerced voter deliberately trigger the evidence channel, which can later be cited as proof of misconduct. The protocol is elegant because it turns leakage into accountability, but it does not reduce the leakage itself: the truncated PET phase still reveals how quickly a match was found (itself a side-channel on the revote distribution), and the evidence is only generated *after* the coercion has already taken effect, which provides no defense at the moment of coercion.

Linear-time JCJ variants, Selections, and re-voting-based schemes. Araujo, Foulle, and Traoré [97] gave the first linear-time JCJ variant by attaching a deterministic public commitment of the credential to each ballot, with subsequent refinements by Araujo et al. [98, 103] and a similar variant by Spycher et al. [99]; however, as Cortier et al. [22] observe and Spycher et al. themselves acknowledge, the deterministic commitment makes per-credential revote counts visible *during* voting, a strict information regression compared to JCJ. Clark and Hengartner [100] introduced *Selections* with an over-the-shoulder coercion-resistance model, equipping each voter with a panic-password mechanism that hides whether a ballot cast under live observation is genuine, yet the scheme inherits JCJ’s cleansing-phase leakage and offers no defence against post-tally interrogation of the voter’s local state. A separate line replaces fake credentials with *deniable revoting*, where the coerced voter quietly recasts after complying, with earlier ballots silently invalidated; this approach is realized by VoteAgain [101], the Estonian e-voting system [104], and the everlasting-privacy variant of Locher et al. [102], but all share the fundamental limitation of assuming the coercer cannot maintain persistent access through the close of voting and provide no protection whatsoever against post-tally interrogation.

CHide. Cortier, Gaudry, and Yang [22] recently proposed CHide, the closest predecessor of our work and the first scheme to explicitly close the JCJ cleansing-leakage channel. CHide replaces JCJ’s PETs and clear-text duplicate detection with a fully encrypted cleansing pipeline built from MPC primitives over exponential ElGamal [105]: equality tests produce encrypted validity bits, an And gadget combines them, a conditional-set-zero gate overwrites invalid ballots without revealing which, and a quasi-linear MPC sort based on Batchier’s odd-even-merge network groups ballots by credential

and selects the survivor per credential, all under threshold ElGamal. The scheme leaks only the size of the ballot box and the final tally, which matches the ideal-world floor, and is accompanied by a strengthened CR definition that quantifies over all behavioral distributions. CHide’s main limitation is its classical-cryptography basis: every primitive relies on the hardness of discrete logarithms, so CHide breaks in polynomial time under a quantum adversary and cannot serve elections that need post-quantum confidentiality.

Existing CR definitions. The original JCJ paper [20] gave a simulation-based CR definition that compares the real protocol to an idealized one where only the result is leaked, asking that no adversary can distinguish whether a coerced voter obeyed or evaded. Haines and Smyth [21] surveyed and unified the literature’s many variants and showed that the JCJ definition is technically unrealizable because the real game inevitably leaks the public board’s size while the ideal one does not; they propose a fix that exposes the board size in both games. Kuesters, Truderung, and Vogt [13] introduced a quantitative δ -CR framework that measures the gap between an honest-evasion world and a full-cooperation world and admits explicit numerical comparisons rather than only asymptotic negligibility. Cortier et al. [22] generalized the JCJ definition to quantify over all voter-behavior distributions and to model revoting and authority-injected fake ballots, which is the strongest definition prior to ours.

All of these definitions share a single decisive limitation: the adversary is assumed to leave the voter alone once the casting window closes, so none of them captures a coercer who returns post-tally and demands the voter’s credential, casting coins, and full local state, which is precisely the gap our PCR notion is designed to close.

3 Coercion Resistance in Electronic Voting

The strongest classical coercion-resistance notion is the one introduced by Cortier et al. alongside CHide [22], which we now restate in the notation of Section 4 so that the gap with PCR becomes precise.

A voting scheme in the CHide framework is a tuple $\Pi = (\text{Setup}, \text{Register}, \text{Vote}, \text{Valid}, \text{fakecred}, \text{Tally}, \text{Verify})$ together with a behavioral distribution $\mathcal{B}$ over sequences of pairs (i, v) . Indices i in the natural voter range $[1, n_V]$ denote ballots cast by registered voters (possibly multiple times, modeling revoting), and fresh *negative* indices denote ballots posted with fabricated credentials by authorities, registrars, or adversaries. The absence of any pair $(i, \star)$ for a given i models abstention. The function `cleanse` resolves multiple ballots per voter according to the protocol’s re-voting policy, and the deterministic `Result` function then maps the cleansed multiset to an election outcome. Compared to PCR, the scheme omits the `Fake` algorithm: under coercion the voter is allowed only to substitute a fresh fake credential via `fakecred`, with no mechanism to lie about the casting coins of any already-posted ballot. The definition takes the form of two experiments, $\text{Exp}_{\Pi, \mathcal{A}}^{\text{CR-Real}}$ and $\text{Exp}_{\Pi, \mathcal{A}}^{\text{CR-Ideal}}$, showed in Figure 3.

Definition 1 (Coercion Resistance [22]). *A voting scheme Π satisfies Coercion Resistance if for every PPT adversary $\mathcal{A}$, every choice of parameters n_T, t, n_V, n_A, n_C , and every behavioural distribution $\mathcal{B}$, there exists a PPT adversary $\mathcal{B}^*$ and a negligible function $\mu(\cdot)$, as demonstrated in Figure 3, such that*

$$\left| \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{CR-Real}}(\lambda, n_T, t, n_V, n_A, n_C, \mathcal{B}) = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{B}^*}^{\text{CR-Ideal}}(\lambda, n_V, n_A, n_C, \mathcal{B}) = 1] \right| \leq \mu(\lambda).$$

$\text{Exp}_{\Pi, \mathcal{A}}^{\text{CR-Real}}(\lambda, n_T, t, n_V, n_A, n_C, \mathcal{B})$	$\mathcal{O}_{\text{board}}()$
1: $BB \leftarrow \emptyset$ 2: $(\text{pk}, \{(s_i, h_i)\}_{i=1}^{n_T}) \leftarrow \mathcal{P}_{\text{Setup}}^A(\lambda, n_T, t)$ 3: $\{\text{cred}_i\}_{i=1}^{n_V}, R \leftarrow \text{Register}(\lambda, \text{pk}, n_V)$ 4: $\mathcal{D} \leftarrow \mathcal{A}(R)$ 5: if $ \mathcal{D} \neq n_A$ then return 0 6: $(j, \alpha) \leftarrow \mathcal{A}(\{\text{cred}_i\}_{i \in \mathcal{D}})$ 7: if $j \in \mathcal{D} \vee \alpha \notin [1, n_C] \cup \{\phi\}$ then return 0 8: $B \leftarrow \mathcal{B}([1, n_V] \setminus \mathcal{D}, n_C)$ 9: for $(i, v) \in B$ with $i \notin [1, n_V]$ do 10: $\text{cred}_i \leftarrow \text{fakecred}(\text{pk})$ 11: $\beta \leftarrow \mathcal{S}\{0, 1\}$ 12: $\widetilde{\text{cred}} \leftarrow \text{cred}_j$ 13: if $\beta = 0$ then 14: $\widetilde{\text{cred}} \leftarrow \text{fakecred}(\text{cred}_j)$ 15: remove all $(j, \star)$ from B but the last, 16: which is replaced by (j, α) if $\alpha \neq \phi$, 17: else removed 18: else 19: remove all $(j, \star)$ from B 20: for $(i, v) \in B$ in order do 21: $M \leftarrow \mathcal{A}^{\mathcal{O}_{\text{board}}}(BB)$ 22: $BB \leftarrow BB \parallel \{m \in M \mid \text{Valid}(\text{pk}, m, BB) = \top\}$ 23: $b \leftarrow \text{Vote}(\text{pk}, \text{cred}_i, v)$ 24: $BB \leftarrow BB \parallel b$ 25: $M \leftarrow \mathcal{A}^{\mathcal{O}_{\text{board}}}(BB, \widetilde{\text{cred}}, \text{"casting closed"})$ 26: $BB \leftarrow BB \parallel \{m \in M \mid \text{Valid}(\text{pk}, m, BB) = \top\}$ 27: $(r^*, \Pi^*) \leftarrow \mathcal{P}_{\text{Tally}}^A(BB, R, \text{pk}, \{(s_i, h_i)\}, t)$ 28: $d \leftarrow \mathcal{A}(r^*, \Pi^*)$ 29: return d	1: return $\text{Publish}(BB)$ $\text{Exp}_{\Pi, \mathcal{A}}^{\text{CR-Ideal}}(\lambda, n_V, n_A, n_C, \mathcal{B})$ 1: $D \leftarrow \emptyset$ 2: $\mathcal{D} \leftarrow \mathcal{A}(\lambda)$ 3: if $ \mathcal{D} \neq n_A$ then return 0 4: $(j, \alpha) \leftarrow \mathcal{A}()$ 5: if $j \in \mathcal{D} \vee \alpha \notin [1, n_C] \cup \{\phi\}$ then return 0 6: $B \leftarrow \mathcal{B}([1, n_V] \setminus \mathcal{D}, n_C)$ 7: $\beta \leftarrow \mathcal{S}\{0, 1\}$ 8: if $\beta = 0$ then 9: remove all $(j, \star)$ from B but the last, 10: which is replaced by (j, α) if $\alpha \neq \phi$, 11: else removed 12: else 13: remove all $(j, \star)$ from B 14: for $(i, v) \in B$ with $i \in [1, n_V] \cup \{\text{neg}\}$ do 15: $D \leftarrow D \parallel (i, v)$ 16: $(\{\nu_i\}_{i \in \mathcal{D}, \gamma}) \leftarrow \mathcal{A}(D)$ 17: for $i \in \mathcal{D}$ do 18: if $\nu_i \in [1, n_C]$ then $D \leftarrow D \parallel (i, \nu_i)$ 19: if $\beta = 1 \wedge \gamma \in [1, n_C]$ then 20: $D \leftarrow D \parallel (j, \gamma)$ 21: $r^* \leftarrow \text{Result}(\text{cleanse}(D))$ 22: $d \leftarrow \mathcal{A}(r^*)$ 23: return d

Fig. 3: In the Real experiment the adversary is given the disclosed credential $\widetilde{\text{cred}}$ during the casting phase, may keep posting ballots on its behalf until casting closes, and finally observes the published tally and proof. In the Ideal experiment, only the board length $|D|$, the corrupted-voter assignments, and the tally are revealed; the disclosed credential is not modeled because no post-tally interrogation is permitted.

The Real experiment runs the protocol honestly. The challenger samples a bit β , builds the multiset B from $\mathcal{B}$, and applies an obey-or-evade adjustment: under $\beta = 1$ the coerced voter is removed from B entirely (the coercer is expected to vote with the disclosed credential); under $\beta = 0$ all of voter j 's entries are removed except the last, which is replaced by (j, α) if $\alpha \neq \phi$. The disclosed credential cred is exposed to the adversary *at the moment casting closes*: it equals the genuine cred_j on $\beta = 1$ and a freshly

sampled $\text{fakecred}(\text{cred}_j)$ on $\beta=0$. The adversary then casts any final batch of ballots before the tally is computed. The Ideal experiment skips the protocol entirely: only $|D|$, the corrupted-voter assignments, and the published r^* are leaked, and no credential is ever disclosed because the model assumes the coercer departs once casting closes.

Limitations and the gap. Definition 1 captures every coercion strategy in which the adversary is content to interact with the voter only during the casting window. After the tally is published, the voter is implicitly assumed to walk away unobserved, which means neither the casting coins ρ nor the proof randomness r_π nor any subsequent disclosure of the local view is part of the adversary’s allowed leakage. As a result, a CR-secure scheme is allowed to embed a verbatim record of the voter’s true vote inside the local randomness of the ballot, because that record is never inspected. Two further limitations follow from the same modelling choice. First, simultaneous coercion of several voters is not first-class: the experiment hands the adversary a single target j , so per-voter accumulation of leakage under joint attack cannot be quantified within CR. Second, persistent revoting offers no extra protection in this model, a voter who revotes ten times under coercion looks identical, from the adversary’s CR-experiment view, to one who revotes once, because the revote chain is never disclosed. Each of these gaps motivates our PCR definition (Section 4): PCR replaces the single-shot credential disclosure of CR with a post-tally $\mathcal{O}_{\text{reveal}}$ oracle that releases the voter’s complete local view, lifts the target to a coerced set $\mathcal{H}_{\text{coerce}}$ of size $q_c \geq 1$, accumulates per-ballot leakage over a chain of k_{vot} casts, and requires the scheme to supply a Fake algorithm that can re-explain any already-cast ballot under any later-claimed vote.

4 Persistent Coercion Resistance

Persistent Coercion Resistance (PCR) is a privacy property of voting schemes that captures a coercer who remains active beyond the casting moment. Beyond all interactions available during the casting phase, the coercer may, at any later time, including after the tally has been published and verified, demand from a coerced voter the complete local view of the voting client: the credential, every random coin consumed during every casting operation, and the full sequence of claimed votes. The voter is permitted a single local algorithm Fake for fabricating this view; PCR demands that no coercer can distinguish, from the disclosed view together with the public election transcript, a voter who genuinely cast as instructed from a voter who used Fake to lie about every coin ever consumed.

Here, we consider that a voting scheme is a tuple $\Pi = (\text{Setup}, \text{Register}, \text{Vote}, \text{Fake}, \text{Valid}, \text{Tally}, \text{Verify})$ together with a deterministic result function Result and a re-voting policy Policy. We assume the standard correctness, verifiability, and tally semantics. We add a component as $\text{Fake}(\text{pk}, \text{cred}, \mathbf{v}, \boldsymbol{\rho}, \mathbf{v}^*) \rightarrow (\text{cred}^*, \boldsymbol{\rho}^*)$, which takes the voter’s true sequence of votes $\mathbf{v}$ together with the genuine coins $\boldsymbol{\rho}$ used to cast them, an arbitrary sequence of claimed votes $\mathbf{v}^*$ of the same length, and returns a (possibly fake) credential cred^* together with fake coins $\boldsymbol{\rho}^*$ to be disclosed under coercion. The definition imposes no implementation constraint on Fake.

Definition 2 (Persistent Coercion Resistance). *A voting scheme Π satisfies Persistent Coercion Resistance if for every PPT adversary $\mathcal{A}$, every choice of parameters n_T, t, n_V, n_A, n_C , and every behavioral distribution $\mathcal{B}$, there exists a PPT adversary $\mathcal{B}^*$*

and a negligible function $\mu(\cdot)$, as demonstrated in Figure 4, such that

$$|\Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Real}}(\lambda, n_T, t, n_V, n_A, n_C, \mathcal{B}) = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{B}^*}^{\text{PCR-Ideal}}(\lambda, n_V, n_A, n_C, \mathcal{B}) = 1]| \leq \mu(\lambda).$$

Figure 4 specifies two experiments and two oracles. Both experiments are parameterized by a security parameter λ , a voter count n_V , a corrupted-voter count n_A , and a candidate count n_C . The Real experiment additionally takes a trustee count n_T , a reconstruction threshold t , and a behavioral distribution $\mathcal{B}$. The distribution $\mathcal{B}(S, n_C)$ samples a sequence of pairs (i, v) : indices $i \in S$ represent genuine registered voters, fresh negative integers represent ballots carrying fabricated credentials (injected by corrupted parties), and $v \in [1, n_C]$ is the chosen candidate; the absence of any pair $(i, \star)$ for some i models abstention.

The real experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Real}}$ maintains three mutable structures: a bulletin board BB (the ordered list of accepted messages), a state map $State$ (recording, per coerced voter, every vote–randomness–ballot triple produced during casting), and a disclosure set $Disc$ (the coerced voters whose local views have been revealed to $\mathcal{A}$).

The joint procedure $\mathcal{P}_{\text{Setup}}^{\mathcal{A}}(\lambda, n_T, t)$ runs the **Setup** algorithm in the presence of the adversary $\mathcal{A}$, producing a public election key pk and per-trustee secret–hint pairs $\{(s_i, h_i)\}_{i=1}^{n_T}$. Registration $\text{Register}(\lambda, \text{pk}, n_V)$ outputs credentials $\{\text{cred}_i\}_{i=1}^{n_V}$ and a public registration record R . The adversary, given R , nominates a set $\mathcal{D}$ of n_A corrupted voters and receives their credentials. It then selects a disjoint set $\mathcal{H}_{\text{coerce}} \subseteq [1, n_V] \setminus \mathcal{D}$ of coerced honest voters, supplying for each $j \in \mathcal{H}_{\text{coerce}}$ a desired claim $\alpha_j \in [1, n_C] \cup \{\phi\}$, where ϕ denotes the special abstention claim. The challenger samples $B \leftarrow \mathcal{B}([1, n_V] \setminus \mathcal{D}, n_C)$. For every voter index i appearing in B but outside $[1, n_V]$, a fake credential $\text{fakecred}(\text{pk})$ replaces the genuine one.

A private bit $\beta \leftarrow \{0, 1\}$ is then drawn. If $\beta = 1$ (obey), each coerced voter’s entries are removed from B and replaced by the single pair (j, α_j) (or nothing if $\alpha_j = \phi$); if $\beta = 0$ (disobey), B is left unchanged. For each pair $(i, v) \in B$ in order, the adversary may first post messages to BB via the board oracle $\mathcal{O}_{\text{board}}$ (defined below). A fresh random tape $\rho \leftarrow \{0, 1\}^{\ell(\lambda)}$ is drawn and $b \leftarrow \text{Vote}(\text{pk}, \text{cred}_i, v; \rho)$ produces a ballot b , which is appended to BB . If $i \in \mathcal{H}_{\text{coerce}}$, the triple (v, ρ, b) is appended to $State[i]$. After all pairs in B are processed, the adversary posts a final batch of messages, validity-filtered by $\text{Valid}(\text{pk}, \cdot, BB)$. The joint tally procedure $\mathcal{P}_{\text{Tally}}^{\mathcal{A}}(BB, R, \text{pk}, \{(s_i, h_i)\}, t)$ produces a result r^* and a zero-knowledge proof Π^* . The adversary, given (r^*, Π^*) , may then call $\mathcal{O}_{\text{reveal}}$ (defined below) on each coerced voter. The experiment is *gated*: if $\mathcal{H}_{\text{coerce}} \not\subseteq Disc$ when the adversary finishes querying, a uniformly random bit is returned in place of the adversary’s guess. Otherwise the adversary’s output bit d is returned.

The oracle $\mathcal{O}_{\text{board}}$, returns $\text{Publish}(BB)$, the current public view of the bulletin board. The oracle $\mathcal{O}_{\text{reveal}}(j)$, $j \in \mathcal{H}_{\text{coerce}}$, adds j to $Disc$. It parses $State[j]$ as the sequence $\{(v^k, \rho^k, b^k)\}_k$ of genuine vote–coin–ballot triples. If $\beta = 1$, it returns the genuine local view $(\text{cred}_j, \{(v^k, \rho^k, b^k)\}_k)$. If $\beta = 0$, it sets $\mathbf{v}^* = (\alpha_j, \dots, \alpha_j)$ of the same length, invokes $(\text{cred}_j^*, \rho^*) \leftarrow \text{Fake}(\text{pk}, \text{cred}_j, \{v^k\}_k, \{\rho^k\}_k, \mathbf{v}^*)$, and returns the fabricated view $(\text{cred}_j^*, \{(\alpha_j, \rho^{*,k}, b^k)\}_k)$. In either case the ballots $\{b^k\}$ are identical; only the credential and coins may differ.

The ideal experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Ideal}}$ address that no protocol algorithm beyond the result function is executed. The adversary nominates $\mathcal{D}$ and $(\mathcal{H}_{\text{coerce}}, \{\alpha_j\})$ as before. The challenger samples β , builds a multi-set D from $\mathcal{B}$, and applies the same obey/disobey adjustment. The adversary then learns only: $|D|$ (the unavoidable leakage of any verifiable scheme), its own chosen votes $\{\nu_i\}_{i \in \mathcal{D}}$ appended to D , the tally

$\text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Real}}(\lambda, n_T, t, n_V, n_A, n_C, \mathcal{B})$	$\mathcal{O}_{\text{board}}()$
1: $BB \leftarrow \emptyset$ 2: $State \leftarrow \text{empty}$ 3: $Disc \leftarrow \emptyset$ 4: $(\text{pk}, \{(s_i, h_i)\}_{i=1}^{n_T}) \leftarrow \mathcal{P}_{\text{Setup}}^A(\lambda, n_T, t)$ 5: $\{\text{cred}_i\}_{i=1}^{n_V}, R \leftarrow \text{Register}(\lambda, \text{pk}, n_V)$ 6: $\mathcal{D} \leftarrow \mathcal{A}(R)$ 7: if $ \mathcal{D} \neq n_A$ then return 0 8: $(\mathcal{H}_{\text{coerce}}, \{\alpha_j\}_{j \in \mathcal{H}_{\text{coerce}}}) \leftarrow \mathcal{A}(\{\text{cred}_i\}_{i \in \mathcal{D}})$ 9: if $\mathcal{H}_{\text{coerce}} \cap \mathcal{D} \neq \emptyset$ then return 0 10: for $j \in \mathcal{H}_{\text{coerce}}$ do 11: if $\alpha_j \notin [1, n_C] \cup \{\phi\}$ then return 0 12: $B \leftarrow \mathcal{B}([1, n_V] \setminus \mathcal{D}, n_C)$ 13: for $(i, v) \in B$ with $i \notin [1, n_V]$ do 14: $\text{cred}_i \leftarrow \text{fakecred}(\text{pk})$ 15: $\beta \leftarrow \mathbb{S}\{0, 1\}$ 16: for $j \in \mathcal{H}_{\text{coerce}}$ do 17: if $\beta = 1$ then 18: remove all $(j, \star)$ from B 19: if $\alpha_j \neq \phi$ then $B \leftarrow B \parallel (j, \alpha_j)$ 20: else 21: keep $(j, \star)$ entries in B unchanged 22: for $(i, v) \in B$ in order do 23: $M \leftarrow \mathcal{A}^{\mathcal{O}_{\text{board}}}(BB)$ 24: $BB \leftarrow BB \parallel \{m \in M \mid \text{Valid}(\text{pk}, m, BB) = \top\}$ 25: $\rho \leftarrow \mathbb{S}\{0, 1\}^{\ell(\lambda)}$ 26: $b \leftarrow \text{Vote}(\text{pk}, \text{cred}_i, v; \rho)$ 27: $BB \leftarrow BB \parallel b$ 28: if $i \in \mathcal{H}_{\text{coerce}}$ then 29: $State[i] \leftarrow State[i] \parallel (v, \rho, b)$ 30: $M \leftarrow \mathcal{A}^{\mathcal{O}_{\text{board}}}(BB, \text{"casting closed"})$ 31: $BB \leftarrow BB \parallel \{m \in M \mid \text{Valid}(\text{pk}, m, BB) = \top\}$ 32: $(r^*, \Pi^*) \leftarrow \mathcal{P}_{\text{Tally}}^A(BB, R, \text{pk}, \{(s_i, h_i)\}_{i \in \mathcal{D}}, t)$ 33: $\mathcal{A}^{\mathcal{O}_{\text{reveal}}, \mathcal{O}_{\text{board}}}(r^*, \Pi^*)$ 34: if $\mathcal{H}_{\text{coerce}} \not\subseteq Disc$ then 35: return $d \leftarrow \mathbb{S}\{0, 1\}$ 36: $d \leftarrow \mathcal{A}()$ 37: return d	1: return $\text{Publish}(BB)$ $\mathcal{O}_{\text{reveal}}(j)$ for $j \in \mathcal{H}_{\text{coerce}}$ 1: $Disc \leftarrow Disc \cup \{j\}$ 2: parse $State[j]$ as $\{(v^k, \rho^k, b^k)\}_k$ 3: if $\beta = 1$ then 4: return $(\text{cred}_j, \{(v^k, \rho^k, b^k)\}_k)$ 5: else 6: $v^* \leftarrow (\alpha_j, \alpha_j, \dots, \alpha_j)$ 7: $(\text{cred}_j^*, \rho^*)$ 8: $\leftarrow \text{Fake}(\text{pk}, \text{cred}_j, \{v^k\}_k, \{\rho^k\}_k, v^*)$ 9: return $(\text{cred}_j^*, \{(\alpha_j, \rho^{*,k}, b^k)\}_k)$ $\text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Ideal}}(\lambda, n_V, n_A, n_C, \mathcal{B})$ 1: $D \leftarrow \emptyset; \mathcal{D} \leftarrow \mathcal{A}(\lambda)$ 2: if $ \mathcal{D} \neq n_A$ then return 0 3: $(\mathcal{H}_{\text{coerce}}, \{\alpha_j\}_{j \in \mathcal{H}_{\text{coerce}}}) \leftarrow \mathcal{A}()$ 4: if $\mathcal{H}_{\text{coerce}} \cap \mathcal{D} \neq \emptyset$ then return 0 5: for $j \in \mathcal{H}_{\text{coerce}}$ do 6: if $\alpha_j \notin [1, n_C] \cup \{\phi\}$ then return 0 7: $B \leftarrow \mathcal{B}([1, n_V] \setminus \mathcal{D}, n_C)$ 8: $\beta \leftarrow \mathbb{S}\{0, 1\}$ 9: for $j \in \mathcal{H}_{\text{coerce}}$ do 10: if $\beta = 1$ then 11: remove all $(j, \star)$ from B 12: if $\alpha_j \neq \phi$ then $B \leftarrow B \parallel (j, \alpha_j)$ 13: else 14: keep $(j, \star)$ entries in B unchanged 15: for $(i, v) \in B$ with $i \in [1, n_V] \cup \{\text{neg}\}$ do 16: $D \leftarrow D \parallel (i, v)$ 17: $(\{\nu_i\}_{i \in \mathcal{D}}, \beta_{\text{adv}}) \leftarrow \mathcal{A}(D)$ 18: for $i \in \mathcal{D}$ do 19: if $\nu_i \in [1, n_C]$ then $D \leftarrow D \parallel (i, \nu_i)$ 20: $A \leftarrow \{(j, \alpha_j) : j \in \mathcal{H}_{\text{coerce}}\}$ 21: $r^* \leftarrow \text{Result}(\text{Policy}(\text{cleanse}(D)))$ 22: $d \leftarrow \mathcal{A}(r^*, A)$ 23: return d

Fig. 4: The real experiment $\text{Exp}^{\text{PCR-Real}}$: After the tally is published, the adversary may issue $\mathcal{O}_{\text{reveal}}$ queries on any coerced voter to obtain a complete (real or faked) local view; the experiment is gated so that the adversary must exhaust this oracle on every coerced voter before guessing. The ideal experiment: The adversary learns the size of D (modelling the unavoidable leakage of board length), the tally r^* , and a claim list A that announces, for every coerced voter, the vote the voter publicly claims to have cast, nothing about whether the claim is genuine or fabricated.

$r^* = \text{Result}(\text{Policy}(\text{cleanse}(D)))$ where Policy enforces the re-voting policy and cleanse resolves multiple ballots per voter, and the claim list $\Lambda = \{(j, \alpha_j) : j \in \mathcal{H}_{\text{coerce}}\}$ recording, for every coerced voter, the vote they publicly claim to have cast, irrespective of β .

The universality of $\mathcal{B}$. Our definition quantifies over *all* behavioral distributions $\mathcal{B}$, not merely a fixed family. This matters for two reasons.

First, the disclosed view in the Real world reveals correlations between the original coins ρ and the cast vote v ; if Fake produces fake coins whose joint distribution with α_j differs measurably from the honest joint distribution under any $\mathcal{B}$, the gap between Real and Ideal becomes non-negligible for that specific $\mathcal{B}$, and the scheme fails. Universal quantification over $\mathcal{B}$ therefore forces Fake to produce coins whose distribution is faithful for *every* possible voter behavior, not merely the uniform or some assumed "natural" one.

Second, by allowing $\mathcal{B}$ to inject ballots with negative indices, i.e. ballots bearing fabricated credentials authored by authorities or external parties, the experiment captures the case in which the public board contains noise the adversary did not produce and the coerced voter did not produce, and demands that even such noise leak nothing beyond $|D|$, r^* , and Λ . This is the post-tally analogue of the cleansing-leakage analysis of [22], extended to the persistent-disclosure setting where the adversary may interrogate voters' coins in arbitrary order after observing the result.

4.1 Persistent Coercion Resistance Implies Coercion Resistance

The intuition is that any CR distinguisher lifts directly to a PCR distinguisher by deferring credential exposure to the post-tally $\mathcal{O}_{\text{reveal}}$ oracle, since PCR's adversary observes a strict superset of CR's allowed leakage, which the following theorem address it; the formal proof of theorems are available in the Appendix of full version [106] of this work.

Theorem 1. *Let Π be a voting scheme. If Π satisfies Persistent Coercion Resistance (Definition 2), then Π satisfies Coercion Resistance (Definition 1).*

PCR's adversary observes every leakage CR's does, plus more: the full local view of every coerced voter disclosed post-tally is a strict superset of CR's bare credential disclosure during voting, and PCR's universal quantification over behavioral distributions $\mathcal{B}$ subsumes every voter-behavior configuration CR considers, including revoting, abstention, and fake-credential injection by authorities. Any CR distinguisher therefore lifts to a PCR distinguisher by deferring CR's credential exposure to the $\mathcal{O}_{\text{reveal}}$ oracle, with the resulting view differing only by the negligible statistical slack of the protocol's faking procedure.

4.2 Quantitative Framework for Persistent Coercion Resistance

Here, we quantify the PCR generally, and for further details refer to the Appendix of full version [106]. The qualitative Definition 2 only asks whether the Real-Ideal gap is negligible. Practical voting schemes meet this bar only up to some residual leakage, so to compare schemes meaningfully we develop a quantitative framework, in the spirit of Kuesters et al. [13] and its application to coercion resistance in [22], extended to our persistent-disclosure setting where the coercer interrogates the voter's local view *after* the tally.

Three structural features of the persistent setting prevent direct reuse of the CR analysis in [13, 22]. (i) The adversary's view contains the complete local state of every

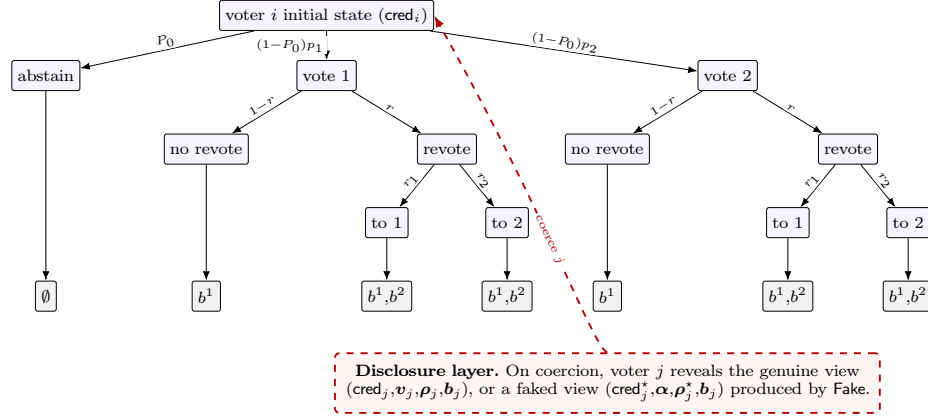


Fig. 5: Per-voter behavioral tree with persistent-disclosure layer ($n_C = 2$, at most one revote). Black solid arcs are the honest casting probabilities; the red dashed arc denotes the post-tally disclosure event in PCR. The leaf records both the final-vote multiset and the sequence of cast ballots $\mathbf{b}_j$ that the coercer obtains under disclosure.

coerced voter: credential, every random tape, every ballot. (ii) The experiment admits simultaneous coercion of $q_c \geq 1$ voters whose disclosures may be interrogated in arbitrary post-tally order. (iii) Each coerced voter may have cast $k_{\text{vot}} \geq 1$ ballots before disclosure, so persistent revoting is first-class. None of these features is captured by [22] work.

Behavioral model and the persistent gap. We model each honest voter as a random walk on a probability tree, extended with an explicit *disclosure* branch: the local state generated along the path is recorded and may be exposed under coercion. This is the formal device that distinguishes our analysis from prior work. Figure 5 shows the per-voter tree for $n_C = 2$ candidates with at most one revote, annotated with the disclosure point. Each voter abstains with probability P_0 , otherwise casts a first vote with distribution $(p_1, \dots, p_{n_C})$; given a first vote, revoting occurs with probability r under distribution $(r_1, \dots, r_{n_C})$. Figure 5 traces one honest voter’s path through the protocol: solid black arcs are the casting probabilities (P_0 for abstention, p_ℓ for a first vote on candidate $\ell \in \{1, 2\}$, r for revoting and r_ℓ for the revote target), the leaves record the resulting ballot sequence $\mathbf{b}_j$ on the public board, and the red dashed arc marks the post-tally event in which a coercer interrogates voter j and obtains either the genuine local view or a Fake-produced one.

We summarize the asymmetry between first-round and revote behavior by the *revoting bias* $\text{Bias}(\mathbf{p}, \mathbf{r}) = \frac{1}{2} \sum_\ell |p_\ell - r_\ell| \in [0, 1]$, with 0 denoting symmetric revoting and 1 complete preference inversion (e.g., a candidate becoming discredited mid-election). Integrating out the revote branch gives a *marginal final-vote distribution* $\mathbf{P}^\circ$ which we use throughout to quote asymptotic rates; both the marginal and the bias generalize the quantities of [22, App. D] to a setting where the voter’s disclosure is part of the leakage.

For an intention vector α and a behavioural distribution $\mathcal{B}$, let $\delta_\alpha^{\text{Real}}(\Pi)$ and $\delta_\alpha^{\text{Ideal}}(\Pi)$ denote the total-variation distance between the $\beta=0$ and $\beta=1$ joint views in $\text{Exp}^{\text{PCR-Real}}$ and $\text{Exp}^{\text{PCR-Ideal}}$ respectively. By the Neyman–Pearson lemma [107], these are exactly the optimal distinguishing advantages of any PPT coercer, and Π satisfies PCR iff the

persistent gap $\Delta^{\text{pers}}(\Pi) := \delta^{\text{Real}}(\Pi) - \delta^{\text{Ideal}}(\Pi)$ (maximised over $\mathcal{B}, \alpha$) is negligible. The gap isolates the leakage attributable to persistent disclosure beyond the unavoidable [108] tally leakage.

A standard sufficient-statistic argument reduces the optimal coercer to a likelihood-ratio test on the disclosed view; for the ideal experiment with $n_C = 2$ and $q_c = 1$, the test admits a closed-form expression as a Total Variation TV-distance between two multinomials differing by a unit shift on coordinate α , computable in $O((n_V + 1)^2)$ floating-point operations via log-gamma arithmetic. Universal quantification over $\mathcal{B}$ is essential here: any subtle correlation between the original coins ρ and the cast vote v that Fake fails to reproduce for *some* $\mathcal{B}$ makes the gap non-negligible for that $\mathcal{B}$ and breaks the scheme. Quantification over all $\mathcal{B}$, including distributions that inject ballots with fabricated credentials authored by authorities, also captures the post-tally analogue of the cleansing-leakage analysis of [22] work.

Per-ballot deniability and the channel decomposition. To bound the gap independently of any concrete construction we parameterize disclosure leakage abstractly. For a scheme Π whose ballot algorithm consumes randomness of length $\ell(\lambda)$ and whose Fake algorithm produces fake coins $\rho^* \leftarrow \text{Fake}(\text{pk}, \text{cred}, v, \rho, v^*)$, the *per-ballot deniability advantage* $\varepsilon_{\text{ball}}(\Pi, \lambda)$ is the maximum advantage of any PPT adversary in distinguishing a genuine pair (v^*, ρ, b^*) from a faked pair (v^*, ρ^*, b) . This single quantity captures everything a coercer can extract from one disclosed ballot beyond the public board.

Our primary quantitative finding is that the persistent gap decomposes additively into four orthogonal information channels. For every PPT adversary issuing at most q_c disclosure queries on voters casting at most k_{vot} ballots each, every behavioral distribution, and every claim vector,

$$\delta_{\alpha}^{\text{Real}}(\Pi) \leq \delta_{\alpha}^{\text{Ideal}}(\Pi) + \delta^{\text{tally}}(\Pi, \lambda) + q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}(\Pi, \lambda) + \delta^{\text{board}}(\Pi, \lambda),$$

where δ^{tally} and δ^{board} are the tally simulator and board CPA advantages of Π respectively. The adversary's view factorises into four independent components: a tally channel (closed by the tally simulator), a board channel (closed by encryption), $q_c \cdot k_{\text{vot}}$ disclosure channels (closed by Fake), and an unavoidable claim channel (which yields the ideal advantage). No single primitive needs to absorb the entire distinguishing power, and persistent revoting amplifies leakage by exactly the factor k_{vot} . This bound is tight up to a factor of 4: a matching lower bound is witnessed by explicit adversaries targeting each channel separately, so no constant-factor improvement is achievable without strengthening at least one underlying primitive. When all three of $\delta^{\text{tally}}, \delta^{\text{board}}, \varepsilon_{\text{ball}}$ are negligible, Π satisfies qualitative PCR. In the asymptotic regime the ideal advantage follows the central-limit-theorem rate

$$\delta_{\alpha}^{\text{Ideal}}(\Pi, n_V) = \frac{1 + O(n_V^{-1/2})}{\sqrt{2\pi P_{\alpha}^{\circ}(1 - P_{\alpha}^{\circ})n_V}},$$

which recovers the $1/\sqrt{n_V}$ regime of [109], with the pre-factor showing that rare candidates leak more per vote. The persistent contribution $q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}$ scales linearly and dominates the ideal floor exactly when $\varepsilon_{\text{ball}} \cdot q_c \cdot k_{\text{vot}} > 1/\sqrt{n_V}$, a threshold any scheme aiming at PCR must beat.

The decomposition also reveals why PCR is stronger than CR. A scheme without a Fake algorithm has $\varepsilon_{\text{ball}} = \Theta(1)$, under disclosure the coercer simply checks whether the revealed plaintext matches the claim α^* , which holds deterministically under obey and

only with marginal probability $P_{\alpha^*}^o$ under evade. The persistent gap then saturates at

$$\Delta^{\text{pers}}(\Pi) \geq 1 - \min_{\alpha} P_{\alpha}^o - \delta^{\text{ideal}}(\Pi, n_V) = 1 - \frac{1}{n_C} - O(1/\sqrt{n_V})$$

for an explicitly constructible $\mathcal{B}^*$. Conventional credential faking does not rescue such schemes because the credential is independent of the disclosed vote. Finally, under independent coercion of multiple voters the gap composes linearly, $\delta_{\alpha}^{\text{ideal}}(\Pi, q_c) \leq \sum_j \delta_{\alpha_j}^{\text{ideal}}(\Pi, 1)$, tight up to factor 2 in the worst case.

Computational evaluation. The closed forms above are computationally tractable. The exact ideal advantage is computable in $O((n_V + 1)^{n_C})$ floating-point operations via direct enumeration; for $n_V \gtrsim 10^3$ or $n_C \geq 3$ we replace exact enumeration by Monte Carlo with Hoeffding bound

$$\Pr[|\hat{\delta}^{\text{ideal}} - \delta^{\text{ideal}}| \geq \varepsilon] \leq 2e^{-2N\varepsilon^2}, \quad (1)$$

which yields $\varepsilon \leq 4 \cdot 10^{-3}$ at $N = 10^5$.

Quantifying experiments. Our experiments demonstrate the four independent channels of the decomposition, validate the $1/\sqrt{n_V}$ asymptotic rate, witness the strict separation between PCR and CR for no-fake schemes, and exhibit the linear composition under multiple coerced voters. We initialize everything from the behavioral tree of Figure 5, evaluate the ideal closed form and the decomposition bound at varying parameter sweeps, and compare against the impossibility lower bound for no-fake schemes which witnesses the strict $\text{PCR} \neq \text{CR}$ separation. All computations are exact via log-gamma arithmetic for $n_V \leq 256$ and Monte Carlo with bound (1) otherwise.

In simple words, across our new experiments (Figure 18), Figure 12 shows that the persistent gap shrinks exactly in proportion to how good Fake is at hiding randomness, Figure 11 shows that letting a coerced voter cast more ballots before disclosure amplifies the leakage by exactly that ballot count, Figure 14 shows that coercing several voters at once raises the leakage by roughly the number of coerced voters, Figure 15 shows that a Fake algorithm with a $2^{-\lambda}$ bound buys exponentially strong deniability as λ grows, Figure 17 maps out the parameter region in which a scheme without Fake leaks far more than the unavoidable floor, Figure 16 confirms on a log-log scale that doubling either the number of coerced voters or the ballots they cast doubles the leakage, and Figure 13 shows that adding more candidates makes the result harder to read but widens the gap between PCR-secure and no-fake schemes.

More detailed, Figures 6–10 reproduce, in the persistent-disclosure setting, the classical CR sweeps of [22, Figs. 2, 4, 5, 6, 14]: the Real bound stays within $\varepsilon_{\text{ball}}$ of the Ideal curve everywhere, whereas the no-Fake bound sits at $\delta = 1$ for every revote rate, every initial proportion, and every abstention level, which witnesses the strict $\text{PCR} \neq \text{CR}$ separation across all parameter regimes. Figure 10 additionally tracks the $\Theta(1/\sqrt{n_V})$ asymptotic reference curve to within 3% across $n_V \in \{16, \dots, 256\}$, which confirms the central-limit-theorem rate of the ideal advantage. Figures 12–14 validate the three controllable channels of the decomposition: sweeping $\log_2(1/\varepsilon_{\text{ball}})$ from 4 to 20, the persistent gap tracks the linear bound $q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}$ to within 1% on a log-log scale, and the linear factors in k_{vot} and q_c are saturated exactly, which confirms that persistent revoting and simultaneous coercion amplify leakage at the predicted rates. Figures 15–13 address another regime: $\varepsilon_{\text{ball}}$ decays exponentially in λ for schemes whose Fake inherits a $2^{-\lambda}$ statistical-distance bound, which validates the negligibility hypothesis of the decomposition; the disclosure workload $q_c \cdot k_{\text{vot}}$ exhibits log-log slope 1 across three $\varepsilon_{\text{ball}}$ regimes, in line with the

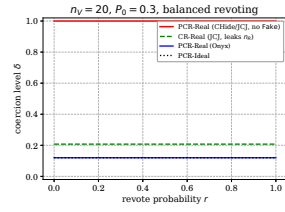


Fig. 6: PCR vs. re-vote rate

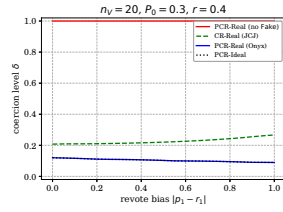


Fig. 7: PCR vs. re-vote bias

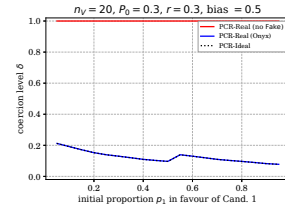


Fig. 8: PCR vs. initial proportion

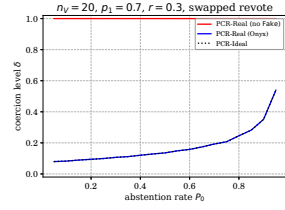


Fig. 9: PCR vs. abstention rate

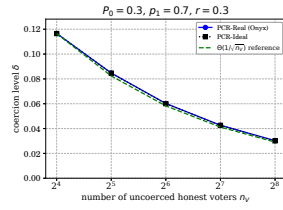


Fig. 10: PCR vs. number of voters

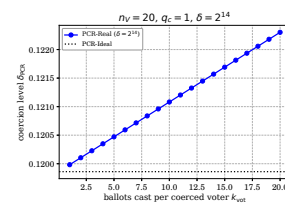


Fig. 11: Persistent gap vs. ballots per voter

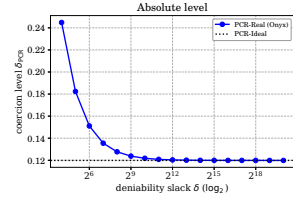


Fig. 12: Persistent gap vs. deniability of Fake

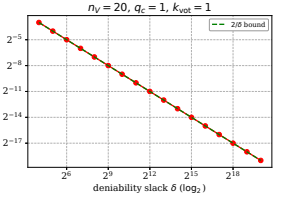


Fig. 13: PCR vs. number of candidates

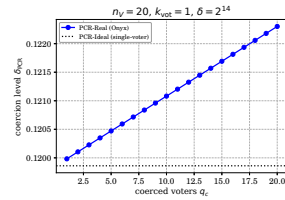


Fig. 14: Persistent gap vs. number of coerced voters

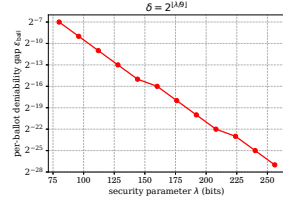


Fig. 15: Deniability vs. security parameter

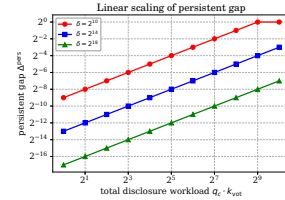


Fig. 16: Linear scaling of disclosure workload

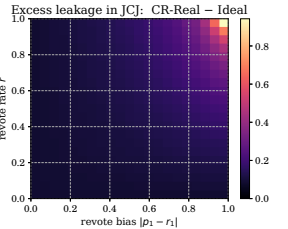
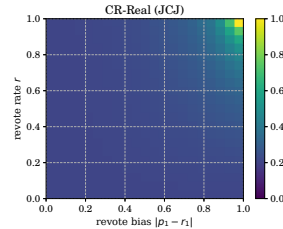
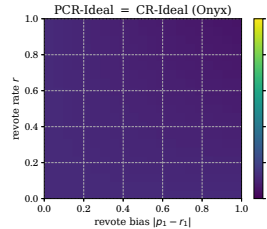


Fig. 17: Phase space of leakage

Fig. 18: Quantifying experiments of Persistent Coercion Resistance.

joint tightness of the additive bound and its matching lower bound; and increasing n_C dilutes the per-vote signal in the Ideal world while strengthening the PCR $\neq$ CR separation, since $\min_{\alpha} P_{\alpha}^{\circ} \rightarrow 1/n_C$. Finally, Figure 17 maps the two-dimensional phase space of Ideal-vs-Real-CR advantages over (r, Bias) , which visualizes the regime in which a CR scheme without Fake function leaks substantially beyond the ideal floor.

5 Onyx: Persistent Coercion-Resistant Voting

Here, we present Onyx, a post-quantum voting system satisfying Persistent Coercion Resistance (Definition 2). Onyx combines a CHide-style [22] cleansing-hiding pipeline operating over a threshold deniable FHE layer, with persistent randomness deniability propagated through every layer of voter-side.

5.1 Building Blocks

In the following we go through the construction primitives and for formal building blocks properties, concrete parameters and instantiations refer to the Appendix of full version [106].

Definition 3 (Deniable Fully Homomorphic Encryption [93]). A compact public-key deniable fully homomorphic encryption (dFHE) scheme for message space $\mathcal{M}$, based on the polynomial hardness of LWE, is a tuple of PPT algorithms $\text{dFHE} = (\text{Gen}, \text{Enc}, \text{Eval}, \text{Dec}, \text{Fake})$:

- $(\text{dpk}, \text{dsk}) \leftarrow \text{Gen}(1^\lambda)$: Takes security parameter 1^λ ; outputs a public key dpk and a secret key dsk .
- $\text{dct} \leftarrow \text{Enc}(\text{dpk}, m; r)$: Takes public key dpk , message $m \in \mathcal{M}$, and ℓ -bit randomness r ; outputs a ciphertext dct .
- $\text{dct} \leftarrow \text{Eval}(\text{dpk}, C, \text{dct}_1, \dots, \text{dct}_k)$: Takes public key dpk , a polynomial-time circuit $C : \mathcal{M}^k \rightarrow \mathcal{M}$, and ciphertexts $\text{dct}_1, \dots, \text{dct}_k$ encrypting $m_1, \dots, m_k$; outputs a ciphertext dct encrypting $C(m_1, \dots, m_k)$.
- $m \leftarrow \text{Dec}(\text{dsk}, \text{dct})$: Takes secret key dsk and ciphertext dct ; outputs a message $m \in \mathcal{M}$.
- $r^* \leftarrow \text{Fake}(\text{dpk}, m, r, m^*)$: Takes public key dpk , original message $m \in \mathcal{M}$, randomness $r \in \{0, 1\}^\ell$, and a target fake message $m^* \in \mathcal{M}$; outputs fake randomness r^* such that $\text{Enc}(\text{dpk}, m; r) = \text{Enc}(\text{dpk}, m^*; r^*)$.

Definition 4 (Universal Thresholdizer [94]). Consider the efficient access structure $(\mathbb{S}, N)$ and circuit class $(\mathbb{C}, d)$. A universal thresholdizer UT consists of the algorithms:

- $(\text{pp}, s_1, \dots, s_N) \leftarrow \text{Setup}(1^\lambda, 1^d, (\mathbb{S}, N), x)$: Given the security parameter 1^λ , a depth bound 1^d , an efficient access structure $(\mathbb{S}, N)$, and a message $x \in \{0, 1\}^k$, this algorithm outputs public parameters pp and a set of shares $s_1, \dots, s_N$.
- $y_i \leftarrow \text{Eval}(\text{pp}, s_i, \mathbb{C})$: On input of the public parameters pp , a share s_i , and a circuit $\mathbb{C} : \{0, 1\}^k \rightarrow \{0, 1\}$ of depth at most d , the algorithm produces a partial evaluation y_i .
- $\{0, 1\} \leftarrow \text{Verify}(\text{pp}, y_i, \mathbb{C})$: This algorithm takes as input pp , a purported partial evaluation y_i , and a circuit $\mathbb{C}$, and outputs a bit indicating whether the evaluation is valid.
- $y \leftarrow \text{Combine}(\text{pp}, B)$: Outputs the final evaluation y , given the collection of partial evaluations $B = \{y_i\}_{i \in S}$ from an authorized set S .

Definition 5 (Verifiable Secret Sharing [110]). A hash-based (t, n) -Verifiable Secret Sharing (VSS) scheme Π , in the quantum random oracle model with $n \geq 2t+1$, consists of the algorithms:

- $(\{s_i, \gamma_i\}_{i=1}^n, \pi_{\text{share}}) \leftarrow \text{Share}(s, t, n)$: Takes a secret $s \in \mathbb{Z}_N$, threshold t , and party count n ; outputs n share-randomizer pairs $\{s_i, \gamma_i\}_{i=1}^n$ and a public proof $\pi_{\text{share}} = (z(X), c_1, \dots, c_n)$, where $c_i = H(s_i \| \gamma_i)$ are hash-based commitments and $z(X) = r(X) + d \cdot f(X)$ is the masked polynomial for challenge $d = H(c_1 \| \dots \| c_n)$.
- $\{0, 1\} \leftarrow \text{VerifyShare}(i, s_i, \gamma_i, \pi_{\text{share}})$: Takes index i , share s_i , randomizer γ_i , and proof π_{share} ; outputs 1 if $\deg(z(X)) \leq t$ and $c_i = H(s_i \| z(i) - d \cdot s_i)$, else 0.
- $s \leftarrow \text{Reconstruct}(\{(i, s_i, \gamma_i)\}_{i \in S}, \pi_{\text{share}})$: Takes a set of verified share tuples with $|S| \geq t+1$ and proof π_{share} ; outputs the secret $s = \sum_{i \in S} \lambda_i^S \cdot s_i \pmod{N}$ via Lagrange interpolation.

Definition 6 (Lattice-Based Zero-Knowledge Proof System [95]). The proof system ZKP, based on the hardness of MSIS and MLWE, proves knowledge of a witness satisfying a specified relation $\mathcal{R}$. $\text{ZKP} = (\text{Setup}, \text{Prove}, \text{Verify})$ consists of:

- $pp \leftarrow \text{Setup}(1^\lambda)$: Outputs public parameters pp defining the relation $\mathcal{R}$.
- $\pi \leftarrow \text{Prove}(pp, x, w)$: Given a statement x and witness w such that $(x, w) \in \mathcal{R}$, outputs a proof π via the Fiat–Shamir transform.
- $\{0, 1\} \leftarrow \text{Verify}(pp, x, \pi)$: Accepts if π is a valid proof that the prover knows a witness w such that $(x, w) \in \mathcal{R}$, and rejects otherwise.

Definition 7 (Verifiable Mix-Net [111]). A verifiable mix-net MixNet is a multi-server protocol for BGV ciphertexts whose security relies on the hardness of the DKS^∞ and SKS^2 lattice problems. $\text{MixNet} = (\text{Setup}, \text{Mix}, \text{MixVerify}, \text{DDec}, \text{DDecVerify}, \text{Comb})$ consists of:

- $pp \leftarrow \text{Setup}(1^\kappa, \xi_1, \xi_2)$: On security parameter 1^κ , shuffle-server count ξ_1 , and decryption-server count ξ_2 ; outputs public parameters pp comprising a BGV public key pk , a BDLOP commitment key $\mathbf{A}_C$, and key shares $\{sk_j\}_{j \in [\xi_2]}$ secret-shared among decryption servers with published commitments $\{\llbracket s_j \rrbracket\}_{j \in [\xi_2]}$.
- $(\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \pi_{\text{Mix}}) \leftarrow \text{Mix}(pp, \{\mathbf{c}_i\}_{i \in [\tau]})$: Executed by shuffle server $\mathcal{S}$ on τ input BGV ciphertexts; samples fresh encryptions of zero $\mathbf{c}'_i = (ar'_i + pe'_{i,1}, br'_i + pe'_{i,2})$, commits to each as $\llbracket \mathbf{c}'_i \rrbracket$, forms re-randomized ciphertexts $\hat{\mathbf{c}}_i = \mathbf{c}_i + \mathbf{c}'_i$, and outputs the randomly permuted list $L = (\hat{\mathbf{c}}_{\pi(i)})_{i \in [\tau]}$ together with $\pi_{\text{Mix}} = (\{\mathbf{t}_i\}_{i \in [\tau]}, \pi_{\text{Small}}, L, \pi_{\text{Shuf}})$, where π_{Small} is an amortized zero-knowledge proof that each $\mathbf{c}'_i$ is a well-formed encryption of zero and π_{Shuf} is a verifiable shuffle proof that L is a correct permuted opening of the committed ciphertexts.
- $\{0, 1\} \leftarrow \text{MixVerify}(pp, \{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \{\mathbf{c}_i\}_{i \in [\tau]}, \pi_{\text{Mix}})$: Verifies π_{Small} for the committed re-randomizations, recomputes $\llbracket \hat{\mathbf{c}}_i \rrbracket = \llbracket \mathbf{c}_i \rrbracket_0 + \mathbf{t}_i$, and verifies the shuffle proof π_{Shuf} ; accepts iff all checks pass.
- $(\{t_{i,j}\}_{i \in [\tau]}, \pi_{\text{Dec}_j}) \leftarrow \text{DDec}(sk_j, \{\hat{\mathbf{c}}_i\}_{i \in [\tau]})$: Executed by decryption server $\mathcal{D}_j$ holding share $sk_j = s_j$; for each $\hat{\mathbf{c}}_i = (u_i, v_i)$ computes partial decryption $t_{i,j} = s_j u_i + p E_{i,j}$ with drowning noise $\|E_{i,j}\|_\infty \leq 2^{\text{sec}}(B_{\text{Dec}}/p\xi_2)$, and outputs $\pi_{\text{Dec}_j} = (\{\llbracket E_{i,j} \rrbracket\}_{i \in [\tau]}, \{\pi_{L_{i,j}}\}_{i \in [\tau]}, \pi_{\text{Bnd}_j})$, comprising zero-knowledge proofs of the linear relation $t_{i,j} = s_j u_i + p E_{i,j}$ and an amortized boundedness proof that all $E_{i,j}$ are correctly bounded.

- $\{0,1\} \leftarrow \text{DDecVerify}(pp, \{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \{t_{i,j}\}_{i \in [\tau]}, \pi_{\text{Dec}_j})$: Verifies the linearity proofs $\{\pi_{L_{i,j}}\}$ against the public commitment $\llbracket s_j \rrbracket$ and committed noise $\{\llbracket E_{i,j} \rrbracket\}$, and verifies the amortized boundedness proof π_{Bnd_j} ; accepts iff all proofs verify.
- $\{m_i\}_{i \in [\tau]} \cup \{\perp\} \leftarrow \text{Comb}(\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \{\{t_{i,j}\}_{i \in [\tau]}\}_{j \in [\xi_2]})$: Runs DDecVerify for every $j \in [\xi_2]$; outputs $\perp$ if any proof fails, else recovers each plaintext as $m_i = (v_i - t_i \bmod q) \bmod p$ where $t_i = \sum_{j=1}^{\xi_2} t_{i,j}$, and returns $\{m_i\}_{i \in [\tau]}$.

5.2 Non-Interactive Prover-Deniable Zero-Knowledge Proof

In this work, we need a sender-deniable proof system to construct a PCR voting scheme. There are already designated-verifier proofs (DVPs) [35, 36], online/offline deniable proofs [37], transferable zero-knowledge [38], and many other types of proof systems, but none of the existing ones simultaneously achieves non-interactive prover-side randomness deniability in the post-quantum setting under standard assumptions without requiring a random oracle for the deniability guarantee.

So, here we use LNP [39] to construct a non-interactive argument of knowledge for relations $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ quadratically expressible over $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle X^d + 1 \rangle$, together with a prover-only Fake algorithm that produces alternative randomness consistent with any published proof under any other valid witness. The four-tuple $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify}, \text{Fake})$ has the standard non-interactive syntax, with the additional algorithm $r^* \leftarrow \text{Fake}(\text{crs}, \text{tdkey}, x, w, r, w^*)$ returning r^* such that $\text{Prove}(\text{crs}, \text{tdkey}, x, w^*; r^*)$ yields the same proof as the genuine run $\text{Prove}(\text{crs}, \text{tdkey}, x, w; r)$.

Our construction combines three primitives. First, a *dual-mode equivocable commitment* of the BDLOP family [112] placed in its equivocation mode via a Micciancio–Peikert trapdoor [40]: given matrices $\mathbf{A}_{\text{com}} \leftarrow \mathcal{R}_q^{n \times m}$ and a short $\mathbf{T} \leftarrow \chi^{\ell \times m}$, the publicly visible commitment matrix is $\mathbf{B}'_{\text{com}} = \mathbf{B}_{\text{com}} - \mathbf{A}_{\text{com}} \mathbf{T}^\top$ for an independent uniform $\mathbf{B}_{\text{com}}$; the pair $(\mathbf{A}_{\text{com}}, \mathbf{B}'_{\text{com}})$ is computationally indistinguishable from uniform under $\text{MLWE}_{n,m,q,\chi}$, hiding $\mathbf{T}$ from the verifier. Second, the relation-proof component of LNP [39] for quadratically expressible relations, which supplies the verifier matrix $\mathbf{C}$ and the response check on $(\mathbf{u}, c, \mathbf{z})$. Third, a SIS-keyed collision-resistant hash $H_{\text{chal}}: \{0,1\}^* \rightarrow \mathcal{R}_q$ used as a deterministic challenge derivation function. We work in the common-reference-string model with a prover-only equivocation parameter. We assume a setup authority that generates and distributes tdkey honestly and then erases its retained state; provers receive $(\text{crs}, \text{tdkey})$, verifiers receive crs only, and a malicious prover even in possession of tdkey cannot prove false statements (knowledge soundness holds against trapdoor-aware adversaries).

Setup and common reference string. **Setup** samples the commitment matrices $\mathbf{A}_{\text{com}}, \mathbf{B}_{\text{com}}$, the verifier matrix $\mathbf{C} \leftarrow \mathcal{R}_q^{k \times (m+\ell)}$, and the SIS-keyed hash H_{chal} , together with the short trapdoor $\mathbf{T}$; the verifier-visible crs consists of $(\mathbf{A}_{\text{com}}, \mathbf{B}'_{\text{com}}, \mathbf{C}, H_{\text{chal}})$ where $\mathbf{B}'_{\text{com}} = \mathbf{B}_{\text{com}} - \mathbf{A}_{\text{com}} \mathbf{T}^\top$. The witness w is mapped to a ring vector $\mathbf{s} = \text{Encode}(w) \in \mathcal{R}_q^\ell$ with $\|\mathbf{s}\|_\infty \leq \beta_w$, where Encode is the relation-specific embedding fixed by $\mathcal{R}$. Throughout, n, m, ℓ, k are lattice dimensions, q is a prime modulus, χ is a discrete-Gaussian distribution of width σ , and β_z, β_w are norm bounds on the response and the witness respectively.

Prove and verify. The prover's randomness tape r is partitioned as $r = (r_{\text{com}}, r_{\text{samp}})$, where $r_{\text{com}} \in \mathcal{R}_q^m$ is the commitment randomness and r_{samp} is the raw uniform tape consumed by the rejection sampler RejSamp , which produces a discrete-Gaussian

masking vector $\mathbf{y}$ with rejection probability bounded by a fixed factor M . The prover computes the commitment $\mathbf{u} = \mathbf{A}_{\text{com}} r_{\text{com}} + \mathbf{B}'_{\text{com}} \mathbf{s}$, derives the deterministic challenge $c = H_{\text{chal}}(\text{crs}, x, \mathbf{u})$, runs **RejSamp** on r_{samp} with shift $c \cdot \mathbf{s}$, produces the response $\mathbf{z} = r_{\text{com}} + \mathbf{y}$, and attaches the relation-proof sub-component π_{rel} tying $(\mathbf{u}, c, \mathbf{z})$ to the quadratic constraints of $\mathcal{R}$ via $\mathbf{C}$. The verifier recomputes c , checks $\|\mathbf{z}\|_2 \leq \beta_z$, and runs the LNP [39] relation-proof verifier on $(\mathbf{u}, c, \mathbf{z}, \pi_{\text{rel}})$.

Fake. The **Fake** utilizes the equivocation property of the trapdoor commitment. Given the genuine $(r_{\text{com}}, r_{\text{samp}})$ used to produce π with witness encoding $\mathbf{s}$, and a target encoding $\mathbf{s}^*$, the prover computes $r_{\text{com}}^* = r_{\text{com}} + \mathbf{T}^\top (\mathbf{s}^* - \mathbf{s})$ and adjusts the masking residue accordingly. The commitment is preserved because $\mathbf{A}_{\text{com}} r_{\text{com}}^* + \mathbf{B}'_{\text{com}} \mathbf{s}^* = \mathbf{A}_{\text{com}} r_{\text{com}} + (\mathbf{A}_{\text{com}} \mathbf{T}^\top + \mathbf{B}'_{\text{com}}) \mathbf{s}^* - \mathbf{A}_{\text{com}} \mathbf{T}^\top \mathbf{s}^* + \mathbf{A}_{\text{com}} \mathbf{T}^\top \mathbf{s} - \mathbf{A}_{\text{com}} \mathbf{T}^\top \mathbf{s} + \mathbf{B}'_{\text{com}} \mathbf{s} - \mathbf{B}'_{\text{com}} \mathbf{s} = \mathbf{u}$ after cancellation; since $\mathbf{B}'_{\text{com}} + \mathbf{A}_{\text{com}} \mathbf{T}^\top = \mathbf{B}_{\text{com}}$, the equivocated commitment lies on the same syndrome as the original. The masking inverse RejSamp^{-1} recovers a uniform tape consistent with the adjusted shift $c \mathbf{s}^*$ whenever the norm gap $\|\mathbf{T}^\top (\mathbf{s}^* - \mathbf{s})\|$ falls within the smudging slack of σ , which holds for all $\mathbf{s}^*$ of norm $\leq \beta_w$.

Setup _{crs} (1^λ)	Verify(crs, x, π)
1: $\mathbf{A}_{\text{com}} \leftarrow \mathcal{R}_q^{n \times m}, \mathbf{T} \leftarrow \mathcal{X}^{\ell \times m}, \mathbf{B}_{\text{com}} \leftarrow \mathcal{R}_q^{n \times \ell}$ 2: $\mathbf{B}'_{\text{com}} \leftarrow \mathbf{B}_{\text{com}} - \mathbf{A}_{\text{com}} \mathbf{T}^\top$ 3: $\mathbf{C} \leftarrow \mathcal{R}_q^{k \times (m + \ell)}$ 4: $H_{\text{chal}}: \{0, 1\}^* \rightarrow \mathcal{R}_q$ 5: $\text{crs} \leftarrow (\mathbf{A}_{\text{com}}, \mathbf{B}'_{\text{com}}, \mathbf{C}, H_{\text{chal}}); \text{td} \leftarrow \mathbf{T}$ 6: return (crs, td)	1: parse π as $(\mathbf{u}, \mathbf{z}, \pi_{\text{rel}})$ 2: $c \leftarrow H_{\text{chal}}(\text{crs}, x, \mathbf{u})$ 3: if $\ \mathbf{z}\ _2 > \beta_z$ then return 0 4: if $\text{Verify}_{\text{rel}}(\text{crs}, x, \mathbf{u}, c, \mathbf{z}, \pi_{\text{rel}}) = 0$ then return 0 5: return 1
Fake(crs, td, x, w, r, w^*)	Prove(crs, td, x, w, r)
1: parse r as $(r_{\text{com}}, r_{\text{samp}})$ 2: $\mathbf{s} \leftarrow \text{Encode}(w); \mathbf{s}^* \leftarrow \text{Encode}(w^*)$ 3: if $(x, w^*) \notin \mathcal{R}$ then return $\perp$ 4: $r_{\text{com}}^* \leftarrow r_{\text{com}} + \mathbf{T}^\top (\mathbf{s}^* - \mathbf{s})$ 5: $(\mathbf{y}, \cdot) \leftarrow \text{RejSamp}(r_{\text{samp}}; c \mathbf{s})$ 6: $\mathbf{y}^* \leftarrow \mathbf{y} - c \cdot (\mathbf{s}^* - \mathbf{s}) - \mathbf{T}^\top (\mathbf{s}^* - \mathbf{s})$ 7: $r_{\text{samp}}^* \leftarrow \text{RejSamp}^{-1}(\mathbf{y}^*; c \mathbf{s}^*)$ 8: if $r_{\text{samp}}^* = \perp$ then return $\perp$ 9: return $(r_{\text{com}}^*, r_{\text{samp}}^*)$	1: parse r as $(r_{\text{com}}, r_{\text{samp}})$ 2: $\mathbf{s} \leftarrow \text{Encode}(w)$ 3: $\mathbf{u} \leftarrow \mathbf{A}_{\text{com}} \cdot r_{\text{com}} + \mathbf{B}'_{\text{com}} \cdot \mathbf{s}$ 4: $c \leftarrow H_{\text{chal}}(\text{crs}, x, \mathbf{u})$ 5: $(\mathbf{y}, \text{tape}_{\text{used}}) \leftarrow \text{RejSamp}(r_{\text{samp}}; c \cdot \mathbf{s})$ 6: if $\mathbf{y} = \perp$ then return $\perp$ 7: $\mathbf{z} \leftarrow r_{\text{com}} + \mathbf{y}$ 8: $\pi_{\text{rel}} \leftarrow \text{Prove}_{\text{rel}}(\text{crs}, x, \mathbf{u}, c, \mathbf{z}; \text{tape}_{\text{used}})$ 9: $\pi \leftarrow (\mathbf{u}, \mathbf{z}, \pi_{\text{rel}})$ 10: return π

Fig. 19: Lattice-based deniable NIZK with a dual-mode equivocable commitment over $\mathcal{R}_q$ in equivocation mode, a deterministic-challenge prove/verify pair, and a prover-side randomness-faking procedure using the trapdoor $\mathbf{T}$.

5.3 Entities and Trust Assumptions

Our Onyx involves several roles. The *Setup Authority* is a one-shot trusted party that publishes the deniable-FHE public key and equivocation trapdoor, the NIZK reference string, and the universal-thresholdizer public parameters; it is removed after setup and plays no further role. The *Registrars* jointly issue voter credentials via a verifiable secret-sharing protocol; we assume at least one honest registrar and an untappable channel to each voter at registration time. The *Trustees*, of which there are n_T with threshold $t < n_T$, jointly hold a share of the deniable-FHE secret key under the universal thresholdizer; we assume a strict minority is corrupt. The *Voters* cast ballots through an anonymous channel to the bulletin board and may, at any later time including post-tally, be coerced into disclosing their complete voting state. Finally the *Auditors* verify all public proofs (public bulletin board BB is append-only and publicly readable.); we assume at least one honest auditor exists.

5.4 Voting Protocol of Onyx

Setup phase. The Setup Authority runs the dFHE generator $(dpk, dsk) \leftarrow \text{dFHE.Gen}(1^\lambda)$ to obtain a deniable encryption keypair, then immediately thresholdises the secret key by invoking $(pp_{UT}, s_1, \dots, s_{n_T}) \leftarrow \text{UT.Setup}(1^\lambda, 1^d, (\mathbb{S}, n_T), dsk)$ for an access structure $\mathbb{S}$ encoding t -out-of- n_T threshold and circuit depth d sufficient for cleansing-hiding plus tally. The dFHE secret key is then erased; only the trustee shares $\{s_i\}$ persist. The Authority generates the lattice NIZK reference string by running $(crs, tdkey) \leftarrow \text{NIZK.Setup}(1^\lambda)$, publishing crs and retaining $tdkey$ for distribution to voters at registration. It also generates the mixnet parameters $pp_{MN} \leftarrow \text{MixNet.Setup}(1^\kappa, \xi_1, \xi_2)$ and posts the public election parameters $ep = (dpk, pp_{UT}, crs, pp_{MN}, n_C, \mathcal{M})$ to the bulletin board. Each trustee $\mathcal{T}_i$ stores s_i privately. The Setup Authority erases all retained state apart from $tdkey$, which is handed to the Registrars for distribution.

Registration phase. For each registered voter id , the Registrars jointly generate a credential $c_{id} \in \mathcal{C} \subseteq \mathcal{M}$ via the verifiable secret-sharing protocol $(\{c_{id,k}, \gamma_{id,k}\}_{k=1}^{n_R}, \pi_{\text{share}}) \leftarrow \text{VSS.Share}(c_{id}, t_R, n_R)$ to ensure no single registrar learns c_{id} in isolation. The shares are routed through the registrars' untappable channels to reconstruct a single credential delivered to the voter, together with the prover trapdoor $tdkey$ for use in NIZK. The Registrars also publish to the bulletin board the encrypted public roster $R = (\text{dFHE.Enc}(dpk, c_{id}; \rho_{\text{reg}, id}))_{id}$ together with the VSS public material π_{share} . The roster is permuted in random order so that the public position of a credential ciphertext is unlinked from the voter identity. Each voter's local state after registration is $vs_{id} = (c_{id}, tdkey, ep)$.

Voting phase. A voter casts a ballot for $v \in [1, n_C]$ as follows. Sample fresh randomness r_v, r_c , and produce the two voter-side ciphertexts $C_v \leftarrow \text{dFHE.Enc}(dpk, v; r_v)$ and $C_c \leftarrow \text{dFHE.Enc}(dpk, c_{id}; r_c)$. Construct the ballot relation statement $stmt = (C_v, C_c, ep)$ with witness $wit = (v, c_{id}, r_v, r_c)$ asserting membership in $\mathcal{R}_{\text{bal}} = \{(stmt, wit) : v \in [1, n_C], c_{id} \in \mathcal{C}, C_v = \text{dFHE.Enc}(dpk, v; r_v), C_c = \text{dFHE.Enc}(dpk, c_{id}; r_c)\}$.

Sample tape r_π and produce the deniable NIZK $\pi \leftarrow \text{NIZK.Prove}(crs, tdkey, stmt, wit; r_\pi)$. The complete ballot is $b = (C_v, C_c, \pi)$, posted to BB via an anonymous channel. The voter retains the local state $ls = (v, c_{id}, r_v, r_c, r_\pi, b)$ for future disclosure.

Validity check. The Server algorithm $\text{Valid}(BB, b, ep)$ parses $b = (C_v, C_c, \pi)$, checks $\text{NIZK.Verify}(\text{crs}, \text{stmt}, \pi) = 1$, checks that no ballot in BB pairs with a duplicate (C_v, C_c, π) to prevent exact replay, and accepts iff both checks pass. Note that we do *not* check whether C_c is in the registered roster R at this stage — that is the cleansing-hiding step’s job, and exposing it here would leak which ballots are cast under fake credentials, the very leakage Onyx is designed to prevent.

Cleansing-hiding tally. The Trustees jointly run the tally. Let $\{b_i\}_{i \in [n_B]}$ denote the accepted ballots and $\{R_j\}_{j \in [n_V]}$ the roster entries. Construct a working list of pairs $(V_k, K_k)_{k \in [n_B + n_V]}$ where V_k holds the encrypted vote and $K_k = (K_k^\perp, K_k^\top)$ holds an encrypted board position $K_k^\perp$ and an encrypted credential $K_k^\top$ — for ballot entries, $V_k = C_v^k$, $K_k^\perp = \text{dFHE.Enc}(k-1)$ under fixed null randomness, $K_k^\top = C_c^k$; for roster entries, $V_k = \text{dFHE.Enc}(0)$ under null randomness, $K_k^\perp = \text{dFHE.Enc}(n_B)$, $K_k^\top = R_{k-n_B}$. The Trustees execute the encrypted-domain sort $(\{V'_k, K'_k\}, \Pi_{\text{Sort}}) \leftarrow \text{UT.Eval}_{\text{sort}}(pp_{\text{UT}}, \{s_i\}, \text{OddEvenMergeSort})$ applied to the working list keyed lexicographically on $(K_k^\top, K_k^\perp)$, with each comparison realised as a depth- d Boolean circuit evaluated under UT-thresholded BGV. After sorting, valid entries are exactly those whose successor in the sorted list shares the same $K^\top$ and whose successor’s $K^\perp$ encodes n_B ; this validity bit is computed homomorphically as $D_k \leftarrow \text{UT.Eval}_{\text{eq}}(\{s_i\}, K'_k{}^\top, K'_{k+1}{}^\top)$ and $F_k \leftarrow \text{UT.Eval}_{\text{eq}}(\{s_i\}, K'_{k+1}{}^\perp, \text{Enc}(n_B))$, combined as $D_k \wedge F_k$ via homomorphic AND. Each V'_k is then conditionally zeroed by $V'_k \leftarrow \text{UT.Eval}_{\text{cgate}}(\{s_i\}, V'_k, D_k \wedge F_k)$, implementing the cleansing-hiding step. Each operation produces a verifiable transcript via UT.Verify which is appended to the proof bundle Π_{tally} . The cleaned encrypted votes are then shuffled through the verifiable mixnet, $(\{\hat{V}_k\}, \pi_{\text{Mix}}) \leftarrow \text{MixNet.Mix}(pp_{\text{MN}}, \{V'_k\})$, and finally distributively decrypted by the trustees through $(\{t_{k,j}\}, \pi_{\text{Dec}_j}) \leftarrow \text{MixNet.DDec}(s_j, \{\hat{V}_k\})$ for each j , combined via $\{m_k\} \cup \{\perp\} \leftarrow \text{MixNet.Comb}(\{\hat{V}_k\}, \{\{t_{k,j}\}\})$ to produce the cleartext votes. The result r is the multiset of non-zero m_k tallied per candidate.

Verification. The auditor algorithm $\text{Verify}((ep, BB, r), \Pi_{\text{tally}})$ runs UT.Verify on every cleansing transcript, MixNet.MixVerify on the shuffle, MixNet.DDecVerify on each partial decryption, and recomputes the result r from the combined plaintexts. Output 1 iff all checks succeed.

Persistent faking. The voter’s **Fake** algorithm, invoked under coercion to claim a vote $v^* \in [1, n_C]$ with credential c^* for an already-cast ballot $b = (C_v, C_c, \pi)$, computes:

$$\begin{aligned} r_v^* &\leftarrow \text{dFHE.Fake}(dpk, v, r_v, v^*), r_c^* \leftarrow \text{dFHE.Fake}(dpk, c_{id}, r_c, c^*), \\ r_\pi^* &\leftarrow \text{NIZK.Fake}(\text{crs}, \text{tdkey}, \text{stmt}, (v, c_{id}, r_v, r_c), r_\pi, (v^*, c^*, r_v^*, r_c^*)). \end{aligned}$$

The voter discloses $(c^*, v^*, r_v^*, r_c^*, r_\pi^*)$ to the coercer. The coercer can re-run **Vote** on these values and obtain exactly b — the ballot present on the board — by construction of all three **Fake** algorithms. By the deniability of dFHE (Section ??) and of NIZK (Section ??), the disclosed tuple is statistically/computationally close to a tuple drawn from an honest casting of v^* with credential c^* .

5.5 Privacy and Verifiability Analysis of Onyx

Remark 1. Let Onyx be constructed as in Figure 20 under the following trust assumptions: (i) the Setup Authority honestly executes **Setup** and erases ds_k together with

Setup ($1^\lambda, 1^d, n_T, t$) 1: $(dpk, dsk) \leftarrow \text{dFHE.Gen}(1^\lambda)$ 2: $(pp_{\text{UT}}, s_1, \dots, s_{n_T}) \leftarrow \text{UT.Setup}(1^\lambda, 1^d, s_t, n_T, dsk)$ 3: $(\text{crs}, \text{tdkey}) \leftarrow \text{NIZK.Setup}(1^\lambda)$ 4: $pp_{\text{MN}} \leftarrow \text{MixNet.Setup}(1^\lambda, \xi_1, \xi_2)$ 5: $ep \leftarrow (dpk, pp_{\text{UT}}, \text{crs}, pp_{\text{MN}}, n_C)$ 6: erase dsk 7: return $(ep, \{s_i\}_{i=1}^{n_T}, \text{tdkey})$	Vote (v, vs_{id}) 1: $(c_{id}, \text{tdkey}, ep) \leftarrow vs_{id}$ 2: $r_v, r_c \leftarrow \mathbb{S}\{0, 1\}^\ell; \quad r_\pi \leftarrow \mathbb{S}\{0, 1\}^{\ell_\pi}$ 3: $C_v \leftarrow \text{dFHE.Enc}(dpk, v; r_v)$ 4: $C_c \leftarrow \text{dFHE.Enc}(dpk, c_{id}; r_c)$ 5: $\text{stmt} \leftarrow (C_v, C_c, ep)$ 6: $\text{wit} \leftarrow (v, c_{id}, r_v, r_c)$ 7: $\pi \leftarrow \text{NIZK.Prove}(\text{crs}, \text{tdkey}, \text{stmt}, \text{wit}; r_\pi)$ 8: $b \leftarrow (C_v, C_c, \pi)$ 9: $ls \leftarrow (v, c_{id}, r_v, r_c, r_\pi, b)$ 10: return (b, ls)
Register (id, ep, tdkey) 1: $c_{id} \leftarrow \mathbb{S}\mathcal{C}$ 2: $(\{c_{id,k}, \gamma_{id,k}\}_{k=1}^{n_R}, \pi_{\text{share}}) \leftarrow \text{II.Share}(c_{id}, t_R, n_R)$ 3: $\rho_{\text{reg}} \leftarrow \mathbb{S}\{0, 1\}^\ell; \quad R_{id} \leftarrow \text{dFHE.Enc}(dpk, c_{id}; \rho_{\text{reg}})$ 4: $vs_{id} \leftarrow (c_{id}, \text{tdkey}, ep)$ 5: return $(vs_{id}, R_{id}, \pi_{\text{share}})$	Tally ($BB, R, ep, \{s_i\}$) 1: for $b_k \in BB$ do 2: $V_k \leftarrow b_k.C_v$ 3: $K_k^\perp \leftarrow \text{dFHE.Enc}(k-1; 0)$ 4: $K_k^\top \leftarrow b_k.C_c$ 5: for $R_j \in R$ do 6: $V_{n_B+j} \leftarrow \text{dFHE.Enc}(0; 0)$ 7: $K_{n_B+j}^\perp \leftarrow \text{dFHE.Enc}(n_B; 0)$ 8: $K_{n_B+j}^\top \leftarrow R_j$ 9: $(\{(V'_k, K'_k)\}, \text{II}_{\text{Sort}}) \leftarrow \text{UT.Eval}(pp_{\text{UT}}, \{s_i\}, \text{Sort}_{K^\top, K^\perp})$ 10: for $k \in [n_B + n_V - 1]$ do 11: $D_k \leftarrow \text{UT.Eval}(\{s_i\}, \text{Eq}(K_k'^\top, K_{k+1}'^\top))$ 12: $F_k \leftarrow \text{UT.Eval}(\{s_i\}, \text{Eq}(K_{k+1}'^\perp, n_B))$ 13: $V'_k \leftarrow \text{UT.Eval}(\{s_i\}, \text{CGate}(V'_k, D_k \wedge F_k))$ 14: $(\{\hat{V}_k\}, \pi_{\text{Mix}}) \leftarrow \text{MixNet.Mix}(pp_{\text{MN}}, \{V'_k\})$ 15: for $j \in [n_T]$ do 16: $(\{t_{k,j}\}, \pi_{\text{Dec}_j}) \leftarrow \text{MixNet.DDec}(s_j, \{\hat{V}_k\})$ 17: $\{m_k\} \leftarrow \text{MixNet.Comb}(\{\hat{V}_k\}, \{\{t_{k,j}\}\})$ 18: $r \leftarrow \text{result}(\{m_k \mid m_k \neq 0\})$ 19: $\text{II}_{\text{Tally}} \leftarrow (\text{II}_{\text{Sort}}, \pi_{\text{Mix}}, \{\pi_{\text{Dec}_j}\}_j)$ 20: return $(r, \text{II}_{\text{Tally}})$
Valid (BB, b, ep) 1: $(C_v, C_c, \pi) \leftarrow b; \quad \text{stmt} \leftarrow (C_v, C_c, ep)$ 2: $e_1 \leftarrow \text{NIZK.Verify}(\text{crs}, \text{stmt}, \pi)$ 3: $e_2 \leftarrow b \notin BB$ 4: return $(e_1 \wedge e_2)$	
Fake (ls, v^*, c^*, vs_{id}) 1: $(v, c, r_v, r_c, r_\pi, b) \leftarrow ls$ 2: $r_v^* \leftarrow \text{dFHE.Fake}(dpk, v, r_v, v^*)$ 3: $r_c^* \leftarrow \text{dFHE.Fake}(dpk, c, r_c, c^*)$ 4: $\text{stmt} \leftarrow b.\text{stmt}; \quad \text{wit}^* \leftarrow (v^*, c^*, r_v^*, r_c^*)$ 5: $r_\pi^* \leftarrow \text{NIZK.Fake}(\text{crs}, \text{tdkey}, \text{stmt}, (v, c, r_v, r_c), r_\pi, \text{wit}^*)$ 6: return $(c^*, v^*, r_v^*, r_c^*, r_\pi^*)$	
Verify ($ep, BB, R, r, \text{II}_{\text{Tally}}$) 1: $e_1 \leftarrow \text{UT.Verify}(pp_{\text{UT}}, \text{II}_{\text{Sort}}, \text{Sort})$ 2: $e_2 \leftarrow \text{MixNet.MixVerify}(pp_{\text{MN}}, \{\hat{V}_k\}, \{V'_k\}, \pi_{\text{Mix}})$ 3: $e_3 \leftarrow \bigwedge_j \text{MixNet.DDecVerify}(\{\hat{V}_k\}, \{t_{k,j}\}, \pi_{\text{Dec}_j})$ 4: return $(e_1 \wedge e_2 \wedge e_3)$	

Fig. 20: The Onyx voting system built upon $\text{dFHE} = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval}, \text{Fake})$, $\text{UT} = (\text{Setup}, \text{Eval}, \text{Verify}, \text{Combine})$, $\text{II} = (\text{Share}, \text{VerifyShare}, \text{Reconstruct})$, $\text{MixNet} = (\text{Setup}, \text{Mix}, \text{MixVerify}, \text{DDec}, \text{DDecVerify}, \text{Comb})$, and $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify}, \text{Fake})$.

all internal state thereafter; (ii) at most $n_T - t - 1$ trustees and at most $n_R - t_R - 1$ registrars are corrupted; (iii) the registration channel from each honest registrar to each voter is untappable; (iv) the casting channel to the bulletin board is anonymous; (v) the bulletin board is honest and append-only; and (vi) at least one auditor is honest. The threat model permits the adversary $\mathcal{A}$ to (a) corrupt up to n_A voters at registration time and read their credentials, (b) corrupt up to the stated number of trustees and registrars, (c) coerce honest voters during the voting phase by issuing vote instructions and observing them, (d) inject ballots through the $\mathcal{O}_{\text{cast}}$ oracle, (e)

observe $\text{Publish}(BB)$ at every step via $\mathcal{O}_{\text{board}}$, and (f) at any time after tally publication, demand the complete local state of any coerced voter via $\mathcal{O}_{\text{reveal}}$; $\mathcal{A}$ is PPT and may neither break the underlying lattice assumptions, nor corrupt the bulletin board, nor eavesdrop on registration channels, nor de-anonymise the casting channel.

Theorem 2. *Considering Remark 1, for any PPT adversary $\mathcal{A}$ issuing at most q_b total ballot castings and q_d disclosure queries across at most n_V coerced voters, there exist PPT reductions $\mathcal{B}_D, \mathcal{B}_N, \mathcal{B}_C, \mathcal{B}_Z, \mathcal{B}_U, \mathcal{B}_M, \mathcal{B}_{DD}, \mathcal{B}_H$ such that*

$$\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{PCR}}(\lambda) \leq q_d \cdot (2 \cdot \text{Adv}_{\mathcal{B}_D, \text{dFHE}}^{\text{Den}}(\lambda) + \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{RD}}(\lambda)) + 2q_b \cdot \text{Adv}_{\mathcal{B}_C, \text{dFHE}}^{\text{IND-CPA}}(\lambda) + q_b \cdot \text{Adv}_{\mathcal{B}_Z, \text{NIZK}}^{\text{ZK}}(\lambda) \\ + \text{Adv}_{\mathcal{B}_U, \text{UT}}^{\text{SimSec}}(\lambda) + \text{Adv}_{\mathcal{B}_M, \text{MixNet}}^{\text{Shuf}}(\lambda) + n_T \cdot \text{Adv}_{\mathcal{B}_{DD}, \text{MixNet}}^{\text{DDec}}(\lambda) + \text{Adv}_{\mathcal{B}_H, H}^{\text{CR}}(\lambda) + \text{negl}(\lambda).$$

In fact, if dFHE is plan-ahead $1/\delta$ -deniable with $\delta = 2^{\omega(\log \lambda)}$ and IND-CPA under LWE, NIZK is randomness-deniable and zero-knowledge under MLWE and MSIS, UT is simulation-secure under LWE, MixNet is zero-knowledge for both its shuffle and its distributed-decryption proofs under DKS $^\infty$ and SKS 2 , and H is quantum-collision-resistant, then $\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{PCR}}(\lambda)$ is negligible in λ , i.e., Onyx is Persistent Coercion Resistant.

Onyx achieves PCR because every component of the local view that a coerced voter must disclose is independently deniable: the credential through the secret-shared registration and the uniformity of `fakecred`, the vote and credential ciphertexts through dFHE δ -deniability, and the ballot's zero-knowledge proof through randomness-deniability of the lattice NIZK. The cleansing-hiding tally, performed homomorphically under the universal thresholdizer and finalized through a verifiable mix-net, which releases no intermediate information about which ballots used fake credentials or which were superseded by revotes. Moreover, the bulletin board, the disclosed local views, and the published result collectively reveal only the board length and the cleansed outcome, exactly what the PCR ideal experiment permits.

Apart from our formal proofs in Appendix of ??, we also provide machine-checked proofs for privacy of Onyx in EasyCrypt [113]. As an interactive theorem prover with support for expressing proofs in the computational model of cryptography, EasyCrypt [113] has been used extensively for checking security mechanisms. The machine-checked privacy proof of Onyx in EasyCrypt is available at [114], where Theorem 2 corresponds to lemma `onyx_pcr` in `OnyxPCR.ec`, which is shown in Figure 21 below, and the proof model follows [115]. The Theorem 1 corresponds to lemma `pcr_implies_cr` inside the `PCRImpliesCR.ec`. Each building block (dFHE [93], UT [94], MixNet [52], NIZK [95]) is a formal abstract `module type` paired with its security game, composed into the protocol in `Onyx.ec`. Theorem 1 is demonstrated in EasyCrypt via a contrapositive black-box reduction $\mathcal{A}_{\text{PCR}}$ which defers credential disclosure to $\mathcal{O}_{\text{reveal}}$, and Theorem 2 via a machine-checked nine-point triangle inequality summing the eight per-hop bounds $|\Pr[G_i] - \Pr[G_{i+1}]|$ along the hybrid chain $G_0 \rightarrow \dots \rightarrow G_8$ in [114] code.

Definition 8 (Verifiability [10]). *A voting protocol (Setup, Register, Vote, Check, isVal, Tally, Verify) is verifiable if, for every PPT adversary $\mathcal{A}$ and every parameter tuple (n_T, t, n_V, n_C) , there exists a negligible function μ such that*

$$\Pr[\text{Ver}(\mathcal{A}, \lambda, n_T, t, n_V, n_A, n_C) = 1] \leq \mu(\lambda).$$

The oracles $\mathcal{O}_{\text{vote}}$, $\mathcal{O}_{\text{cast}}$, and $\mathcal{O}_{\text{check}}$ are defined in Figure 22. An honest voter who calls $\mathcal{O}_{\text{vote}}$ and whose subsequent $\mathcal{O}_{\text{check}}$ query succeeds is called happy; the adversary wins only if all honest voters who voted are happy and the announced result X cannot be explained by any combination of honest votes and adversarially chosen votes for the corrupted voters.

```

1 lemma onyx_pcr (A <: PCR_REAL_ADV
2   {-G1_UT, -G2_MixShuf, -G3_MixDDec, -G4_Hash, -G5_NIZK_ZK,
3   -G6_dFHE_CPA, -G7_DiscRand, -OnyxSim,
4   -PCR_Real_Experiment, -PCR_Ideal_Experiment})
5   &m λ nT t nV nA nC :
6   ' | Pr[PCR_Real_Experiment(Onyx,
7     A).main(λ, nT, t, nV, nA, nC) @ &m : res]
8     - Pr[PCR_Ideal_Experiment(OnyxSim(A)).main(λ, nV, nA, nC) @ &m : res] |
    ≤ total_bound λ.

```

Fig. 21: `onyx_pcr` bounding real-ideal PCR experiments' distinguishing advantage.

$\text{Ver}(\mathcal{A}, \lambda, n_T, t, n_V, n_A, n_C)$	$\mathcal{O}_{\text{vote}}(i, \omega)$
1: $(\text{pk}, \{(s_i, h_i)\}_{i \in [n_T]}) \leftarrow \text{Setup}(1^\lambda, n_T, t)$	1: if $i \notin [n_V] \setminus A \vee \omega \notin [n_C]$ then return $\perp$
2: $\{c_i\}_{i \in [n_V]}, R \leftarrow \text{Register}(1^\lambda, \text{pk}, n_V)$	2: $H_i \leftarrow H_i + 1$
3: $A \leftarrow \mathcal{A}$	3: $b \leftarrow \text{Vote}_{\text{pk}}(c_i, \omega)$
4: $\text{BB} \leftarrow \emptyset$; for $i \in [n_V] \setminus A$ do $H_i \leftarrow 1$; $\text{HV}_i \leftarrow \perp$	4: $\text{HV}_i \leftarrow (\text{pk}, b, c_i, \omega)$
5: $\mathcal{A}^{\mathcal{O}_{\text{vote}}, \mathcal{O}_{\text{cast}}, \mathcal{O}_{\text{check}}}(\{c_i\}_{i \in A})$	5: return b
6: if $ A > n_A \vee \exists i \in [n_V] \setminus A \mid H_i < 1$ then return 0	$\mathcal{O}_{\text{cast}}(b)$
7: $(X, \tau) \leftarrow \text{Tally}(\text{BB}, R, \text{pk}, \{h_i, s_i\}_{i \in [n_T]}, t)$	1: if $\text{isVal}(\text{pk}, \text{BB}, b) = 1$ then
8: if $\text{Verify}(\text{BB}, \tau, X) = 0$ then return 0	2: $\text{BB} \leftarrow \text{BB} \cup \{b\}$
9: $\text{honest} \leftarrow \text{cleanse}(\{(i, \text{HV}_i[4]) \mid i \in [n_V] \setminus A, \text{HV}_i \neq \perp\})$	$\mathcal{O}_{\text{check}}(i)$
10: if $\#\{(cor_i, \varpi_i)\}_{i=1}^{n_C} \in (A \cup [n_C])$:	1: $(\text{pk}, b, c_i, \omega) \leftarrow \text{HV}_i$; $\text{HV}_i[2] \leftarrow \perp$
11: $X = \text{tally}(\text{honest} \cup \{(cor_i, \varpi_i)\}_i)$ then return 0	2: if $\text{Check}(\text{BB}, \text{pk}, b, c_i, \omega) = 1$ then
12: return 1	3: $H_i \leftarrow H_i + 1$

Fig. 22: Verifiability game Ver for a voting protocol (Setup , Register , Vote , Check , isVal , Tally , Verify). Here n_T is the number of tallies, t the decryption threshold, n_V the number of voters, n_A the maximum number of corrupted voters, and n_C the number of voting options (excluding abstention).

The Definition 8 assumes that the registrar is honest, and that up to t decryption trustees can be corrupted.

Remark 2. Let *Onyx* be built as in Figure 20 under the following trust assumptions for verifiability: (i) the Registrar is honest; (ii) at most $n_T - t - 1$ decryption trustees are corrupt; (iii) the bulletin board is honest and append-only; (iv) the registration channel to each honest voter is untappable. The adversary $\mathcal{A}$ is PPT and may corrupt up to n_A voters at registration time, fully control the votes of honest voters via $\mathcal{O}_{\text{vote}}$, inject ballots via $\mathcal{O}_{\text{cast}}$, drive $\mathcal{O}_{\text{check}}$, and participate adversarially as a corrupt minority of trustees during the tally; $\mathcal{A}$ cannot break the underlying lattice assumptions, nor corrupt the bulletin board, nor eavesdrop on registration.

Theorem 3. Assuming Remark 2 holds, then for any PPT adversary $\mathcal{A}$ producing at most q_b ballots on BB , there exist PPT reductions $\mathcal{B}_N, \mathcal{B}_U, \mathcal{B}_M, \mathcal{B}_{DD}, \mathcal{B}_C, \mathcal{B}_H$ such that

$$\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{Ver}}(\lambda) \leq q_b \cdot \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{KS}}(\lambda) + \text{Adv}_{\mathcal{B}_U, \text{UT}}^{\text{Sound}}(\lambda) + \text{Adv}_{\mathcal{B}_M, \text{MixNet}}^{\text{ShufSound}}(\lambda) + n_T \cdot \text{Adv}_{\mathcal{B}_{DD}, \text{MixNet}}^{\text{DDecSound}}(\lambda) \\ + n_V \cdot \text{Adv}_{\mathcal{B}_C, \text{dFHE}}^{\text{IND-CPA}}(\lambda) + \text{Adv}_{\mathcal{B}_H, H}^{\text{CR}}(\lambda) + \text{negl}(\lambda).$$

	Setup	Register	Vote	Valid	Tally	Verify	Fake	Total
M2	0.0379	0.0039	0.0314	0.0047	0.0903	0.0033	0.0102	0.1817
Snapdragon	0.0493	0.0051	0.0408	0.0061	0.1174	0.0043	0.0133	0.2362
Dell Precision	0.0436	0.0045	0.0361	0.0054	0.1038	0.0038	0.0117	0.2090
Core i7	0.0409	0.0042	0.0339	0.0051	0.0975	0.0036	0.0110	0.1962

Table 2: Runtime (seconds) of **Onyx**’s procedures for an election with $n_V=4$ voters, $n_C=3$ candidates, $n_T=4$ trustees ($t=2$), and $n_R=2$ registrars at $\lambda=128$ -bit PQ security.

Actually, under the lattice assumptions of Theorem 2 (Module-LWE/SIS for NIZK, LWE for dFHE, DKS^∞ and SKS^2 for MixNet, simulation-soundness for UT, and quantum collision resistance for H), $\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{Ver}}(\lambda)$ is negligible in λ , i.e., **Onyx** is verifiable.

Onyx is universally verifiable because every public step is accompanied by a publicly checkable transcript whose soundness relies on a standard lattice or hash assumption. Each ballot carries a deniable NIZK whose knowledge soundness, under MSIS, forces a unique well-formed vote-and-credential pair behind the ciphertexts; the universal thresholdizer’s evaluation proofs guarantee that the encrypted-domain sort, equality tests, and conditional zeroing in the cleansing-hiding pipeline were executed correctly; the verifiable mix-net’s shuffle proof binds the permutation to the BDLOP commitments [112]; and the distributed-decryption proofs tie each plaintext to its trustee’s secret share. An honest auditor accepts only when every check succeeds, which rules out undetected result manipulation.

5.6 Performance and Efficiency Analysis

None of the existing PQ voting schemes [4, 15, 23, 25–31] provided any implementation, except [24]. The core implementation of **Onyx** is available in [116], where our **Onyx** system targets $\lambda=128$ bits of post-quantum security, with parameters derived from the Lattice Estimator [117]; we benchmarked our protocol pipeline, which encompasses dFHE encryption, NIZK proving with Micciancio–Peikert trapdoor equivocation, VSS-based credential issuance, UT-thresholded cleansing-hiding tally, verifiable mix-net shuffling, and the persistent Fake algorithm, across four different platforms: an Apple M2 (macOS), an Android Qualcomm Snapdragon 8 Gen 2 (aarch64) via PRoot Distro¹ on Termux, a Dell Precision 3581 with an Intel Core i7-1355U (Windows 11), and the same Intel Core i7-1355U with different RAM running Ubuntu 22.04. The benchmark targets the medium-PQ-compact parameters at $\lambda=128$ bits, with dFHE ring degree $N_D=1024$, NIZK ring degree $d'=128$, mix-net ring degree $N_M=512$, $n_T=4$ trustees with threshold $t=2$, $n_R=2$ registrars, $n_V=4$ voters, and $n_C=3$ candidates. As shown in Table 2, the dominant cost is the cleansing-hiding tally (≈ 90 ms on M2), which is independent of any single voter and amortizes over the full election; the per-voter **Vote** and **Fake** operations together complete in under 50 ms on every platform tested, which shows that **Onyx**’s persistent disclosure imposes negligible overhead on the casting client. With a total wall-clock of 0.1817 s on the Apple M2 for a complete election cycle, **Onyx** is practical for elections in high-coercion environments where post-tally voter interrogation is a credible threat, even on mobile hardware. For computational complexities of **Onyx** refer to Appendix of full version [106] of our work.

¹ <https://github.com/termux/proot-distro>

A Appendix

Here, we provide supplementary materials for our work.

A.1 Computational Assumptions

Learning With Errors [118]. Let n, m, q be positive integers with q prime, and let χ be an error distribution over $\mathbb{Z}_q$. The $\text{LWE}_{n,m,q,\chi}$ decision problem requires distinguishing, with non-negligible advantage, the distribution $\{(\mathbf{a}_i, \mathbf{a}_i^\top \mathbf{s} + e_i)\}$ from $\{(\mathbf{a}_i, u_i)\}$, where $\mathbf{a}_i \leftarrow \mathbb{Z}_q^n$, $\mathbf{s} \leftarrow \mathbb{Z}_q^m$, $e_i \leftarrow \chi$, and $u_i \leftarrow \mathbb{Z}_q$.

Module Learning With Errors [119]. Let n, m, k, q be positive integers with q prime, let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and let χ be an error distribution over R_q . The $\text{MLWE}_{n,m,k,q,\chi}$ decision problem requires distinguishing, with non-negligible advantage, the distribution $\{(\mathbf{a}_i, \mathbf{a}_i^\top \mathbf{s} + e_i)\}$ from $\{(\mathbf{a}_i, u_i)\}$, where $\mathbf{a}_i \leftarrow R_q^k$, $\mathbf{s} \leftarrow R_q^k$, $e_i \leftarrow \chi$, and $u_i \leftarrow R_q$.

Module Short Integer Solution [120]. Let n, m, k, q be positive integers with q prime, let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and let β be a positive real bound. The $\text{MSIS}_{n,m,k,q,\beta}$ problem requires finding, given a uniformly random matrix $\mathbf{A} \leftarrow R_q^{k \times m}$, a non-zero vector $\mathbf{z} \in R_q^m$ such that $\mathbf{A}\mathbf{z} = \mathbf{0}$ and $\|\mathbf{z}\| \leq \beta$.

Decisional Knapsack Problem (DKS^∞) [112]. Let n, k, q be positive integers with q prime, let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and let β be a positive integer. The $\text{DKS}_{n,k,\beta}^\infty$ decision problem requires distinguishing, with non-negligible advantage, the distribution $\{[\mathbf{I}_n \mid \mathbf{A}'] \cdot \mathbf{y}\}$ from the uniform distribution over R_q^n , where $\mathbf{A}' \leftarrow R_q^{n \times (k-n)}$ and $\mathbf{y} \leftarrow S_\beta^k$, with $S_\beta = \{x \in R_q \mid \|x\|_\infty \leq \beta\}$. This problem is equivalent to the MLWE [121] when the number of samples is limited.

Search Knapsack Problem (SKS^2) [112]. Let n, k, q be positive integers with q prime, let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, and let β be a positive real bound. The $\text{SKS}_{n,k,\beta}^2$ problem requires finding, given a uniformly random matrix $\mathbf{A}' \leftarrow R_q^{n \times (k-n)}$, a non-zero vector $\mathbf{y} = [y_1, \dots, y_k]^\top \in R_q^k$ such that $[\mathbf{I}_n \mid \mathbf{A}'] \cdot \mathbf{y} = \mathbf{0}^n$ and $\|y_i\|_2 \leq \beta$ for all $i \in [k]$. This problem is equivalent to the MSIS [121] in its Hermite Normal Form.

A.2 Deniable Encryption

Considering deniable encryption is one of the tools that we will use in the following sections, here we go through its background, the intuition behind it, and how it works.

The defining feature of deniable encryption is that an adversary cannot distinguish whether the parties revealed their true randomness and plaintext or provided fabricated alternatives [34, 122]. This guarantee holds even when the adversary observes the entire communication and later coerces the parties. Since its emergence, deniable encryption has shown promising real-world applications such as Smartphones [123], Signal Messenger [124], IoT and wearable devices [125], MobiWear for smartwatches [126], FractalCrypt [127], VeraCrypt [128, 129], and many others.

Definition 9 (Deniable Encryption [130]). A deniable encryption scheme DE for message space $\mathcal{M}$ is a tuple of PPT algorithms $\text{DE} = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Fake})$:

- $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$: Takes security parameter 1^λ ; outputs a public key pk and a secret key sk .
- $ct \leftarrow \text{Enc}(pk, m; r)$: Takes public key pk , message $m \in \mathcal{M}$, and ℓ -bit randomness r ; outputs a ciphertext ct .
- $m \leftarrow \text{Dec}(sk, ct)$: Takes secret key sk and ciphertext ct ; outputs a message $m \in \mathcal{M}$.
- $r^* \leftarrow \text{Fake}(pk, m, r, m^*)$: Takes public key pk , original message $m \in \mathcal{M}$, randomness $r \in \{0, 1\}^\ell$, and a target fake message $m^* \in \mathcal{M}$; outputs fake randomness r^* such that $\text{Enc}(pk, m; r)$ is indistinguishable from $\text{Enc}(pk, m^*; r^*)$ even when (m^*, r^*) is revealed to the coercer.

Deniable encryption was introduced by Canetti et al. [122], establishing foundational definitions and initial constructions; however, their scheme achieved only inverse-polynomial distinguishing advantage, requiring superpolynomially long ciphertexts. Sahai and Waters [131] gave the first sender-deniable scheme with polynomial-length ciphertexts via indistinguishability obfuscation, with receiver-deniable conversion via the generic transformation of [122]. Impossibility results subsequently delimited the design space: Bendlin et al. [132] ruled out two-message receiver-deniable encryption, and Dachman-Soled [133] established black-box barriers for sender-deniable encryption from simulatable PKE. Canetti et al. [130] resolved unrestricted bideniability with a three-message CRS-model protocol based on subexponentially secure iO and one-way functions (Figure 23). Extensions include deniable ABE from LWE [134], deniable functional encryption [34], and deniable ciphertext editing [135]. Beyond encryption, Jiang et al. [136] formalized deniable authenticated key exchange, and Canetti et al. [137] proposed a deniable secret sharing scheme in which shareholders may produce fake-but-plausible shares concealing participation under coercion.

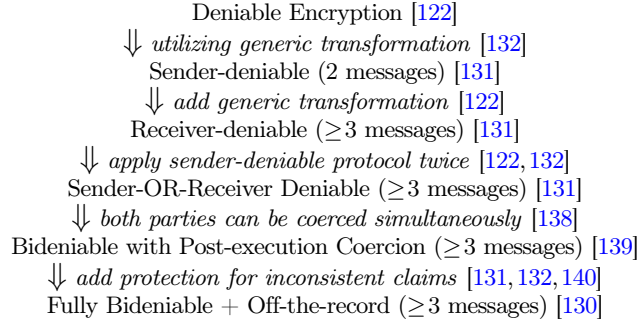


Fig. 23: The hierarchy of deniability by the number of compromised parties.

A.3 Voting System Model

Definition 10 (General Voting Scheme). Let $\mathcal{I}$ be a set of voter identities, $\mathbb{V}$ a set of votes, $\mathcal{R}$ a result space, and $\rho: (\mathcal{I} \times \mathbb{V})^* \rightarrow \mathcal{R}$ a result function. A single-pass voting scheme for $(\mathcal{I}, \mathbb{V}, \rho)$ is a tuple $\mathcal{V} = (\text{Setup}, \text{Vote}, \text{Valid}, \text{Publish}, \text{Tally}, \text{Verify})$ of PPT algorithms with the following syntax:

- $(pk, sk) \leftarrow \text{Setup}(1^\lambda)$: takes a security parameter λ and outputs an election public key pk and a secret tallying key sk . We treat pk as an implicit input to all remaining algorithms.
- $b \leftarrow \text{Vote}(id, v)$: used by voter $id \in \mathcal{I}$ to cast vote $v \in \mathbb{V}$, producing a ballot b .
- $d \leftarrow \text{Valid}(BB, b)$: takes the current ballot box BB and a ballot b ; returns $\top$ if b is valid with respect to BB , and $\perp$ otherwise.
- $PBB \leftarrow \text{Publish}(BB)$: takes the ballot box BB and outputs the public bulletin board PBB , i.e. the publicly visible view of BB .
- $(r, \Pi) \leftarrow \text{Tally}(BB, sk)$: takes the ballot box and the secret key; outputs an election result $r \in \mathcal{R} \cup \{\perp\}$ together with a proof of correct tabulation Π . This algorithm may implement the revote policy specified by ρ .
- $d \leftarrow \text{Verify}(PBB, r, \Pi)$: takes a public bulletin board, a result, and a tabulation proof; returns $\top$ if Π is a valid proof of correct tallying for result r on board PBB , and $\perp$ otherwise.

Generally, a voting protocol built from $\mathcal{V}$ proceeds in three phases: the administrator runs $(pk, sk) \leftarrow \text{Setup}(1^\lambda)$ and publishes pk . Parties submit ballots to BB . On input b , the ballot box appends b to BB iff $\text{Valid}(BB, b) = \top$. Any party may query $\text{Publish}(BB)$ at any time. The administrator computes $(r, \Pi) \leftarrow \text{Tally}(BB, sk)$ and publishes (r, Π) . The ballot box is subsequently closed.

A.4 More Voting Systems Properties

Most voting schemes are designed such that no entity can learn the content of ballots (*ballot privacy* [16]), participation patterns remain hidden from observers (*participation privacy* [141]), voters can verify their ballots were recorded and anyone can verify correct tallying (*verifiability* [12]), the coercer cannot know whether voter followed his instruction or not (*coercion resistance* [19]), ballot secrecy gets preserved against computationally unbounded future adversaries (*everlasting privacy* [?]), only the most recent vote from each voter influences the final tally (*strong non-reusability* [?]), voters can update their choices multiple times (*re-voting* [14]), conflicts between voters and authorities are resolvable through cryptographic evidence (*dispute resolution* [?]), misbehaving parties can be identified and held responsible (*accountability* [142, 143]), the protocol execution is indistinguishable from interaction with a trusted party (*ideal functionality* [9]), voters cannot construct proofs of their choices to third parties (*receipt-freeness* [18]), the election can complete even when some trustees lose their secret keys (*key-loss resilience* [?]), and voters can express preferences by ranking candidates in some cases (*ranked-choice voting* [?]). However, currently, there is no voting scheme that can simultaneously provide all of these properties; there are many schemes that each target one or a very limited number of these properties, not all of them in one place, as it is difficult to provide more than 2 of these properties at the same time [?], and it is challenging.

A.5 More On the Quantitative PCR Framework

Here, we collect the formal statements, definitions, and proofs underlying our quantitative framework of Section 4.2.

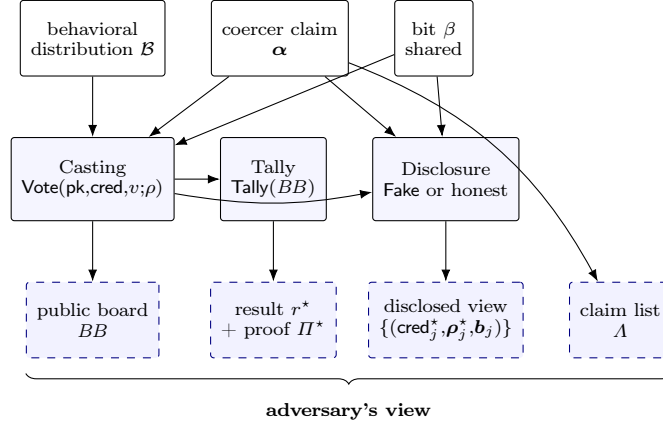


Fig. 24: Factorization of the adversary's view in $\text{Exp}^{\text{PCR-Real}}$ into four components: public board (drives δ^{board}), result+proof (drives δ^{tally}), disclosed local state per coerced voter (drives $\varepsilon_{\text{ball}}$ per disclosure), and claim list (drives δ^{ideal}).

Marginal final-vote distribution. We restrict to at most one revote per voter to match [22, App. D]; the framework extends to bounded revote chains of length k by applying the multinomial argument componentwise across the k ballots. Integrating out the revote branch of Figure 5 yields

$$P_0^\circ = P_0, \quad P_\ell^\circ = (1 - P_0)[p_\ell(1 - r) + rr_\ell], \quad \ell \in [1, n_C]. \quad (2)$$

Real and ideal coercion levels (formal). For an intention vector $\alpha = (\alpha_j)_{j \in \mathcal{H}_{\text{coerce}}}$ and a fixed behavioural distribution $\mathcal{B}$, let $\mathcal{V}_{\alpha, \beta}^{\text{Real}}$ denote the joint distribution of the adversary's entire view in $\text{Exp}^{\text{PCR-Real}}$ when the secret bit is β ; $\mathcal{V}_{\alpha, \beta}^{\text{Ideal}}$ is defined analogously for $\text{Exp}^{\text{PCR-Ideal}}$. By the Neyman–Pearson lemma [107], the optimal coercer's distinguishing advantage equals the total variation distance

$$\delta_\alpha^g(\Pi) := \frac{1}{2} \sum_V |\Pr[\mathcal{V}_{\alpha, 1}^g = V] - \Pr[\mathcal{V}_{\alpha, 0}^g = V]|, \quad g \in \{\text{Real}, \text{Ideal}\}. \quad (3)$$

The *persistent coercion level* of a protocol Π is $\delta^g(\Pi) := \max_{\mathcal{B}, \alpha} \delta_\alpha^g(\Pi)$, and the protocol satisfies Definition 2 iff $|\delta^{\text{Real}}(\Pi) - \delta^{\text{Ideal}}(\Pi)|$ is negligible.

Game decomposition. Figure 24 demonstrates the joint distribution of the adversary's view in $\text{Exp}^{\text{PCR-Real}}$ as a product of four independent factors: a public-board factor produced by the protocol's tally pipeline, a per-voter disclosure factor produced by the Fake algorithm, the tally proof factor, and the unavoidable claim factor announced in Λ . The independence of these factors after the protocol's simulator hops is the source of the additive bound in Theorem 6.

Bayes-optimal persistent coercer

Theorem 4. Fix $\mathcal{B}, \alpha$ and $g \in \{\text{Real}, \text{Ideal}\}$. Let $T_g(V) := \Pr[\mathcal{V}_{\alpha, 1}^g = V] / \Pr[\mathcal{V}_{\alpha, 0}^g = V]$. The optimal coercer's decision rule is $\mathcal{A}^*(V) = \mathbf{1}[T_g(V) \geq 1]$, and the corresponding advantage equals $\delta_\alpha^g = \sum_{V \in M_\alpha^g} (\Pr[\mathcal{V}_{\alpha, 1}^g = V] - \Pr[\mathcal{V}_{\alpha, 0}^g = V])$, where $M_\alpha^g = \{V : T_g(V) \geq 1\}$.

A coercer with full prior knowledge of $\mathcal{B}$ and α faces a binary hypothesis test between obey and evade. By the Neyman–Pearson lemma the most-powerful test compares the likelihood ratio against a threshold, which under the uniform prior on β collapses to threshold 1. Every random coin in the protocol, casting randomness, Fake randomness, the tally proof, contributes to $T_g(V)$ multiplicatively, so the persistent-disclosure setting collapses to the same simple decision rule that drives the CR analysis of [13, 22]; the extra ingredient is that V now contains disclosed local state, not merely the public board.

Closed form for the ideal advantage

Lemma 1 (Ideal advantage, $n_C=2, q_c=1$). *Let $\nu=(\nu_0, \nu_1, \nu_2)$ with $\nu_0+\nu_1+\nu_2=n_V+1$. With marginal $\mathbf{P}^\circ$ given by (2),*

$$\Pr[\nu|\beta=1] = \binom{n_V}{\nu_0, \nu_1-1_{\alpha=1}, \nu_2-1_{\alpha=2}} (P_0^\circ)^{\nu_0} (P_1^\circ)^{\nu_1-1_{\alpha=1}} (P_2^\circ)^{\nu_2-1_{\alpha=2}},$$

$$\Pr[\nu|\beta=0] = \sum_{v=0}^2 P_v^\circ \cdot \binom{n_V}{\nu_0-1_{v=0}, \nu_1-1_{v=1}, \nu_2-1_{v=2}} \prod_{\ell=0}^2 (P_\ell^\circ)^{\nu_\ell-1_{v=\ell}},$$

where any multinomial coefficient with a negative argument is 0. The ideal advantage with claim α is $\delta_\alpha^{\text{ideal}} = \frac{1}{2} \sum_{\nu: \sum_i \nu_i = n_V+1} |\Pr[\nu|\beta=1] - \Pr[\nu|\beta=0]|$, with $O((n_V+1)^2)$ summands, each computable in $O(\log n_V)$ arithmetic operations on log-gamma values.

Proof. Under $\beta=1$ the coerced voter contributes deterministically to coordinate α : $\nu = X + \mathbf{e}_\alpha$ where $X \sim \text{Mult}(n_V, \mathbf{P}^\circ)$. Hence $\Pr[\nu|\beta=1] = \Pr[X = \nu - \mathbf{e}_\alpha]$, the displayed multinomial. Under $\beta=0$ the coerced voter is sampled from $\mathbf{P}^\circ$ and joins the multinomial, yielding $\nu = X + \mathbf{e}_v$ where $v \sim \mathbf{P}^\circ$ independent of X ; marginalising over v gives the displayed mixture. The TV distance is then a direct computation. The complexity follows because each multinomial PMF is exp of an $O(\log n_V)$ sum of pre-tabulated log-gamma values. $\square$

In the ideal experiment, obey and evade both contribute exactly one ballot to the result; they differ only in *which coordinate* the contribution falls on. The $1/\sqrt{n_V}$ scaling of δ^{ideal} visible in our experiments is the central limit theorem at work: shifting a multinomial mean by a single unit becomes harder to detect as the variance grows linearly in n_V .

Definition 11 (Per-ballot deniability advantage). *The per-ballot deniability advantage of Π is*

$$\varepsilon_{\text{ball}}(\Pi, \lambda) := \max_{\mathcal{A}, (v, v^*, \text{cred})} \left| \Pr[\mathcal{A}(\text{pk}, v^*, \rho, b^*) = 1] - \Pr[\mathcal{A}(\text{pk}, v^*, \rho^*, b) = 1] \right|,$$

where the maximum is over PPT adversaries $\mathcal{A}$ and admissible triples (v, v^*, cred) ; $b = \text{Vote}(\text{pk}, \text{cred}, v; \rho)$ with ρ uniform; $b^* = \text{Vote}(\text{pk}, \text{cred}, v^*; \rho'')$ with ρ'' uniform; and $\rho^* = \text{Fake}(\text{pk}, \text{cred}, v, \rho, v^*)$.

We now state formally the four results announced in Section 4.2. The first is the central-limit-theorem rate of the ideal advantage (asymptotic ideal advantage).

Theorem 5. *Fix any $\mathcal{B}$ with marginal $\mathbf{P}^\circ$ satisfying $P_\alpha^\circ \in (0, 1)$, and fix constants q_c, n_C . Then*

$$\delta_\alpha^{\text{ideal}}(\Pi, n_V) = \frac{1}{\sqrt{2\pi P_\alpha^\circ(1-P_\alpha^\circ)}} \cdot \frac{1}{\sqrt{n_V}} (1 + O(n_V^{-1/2})).$$

Proof. Let $X_\alpha := \sum_{i=1}^{n_V} \mathbf{1}[v_i = \alpha] \sim \text{Bin}(n_V, P_\alpha^\circ)$. The coordinate ν_α equals $X_\alpha + B_\alpha$ where $B_\alpha = 1$ under obey and $B_\alpha \sim \text{Bern}(P_\alpha^\circ)$ under evade. By Theorem 4 together with the sufficient statistic argument,

$$\delta_\alpha^{\text{Ideal}} = \frac{1}{2} \text{TV}(\text{Bin}(n_V, P_\alpha^\circ) + 1, \text{Bin}(n_V, P_\alpha^\circ) + \text{Bern}(P_\alpha^\circ)).$$

Let $\sigma^2 := n_V P_\alpha^\circ (1 - P_\alpha^\circ)$. The local CLT for the binomial [144, 145] gives, uniformly in k ,

$$\Pr[\text{Bin}(n_V, p) = k] = \frac{1}{\sqrt{2\pi\sigma}} e^{-(k-np)^2/(2\sigma^2)} (1 + O(n_V^{-1/2})).$$

Applying this and integrating,

$$\begin{aligned} \frac{1}{2} \sum_k |\Pr[\text{Bin} + 1 = k] - \Pr[\text{Bin} + \text{Bern} = k]| &= \frac{1 - P_\alpha^\circ}{2} \sum_k |\Pr[\text{Bin} = k - 1] - \Pr[\text{Bin} = k]| \\ &= \frac{1 - P_\alpha^\circ}{\sqrt{2\pi\sigma}} \cdot \sqrt{\frac{2}{\pi}} \cdot (1 + O(n_V^{-1/2})), \end{aligned}$$

using $\int |\phi'(z)| dz = \sqrt{2/\pi}$. Maximising over claims yields the stated rate. $\square$

The next theorem is the additive channel decomposition, our primary quantitative bound (PCR decomposition).

Theorem 6. *Let Π have a Fake algorithm and a tally simulator. For every PPT adversary issuing at most q_c disclosure queries on voters casting at most k_{vot} ballots each, every behavioural distribution, and every claim vector,*

$$\delta_\alpha^{\text{Real}}(\Pi) \leq \delta_\alpha^{\text{Ideal}}(\Pi) + \delta^{\text{tally}}(\Pi, \lambda) + q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}(\Pi, \lambda) + \delta^{\text{board}}(\Pi, \lambda).$$

Proof. We construct hybrids $H_0, \dots, H_4$ with $H_0 = \text{Exp}^{\text{PCR-Real}}$ and $H_4 = \text{Exp}^{\text{PCR-Ideal}}$.

$H_0 \rightarrow H_1$ (*tally simulation*). Replace (r^*, Π^*) by the simulator's output computed from the shadow record using $\text{Result} \circ \text{Policy} \circ \text{cleanse}$. By tally simulation-security, the gap is $\leq \delta^{\text{tally}}$.

$H_1 \rightarrow H_2$ (*board IND-CPA*). For each honest ballot replace C_v, C_c by canonical encryptions; a $2q_b$ hybrid bounds the gap by δ^{board} .

$H_2 \rightarrow H_3$ (*per-ballot Fake replacement*). For each disclosure and each of k_{vot} ballots, replace the disclosed (cred^*, ρ^*) by a fresh independent honest casting of α_j ; by Definition 11, a $q_c \cdot k_{\text{vot}}$ hybrid contributes $q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}$.

$H_3 \rightarrow H_4$ (*ideal reduction*). The only β -dependent quantity is r^* , captured by $\text{Exp}^{\text{PCR-Ideal}}$; the gap is $\leq \delta_\alpha^{\text{Ideal}}$. Combining via the triangle inequality and maximizing over $\mathcal{A}$ yields the bound. $\square$

The decomposition is tight up to a factor of 4 and the next theorem address the tightness of the decomposition.

Theorem 7. *There exist a behavioral distribution $\mathcal{B}^*$, a claim vector α^* , and a PPT adversary $\mathcal{A}^*$ such that*

$$\delta_{\alpha^*}^{\text{Real}}(\Pi) \geq \frac{1}{4} \left(\delta_{\alpha^*}^{\text{Ideal}}(\Pi) + \delta^{\text{tally}}(\Pi, \lambda) + q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}(\Pi, \lambda) + \delta^{\text{board}}(\Pi, \lambda) \right) - \text{negl}(\lambda).$$

Proof Sketch. We consider four adversaries $\mathcal{A}^{(I)}, \mathcal{A}^{(T)}, \mathcal{A}^{(B)}, \mathcal{A}^{(D)}$, each saturating one term: an ideal-game distinguisher achieving $\delta_{\alpha^*}^{\text{Ideal}}$; a simulator-gap distinguisher witnessing δ^{tally} ; a CPA adversary witnessing δ^{board} ; and a per-disclosure distinguisher witnessing $\varepsilon_{\text{ball}}$ on some disclosed ballot, which by hybrid converse achieves $q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}$. Since these adversaries target independent channels, no single primitive can simultaneously suppress all four advantages; the best of the four achieves advantage at least $\frac{1}{4}$ of the sum. $\square$

Theorem 8. *Let Π be a voting scheme whose only disclosure is the genuine local state $(\text{cred}, \mathbf{v}, \boldsymbol{\rho}, \mathbf{b})$. Then there exists $\mathcal{B}^*, \alpha^*$ such that for every $n_V \geq 1$, $q_c \geq 1$,*

$$\Delta^{\text{pers}}(\Pi) \geq 1 - \min_{\alpha} P_{\alpha}^{\circ} - \delta^{\text{ideal}}(\Pi, n_V).$$

In particular, by choosing $\mathcal{B}^$ such that $\min_{\alpha} P_{\alpha}^{\circ} \leq 1/n_C$ and applying Theorem 5, for any polynomial n_V , $\Delta^{\text{pers}}(\Pi) \geq 1 - 1/n_C - O(1/\sqrt{n_V})$.*

Proof Sketch. Fix $\mathcal{B}^*$ with support on $[1, n_C]$ and a claim α^* . The adversary $\mathcal{A}^*$ coerces a single voter j , issues $\mathcal{O}_{\text{reveal}}(j)$ after the tally, and outputs $\beta' = 1$ iff $\mathbf{v}_j = (\alpha^*)$ as a sequence. Under $\beta = 1$ this holds with probability 1. Under $\beta = 0$, the voter casts honestly; the probability that all k_{vot} casts are for α^* is at most $P_{\alpha^*}^{\circ}$. The advantage is therefore at least $1 - P_{\alpha^*}^{\circ}$. Choosing $\alpha^* = \text{argmin}_{\alpha} P_{\alpha}^{\circ}$ and applying Theorem 5 yields the claim. $\square$

Theorem 9 (Composition under independent coercion). *Suppose the behavioural distribution factorises as $\mathcal{B} = \bigotimes_{j \in \mathcal{H}_{\text{coerce}}} \mathcal{B}_j$. Then $\delta_{\alpha}^{\text{ideal}}(\Pi, q_c) \leq \sum_{j \in \mathcal{H}_{\text{coerce}}} \delta_{\alpha_j}^{\text{ideal}}(\Pi, 1)$, and equality holds up to a factor of 2 in the worst case.*

Proof Sketch. By the standard TV sub-additivity for product measures sharing one coordinate [146], the TV distance between $\bigotimes_j \mathcal{V}_{\alpha_j, 1}$ and $\bigotimes_j \mathcal{V}_{\alpha_j, 0}$ is bounded by $\sum_j \text{TV}(\mathcal{V}_{\alpha_j, 1}, \mathcal{V}_{\alpha_j, 0})$. When per-voter advantages are small ($\delta_j \ll 1/q_c$), the linear bound is tight up to factor 2 by the binomial approximation to the joint TV distance. $\square$

A.6 Efficient Access Structures ($\mathbb{S}, N$) and Circuit class ($\mathbb{C}, d$)

We consider *access structures* as all sets of authorized parties $\mathbb{A} \subset \mathbb{S}$ with N total parties $|\mathbb{A}| \leq N$. Access structures are *efficient* if they are monotone (i.e., $\forall \mathbb{A} \subset \mathbb{S}$ and $|\mathbb{A}| < N$ implies $\mathbb{A} \cup \{x\} \subset \mathbb{S}$) and they can be described using a boolean circuit $f_{\mathbb{A}}: \{0, 1\}^N \rightarrow \{0, 1\}$. A particular type of efficient access structures is *threshold access structures* where a threshold t defines the minimal size of the authorized set: $t \leq \mathbb{A}$ for $\mathbb{A} \subset \mathbb{S}$. We use circuits of arbitrary inputs $\mathbb{C}: \{0, 1\}^k \rightarrow \{0, 1\}$ that are bounded by depth d . For simplicity in some parts, we use $C \in \mathbb{C}$.

A.7 Formal Properties of dFHE (Definition 3)

Definition 12 (Correctness). *A dFHE scheme is correct if for every security parameter λ , every polynomial-time circuit $C: \mathcal{M}^k \rightarrow \mathcal{M}$, and every tuple of messages $m_1, \dots, m_k \in \mathcal{M}$:*

$$\Pr \left[\text{Dec}(dsk, \text{Eval}(dpk, C, dct_1, \dots, dct_k)) = C(m_1, \dots, m_k) \right] \geq 1 - \text{negl}(\lambda),$$

where the probability is over $(dpk, dsk) \leftarrow \text{Gen}(1^\lambda)$ and $dct_i \leftarrow \text{Enc}(dpk, m_i)$ for $i \in [k]$. We note that simultaneous perfect correctness and deniability are impossible [93]; hence a negligible decryption error is inherent.

Definition 13 (Compactness). *A dFHE scheme is compact if there exists a polynomial $\text{poly}(\cdot)$ such that for every security parameter λ , every polynomial-time circuit $C: \mathcal{M}^k \rightarrow \mathcal{M}$, and every tuple of messages $m_1, \dots, m_k \in \mathcal{M}$:*

$$\Pr \left[|\text{Eval}(dpk, C, dct_1, \dots, dct_k)| \leq \text{poly}(\lambda) \right] = 1,$$

where $(dpk, dsk) \leftarrow \text{Gen}(1^\lambda)$ and $dct_i \leftarrow \text{Enc}(dpk, m_i)$ for $i \in [k]$. That is, the output size of Eval is bounded by a fixed polynomial in λ , independently of the circuit C or the number of inputs k .

Definition 14 (IND-CPA Security). A dFHE scheme is IND-CPA secure if for all PPT adversaries $\mathcal{A}$:

$$\left| \Pr[\text{CPA-Game}_{\mathcal{A}}^{\text{dFHE}}(\lambda)=1] - \frac{1}{2} \right| \leq \text{negl}(\lambda),$$

where $\text{CPA-Game}_{\mathcal{A}}^{\text{dFHE}}(\lambda)$ is the following game between a challenger $\mathcal{C}$ and $\mathcal{A}$:

1. $\mathcal{C}$ samples $(dpk, dsk) \leftarrow \text{Gen}(1^\lambda)$ and sends dpk to $\mathcal{A}$.
2. $\mathcal{A}$ chooses two messages $m_0, m_1 \in \mathcal{M}$ and sends them to $\mathcal{C}$.
3. $\mathcal{C}$ samples a uniform challenge bit $b \leftarrow \{0,1\}$, computes $dct^* \leftarrow \text{Enc}(dpk, m_b)$, and sends dct^* to $\mathcal{A}$.
4. $\mathcal{A}$ outputs a guess $b' \in \{0,1\}$. The game outputs 1 if and only if $b' = b$.

Definition 15 (Deniability Compactness). A $\delta(\lambda)$ -deniable dFHE scheme is deniability compact if there exists a polynomial $\text{poly}(\cdot)$ such that for all security parameters λ and all messages $m \in \mathcal{M}$:

$$\Pr[|\text{Enc}(dpk, m)| \leq \text{poly}(\lambda)] = 1,$$

where $(dpk, dsk) \leftarrow \text{Gen}(1^\lambda)$, and the bound holds regardless of the detection probability $1/\delta(\lambda)$ achieved by the scheme. That is, the ciphertext size is bounded by a fixed polynomial in λ that is entirely independent of $\delta(\lambda)$, even as $1/\delta(\lambda)$ grows. This property distinguishes dFHE from all prior deniable encryption constructions based on polynomial hardness assumptions [93], in which the ciphertext size grows inversely with the detection probability.

Definition 16 ($\delta(\lambda)$ -Deniability). A dFHE scheme is $\delta(\lambda)$ -deniable if for all PPT adversaries $\mathcal{A}$:

$$\left| \Pr[\text{Dnbl-Game}_{\mathcal{A},0}^{\text{dFHE}}(\lambda)=1] - \Pr[\text{Dnbl-Game}_{\mathcal{A},1}^{\text{dFHE}}(\lambda)=1] \right| \leq \delta(\lambda),$$

where, for a challenge bit $b \in \{0,1\}$, the game $\text{Dnbl-Game}_{\mathcal{A},b}^{\text{dFHE}}(\lambda)$ proceeds as follows:

1. $\mathcal{C}$ samples $(dpk, dsk) \leftarrow \text{Gen}(1^\lambda)$ and sends dpk to $\mathcal{A}$.
2. $\mathcal{A}$ chooses an original message $m \in \mathcal{M}$ and a target fake message $m^* \in \mathcal{M}$, and sends them to $\mathcal{C}$.
3. $\mathcal{C}$ samples $r \leftarrow \{0,1\}^\ell$ and computes $r^* \leftarrow \text{Fake}(dpk, m, r, m^*)$.
4. Depending on the challenge bit b :
 - $b=0$: $\mathcal{C}$ sends $(m^*, r, \text{Enc}(dpk, m^*; r))$ to $\mathcal{A}$. (honest encryption of m^*)
 - $b=1$: $\mathcal{C}$ sends $(m^*, r^*, \text{Enc}(dpk, m; r))$ to $\mathcal{A}$. (faked randomness for m)
5. $\mathcal{A}$ outputs a bit b' . The game outputs 1 iff $b' = 1$.

Intuitively, when $b=0$ the adversary sees a genuine encryption of m^* with honest coins, and when $b=1$ it sees the original ciphertext of m presented with fake coins r^* that make it appear to encrypt m^* instead. Deniability requires that no PPT adversary can distinguish these two views with advantage greater than $\delta(\lambda)$. The parameter $\delta(\lambda)$ is an inverse polynomial in λ for the fully deniable scheme, and $\text{negl}(\lambda)$ for the weakly deniable variant [93].

A.8 Formal Properties of UT (Definition 4)

Definition 17 (Correctness). A static universal thresholdizer scheme UT satisfies correctness if for all $\lambda \in \mathbb{N}$, all efficient access structures $(\mathbb{S}, N)$, all $x \in \{0,1\}^k$, all circuits $C: \{0,1\}^k \rightarrow \{0,1\}$ of depth at most d , and all authorized sets $S \in \mathbb{S}$:

$$\Pr \left[\begin{array}{l} (\mathbf{p}, s_1, \dots, s_N) \leftarrow \text{Setup}(1^\lambda, 1^d, (\mathbb{S}, N), x) \\ \forall i \in S: y_i \leftarrow \text{Eval}(\mathbf{p}, s_i, C) \\ y \leftarrow \text{Combine}(\mathbf{p}, \{y_i\}_{i \in S}) \end{array} : y = C(x) \right] = 1 - \text{negl}(\lambda)$$

Moreover, for all honestly generated partial evaluations: $\Pr[\text{Verify}(\mathbf{p}, y_i, C) = 1] = 1$.

Definition 18 (Compactness). A static universal thresholdizer scheme UT satisfies compactness if there exists a polynomial $\text{poly}(\lambda)(\cdot)$ such that for all $\lambda \in \mathbb{N}$, all efficient access structures $(\mathbb{S}, N)$, all $x \in \{0,1\}^k$, and all circuits C of depth at most d : $|y_i| \leq \text{poly}(\lambda)(\lambda, d, N)$, where $y_i \leftarrow \text{Eval}(\mathbf{p}, s_i, C)$ for honestly generated parameters and shares.

Definition 19 (Security). A UT scheme satisfies security with respect to circuit sampler D if for all $\lambda \in \mathbb{N}$, there exists a PPT algorithm $\mathcal{S} = (\mathcal{S}_S, \mathcal{S}_E)$ such that for any PPT adversary $\mathcal{A}$, the experiments in Figure 25 and Figure 26 $\text{Exp}_{\mathcal{A}, \text{UT}}^{\text{real}}(1^\lambda, d, N)$ and $\text{Exp}_{\mathcal{A}, \text{UT}}^{\text{ideal}}(1^\lambda, d, N)$ are computationally indistinguishable.

$\text{Exp}_{\mathcal{A}, \text{UT}}^{\text{real}}(1^\lambda, d, N)$

- 1: $x \leftarrow \mathcal{A}(1^\lambda, d, N)$
- 2: $(\mathbf{pp}, \mathbf{sk}_1, \dots, \mathbf{sk}_N) \leftarrow \text{UT.Setup}(1^\lambda, d, N, x)$
- 3: $\mathcal{C} \leftarrow \mathcal{A}(\mathbf{pp})$, where $|\mathcal{C}| = d - 1$
- 4: $\{\mathbf{sk}_i\}_{i \in \mathcal{C}} \rightarrow \mathcal{A}$
- 5: $\forall C \in \mathcal{Q}: \{y_i \leftarrow \text{UT.Eval}(\mathbf{pp}, \mathbf{sk}_i, C)\}_{i \in [N]} \leftarrow \mathcal{O}_{\text{UT}}^{\text{real}}(C)$
- 6: $b \leftarrow \mathcal{A}^{\mathcal{O}_{\text{UT}}^{\text{real}}}(\mathbf{pp}, \{\mathbf{sk}_i\}_{i \in \mathcal{C}})$
- 7: **return** b

Fig. 25: Real-World Security Experiment for UT

Definition 20 (Robustness). A UT is robust if for all λ , and depth bound d , the following holds. For any PPT adversary $\mathcal{A}$, the experiment $\text{Expt}_{\mathcal{A}, \text{UT}, \text{rb}}(1^\lambda, 1^d)$ in Figure 27 outputs 1 with negligible probability.

$\text{Exp}_{\mathcal{A}, \text{UT}}^{\text{ideal}}(1^\lambda, d, N)$
 1: $x \leftarrow \mathcal{A}(1^\lambda, d, N)$
 2: $(\text{pp}, \text{sk}_1, \dots, \text{sk}_N) \leftarrow \text{Sim}_S(1^\lambda, d, N)$
 3: $\mathcal{C} \leftarrow \mathcal{A}(\text{pp})$, where $|\mathcal{C}| = d-1$
 4: $\{\text{sk}_i\}_{i \in \mathcal{C}} \rightarrow \mathcal{A}$
 5: $\forall C \in \mathcal{Q}: \{y_i\}_{i \in [N]} \leftarrow \text{Sim}_E(\text{trapd}, C, C(x)) \leftarrow \mathcal{O}_{\text{UT}}^{\text{ideal}}(C)$
 6: $b \leftarrow \mathcal{A}^{\mathcal{O}_{\text{UT}}^{\text{ideal}}}(\text{pp}, \{\text{sk}_i\}_{i \in \mathcal{C}})$
 7: **return** b

Fig. 26: Ideal-World Security Experiment for UT

$\text{Exp}_{\mathcal{A}, \text{UT}, \text{rb}}(1^\lambda, 1^d)$
 1: $(x, \mathbb{A}) \leftarrow \mathcal{A}(1^\lambda, 1^d)$
 2: $(\text{pp}, \text{s}_1, \dots, \text{s}_N) \leftarrow \text{UT.Setup}(1^\lambda, 1^d, \mathbb{A}, x)$
 3: $y_i^* \leftarrow \mathcal{A}(\text{pp}, \text{s}_1, \dots, \text{s}_N)$
 4: **return** $y_i^* \neq \text{UT.Eval}(\text{pp}, \text{s}_i, C) \wedge \text{UT.Verify}(\text{pp}, y_i^*, C) = 1$

Fig. 27: Robustness Experiment for Universal Thresholdizer

A.9 Formal Properties of VSS (Definition 5)

We recall the three formal security properties required of a hash-based (t, n) -VSS scheme $\Pi = (\text{Share}, \text{VerifyShare}, \text{Reconstruct})$ as introduced by Baghery [110], operating in the quantum random oracle model with $n \geq 2t+1$ parties and threshold t . The scheme works over $\mathbb{Z}_N$ (where N is a large modulus), uses a hash function H modelled as a quantum random oracle, and produces shares $\{s_i, \gamma_i\}_{i=1}^n$ together with a public proof $\pi_{\text{share}} = (z(X), c_1, \dots, c_n)$ (see Definition 5).

Definition 21 (Correctness). *The VSS scheme Π is correct if, for all $\lambda \in \mathbb{N}$, all $n \geq 2t+1$, all $s \in \mathbb{Z}_N$, and all $S \subseteq [n]$ with $|S| \geq t+1$:*

$$\Pr \left[\begin{array}{l} \text{VerifyShare}(i, s_i, \gamma_i, \pi_{\text{share}}) = 1 \ \forall i \in [n] \\ \text{Reconstruct}(\{(i, s_i, \gamma_i)\}_{i \in S}, \pi_{\text{share}}) = s \end{array} \middle| (\{s_i, \gamma_i\}_{i=1}^n, \pi_{\text{share}}) \leftarrow \text{Share}(s, t, n) \right] = 1.$$

Definition 22 (Verifiability). *The VSS scheme Π is verifiable if, for all integers $n \geq 2t+1$ and $t \geq 0$, and for all PPT adversaries $\mathcal{A}$,*

$$\Pr \left[\begin{array}{l} \exists s^* \in \mathbb{Z}_N \text{ s.t.} \\ \forall V \subseteq [n], |V| \geq t+1: \\ \quad \text{Reconstruct}(\{(i, s_i, \gamma_i)\}_{i \in V}, \pi) = s^* \end{array} \middle| \begin{array}{l} (\{s_i, \gamma_i\}_{i=1}^n, \pi) \leftarrow \mathcal{A}(1^\lambda, n, t) \\ \text{VerifyShare}(i, s_i, \gamma_i, \pi) = 1 \\ \forall i \in V \end{array} \right] \geq 1 - \text{negl}(\lambda),$$

where V denotes the set of honest parties. In other words, if every honest party's share passes verification, then any $t+1$ honest parties will reconstruct the same unique secret s^* , regardless of how the shares were generated.

Definition 23 (Unpredictability). *The VSS scheme Π is unpredictable if, for all integers $n \geq 2t+1$ and $t \geq 0$, for all PPT adversaries $\mathcal{A}$ who non-adaptively corrupt a set $Q \subset [n]$ with $|Q| \leq t$, and for a uniformly random secret $s \xleftarrow{\$} \mathbb{Z}_N$,*

$$\Pr \left[s' = s \mid \begin{array}{l} (\{s_i, \gamma_i\}_{i=1}^n, \pi_{\text{share}}) \leftarrow \text{Share}(s, t, n) \\ s' \leftarrow \mathcal{A}(1^\lambda, n, t, \{s_i, \gamma_i\}_{i \in Q}, \pi_{\text{share}}) \end{array} \right] \leq \text{negl}(\lambda).$$

That is, no computationally bounded adversary controlling fewer than $t+1$ parties can predict the secret s with non-negligible advantage, even given the public proof π_{share} and the shares of all corrupted parties.

A.10 Formal Properties of MixNet (Definition 7)

We briefly recall the setting. BGV ciphertexts are pairs $(u, v) \in R_q^2$, where $R_q = \mathbb{Z}_q[X]/\langle X^N + 1 \rangle$, encrypted under a public key $pk = (a, b) = (a, as + pe)$ for short secret s and noise e . A ciphertext (u, v) is *reliably decryptable* under secret key $sk = s$ if the predicate $P_{sk}(u, v) := [\|v - su\|_\infty < B_{\text{Dec}}]$ equals 1, for a publicly known noise bound $B_{\text{Dec}} < \lfloor q/2 \rfloor$. We write $\text{Dec}(sk, c)$ for standard BGV decryption: $(v - su \bmod q) \bmod p$. The protocol involves ξ_1 shuffle servers and ξ_2 decryption servers; κ is the computational security parameter and sec is the statistical security parameter. Public parameters pp output by $\text{Setup}(1^\kappa, \xi_1, \xi_2)$ are implicit inputs to all algorithms. Proofs π_{Mix} produced by Mix consist of BDLOP commitment openings $\{\mathbf{t}_i\}$, an amortized zero-knowledge proof of shortness π_{Small} (showing each re-randomization ciphertext is well-formed), and a verifiable shuffle proof π_{Shuf} . Proofs π_{Dec_j} produced by DDec consist of BDLOP commitments to noise terms $\{\llbracket E_{i,j} \rrbracket\}$, zero-knowledge proofs of linear relations $\{\pi_{L_{i,j}}\}$, and an amortized boundedness proof π_{Bnd_j} .

Definition 24 (Correctness). *MixNet is correct if for every positive integer τ , every τ -tuple of BGV ciphertexts $\{\mathbf{c}_i\}_{i \in [\tau]}$ satisfying $P_{sk}(\mathbf{c}_i) = 1$ for all i and with total noise bounded so that $(1 + 2^{\text{sec}})B_{\text{Dec}} < \lfloor q/2 \rfloor$, the following holds:*

$$\Pr \left[\text{Comb}(\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \{\{t_{i,j}\}_{i \in [\tau]}\}_{j \in [\xi_2]}) = \{m_i\}_{i \in [\tau]} \right] \geq 1 - \text{negl}(\kappa),$$

where $m_i = \text{Dec}(sk, \mathbf{c}_i)$ for each $i \in [\tau]$, and the probability is over the randomness of: $(\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \pi_{\text{Mix}}) \leftarrow \text{Mix}(pp, \{\mathbf{c}_i\}_{i \in [\tau]})$, $(\{t_{i,j}\}_{i \in [\tau]}, \pi_{\text{Dec}_j}) \leftarrow \text{DDec}(sk_j, \{\hat{\mathbf{c}}_i\}_{i \in [\tau]}) \forall j \in [\xi_2]$.

Definition 25 (Verifiability). *MixNet is verifiable if no PPT adversary $\mathcal{A}$, given $(pp, \{\mathbf{c}_i\}_{i \in [\tau]})$, can produce $(\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}, \pi_{\text{Mix}})$ such that the mix proof verifies yet the multi-set of plaintexts is altered. More formally:*

$$\Pr \left[\begin{array}{l} \text{MixVerify}(pp, \{\hat{\mathbf{c}}_i\}, \{\mathbf{c}_i\}, \pi_{\text{Mix}}) = 1 \\ \wedge \quad \{\text{Dec}(sk, \hat{\mathbf{c}}_i)\}_{i \in [\tau]} \neq \{\text{Dec}(sk, \mathbf{c}_i)\}_{i \in [\tau]} \end{array} \mid (\{\hat{\mathbf{c}}_i\}, \pi_{\text{Mix}}) \leftarrow \mathcal{A}(pp, \{\mathbf{c}_i\}) \right] \leq \text{negl}(\kappa),$$

where the multisets are compared as sets with multiplicity. The probability is taken over the coin tosses of $\mathcal{A}$, and security reduces to the binding property of the BDLOP commitment scheme and the special soundness of Π_{Small} and $\Pi_{\text{Shuf}}^{\ell_c}$ (under the SKS^2 and DKS^∞ assumptions).

Definition 26 (Secrecy). *MixNet is secret (the shuffle is zero-knowledge) if there exists a PPT simulator $\mathcal{S}$ such that for every τ -tuple $\{\mathbf{c}_i\}_{i \in [\tau]}$ and every PPT distinguisher $\mathcal{D}$:*

$$\left| \Pr[\mathcal{D}(pp, \{\mathbf{c}_i\}, \{\hat{\mathbf{c}}_i\}, \pi_{\text{Mix}}) = 1] - \Pr[\mathcal{D}(pp, \{\mathbf{c}_i\}, \{\hat{\mathbf{c}}_i\}, \tilde{\pi}_{\text{Mix}}) = 1] \right| \leq \text{negl}(\kappa),$$

where $(\{\hat{\mathbf{c}}_i\}, \pi_{\text{Mix}}) \leftarrow \text{Mix}(pp, \{\mathbf{c}_i\})$ is a real execution with secret permutation π , and $(\{\hat{\mathbf{c}}_i\}, \tilde{\pi}_{\text{Mix}}) \leftarrow \mathcal{S}(pp, \{\mathbf{c}_i\}, \{\hat{\mathbf{c}}_i\})$ is a simulated transcript produced without knowledge of π . In particular, no PPT adversary can determine the permutation π from $(pp, \{\mathbf{c}_i\}, \{\hat{\mathbf{c}}_i\}, \pi_{\text{Mix}})$ with non-negligible advantage. Security follows from the honest-verifier zero-knowledge property of Π_{Small} and $\Pi_{\text{Shuf}}^{\ell_c}$.

Definition 27 (Threshold Verifiability). *MixNet has threshold verifiability with respect to P_{sk} if for every PPT adversary $\mathcal{A}$ that is given pp and corrupts a set $J \subseteq [\xi_2]$ of decryption key shares $\{sk_j\}_{j \in J}$, the following probability is bounded by $\text{negl}(\kappa)$:*

$$\Pr \left[\begin{array}{l} \text{Comb}(\{\hat{\mathbf{c}}_i\}, \{\{t_{i,j}\}\}_{j \in [\xi_2]}) \neq \perp \\ \wedge \text{Comb}(\{\hat{\mathbf{c}}_i\}, \{\{t_{i,j}\}\}_{j \in [\xi_2]}) \neq \{\text{Dec}(sk, \hat{\mathbf{c}}_i)\}_{i \in [\tau]} \end{array} \middle| \begin{array}{l} \forall i \in [\tau]: P_{sk}(\hat{\mathbf{c}}_i) = 1 \\ \forall j \notin J: (\{t_{i,j}\}_i, \pi_{\text{Dec}_j}) \leftarrow \text{DDec}(sk_j, \{\hat{\mathbf{c}}_i\}) \\ \{t_{i,j}, \pi_{\text{Dec}_j}\}_{j \in J} \leftarrow \mathcal{A}(pp, \{sk_j\}_{j \in J}, \{\hat{\mathbf{c}}_i\}, \{t_{i,j}, \pi_{\text{Dec}_j}\}_{j \notin J}) \end{array} \right],$$

where the probability is taken over **Setup**, **DDec**, and the coin tosses of $\mathcal{A}$. Security reduces to the knowledge soundness of Π_{Lin} (which proves each linear relation $t_{i,j} = s_j u_i + p E_{i,j}$) and Π_{Bnd} (which proves the noise terms $E_{i,j}$ are correctly bounded); specifically, any adversary $\mathcal{A}$ with advantage ϵ_0 implies adversaries against the soundness of Π_{Lin} or Π_{Bnd} with combined advantage at least ϵ_0 .

Definition 28 (Decryption Simulatability). *MixNet is decryption simulatable with respect to P_{sk} if there exists a PPT simulator Sim such that for every PPT adversary $\mathcal{A}$ corrupting a strict subset $J \subsetneq [\xi_2]$ of key shares $\{sk_j\}_{j \in J}$, every positive integer τ , and every τ -tuple $\{\hat{\mathbf{c}}_i\}_{i \in [\tau]}$ with $P_{sk}(\hat{\mathbf{c}}_i) = 1$:*

$$\left| \Pr[\mathcal{A}(pp, \{sk_j\}_{j \in J}, \{t_{i,j}^{\text{real}}, \pi_{\text{Dec}_j}^{\text{real}}\}_{j \in [\xi_2]}) = 1] \right. \\ \left. - \Pr[\mathcal{A}(pp, \{sk_j\}_{j \in J}, \{t_{i,j}^{\text{sim}}, \pi_{\text{Dec}_j}^{\text{sim}}\}_{j \in [\xi_2]}) = 1] \right| \leq \text{negl}(\text{sec}),$$

where, for each $j \notin J$, the real output is

$$(\{t_{i,j}^{\text{real}}\}_{i \in [\tau]}, \pi_{\text{Dec}_j}^{\text{real}}) \leftarrow \text{DDec}(sk_j, \{\hat{\mathbf{c}}_i\}_{i \in [\tau]}),$$

and the simulated output is

$$(\{t_{i,j}^{\text{sim}}\}_{i \in [\tau]}, \pi_{\text{Dec}_j}^{\text{sim}}) \leftarrow \text{Sim}(pp, \{sk_j\}_{j \in J}, \{(\hat{\mathbf{c}}_i, \text{Dec}(sk, \hat{\mathbf{c}}_i))\}_{i \in [\tau]});$$

the adversary's decryption shares for $j \in J$ are set to the honest output of $\text{DDec}(sk_j, \{\hat{\mathbf{c}}_i\})$ for each $j \in J$. The probability is taken over **Setup**, **DDec**, **Sim**, and the coin tosses of $\mathcal{A}$. Security follows from the honest-verifier zero-knowledge of Π_{Lin} and Π_{Bnd} together with the hiding property of the BDLOP commitment scheme.

A.11 Formal Properties of deniable NIZK (Section 5.2)

Definition 29 (Completeness). *For all λ , all $(x, w) \in \mathcal{R}$, and $(\text{crs}, \text{td}) \leftarrow \text{Setup}(1^\lambda)$,*

$$\Pr[\text{Verify}(\text{crs}, x, \text{Prove}(\text{crs}, \text{td}, x, w; r)) = 1] \geq 1 - \text{negl}(\lambda),$$

where the probability is over r and the internal coins of **Setup**. Negligible completeness error is permitted to absorb the rejection-sampling abort probability.

Definition 30 (Knowledge Soundness). *There exists a PPT extractor $\mathcal{E}$ such that for every PPT prover $\mathcal{P}^*$ producing an accepting proof (x, π) with non-negligible probability, $\mathcal{E}^{\mathcal{P}^*}(\text{crs}, x)$ outputs w with $(x, w) \in \mathcal{R}$ with non-negligible probability, assuming $\text{MSIS}_{n,m,q,\beta_{\text{SIS}}}$ is hard. The reduction holds against adversaries that may possess td .*

Definition 31 (Computational Zero-Knowledge). *There exists a PPT simulator $\mathcal{S}$ such that for every PPT distinguisher $\mathcal{D}$ and every $(x, w) \in \mathcal{R}$,*

$$|\Pr[\mathcal{D}(\text{crs}, x, \text{Prove}(\text{crs}, \text{td}, x, w; r)) = 1] - \Pr[\mathcal{D}(\text{crs}, x, \mathcal{S}(\text{crs}, x)) = 1]| \leq \text{negl}(\lambda),$$

under $\text{MLWE}_{n, m, q, \chi}$.

Definition 32 (Randomness Deniability). *For every PPT adversary $\mathcal{A}$ and every (x, w, w^*) with $(x, w), (x, w^*) \in \mathcal{R}$,*

$$|\Pr[\mathcal{A}(\text{crs}, x, w, r, \pi) = 1] - \Pr[\mathcal{A}(\text{crs}, x, w^*, r^*, \pi) = 1]| \leq \mu_{\text{MLWE}}(\lambda) + \mu_{\text{RS}}(\lambda),$$

where $\pi \leftarrow \text{Prove}(\text{crs}, \text{td}, x, w; r)$ with r uniform, $r^ \leftarrow \text{Fake}(\text{crs}, \text{td}, x, w, r, w^*)$, μ_{MLWE} is the Module-LWE distinguishing advantage on the crs matrices, and μ_{RS} is the statistical distance of RejSamp on inputs of norm gap $\leq 2\beta_w$.*

In the following, we state and prove the four security properties of NIZK in the CRS-plus-QROM hybrid model: the structural matrices and H_{chal} 's SIS-key live in the CRS, while H_{chal} 's output distribution is modelled as a quantum random oracle for the purposes of $\text{Prove}_{\text{rel}}$'s Fiat-Shamir compilation.

Theorem 10. *For every $(x, w) \in \mathcal{R}$ and every $(\text{crs}, \text{tdkey}) \leftarrow \text{Setup}(1^\lambda)$,*

$$\Pr[\text{Verify}(\text{crs}, x, \text{Prove}(\text{crs}, \text{tdkey}, x, w; r)) = 1] \geq 1 - \text{negl}(\lambda),$$
where the probability is taken over $r \leftarrow \mathcal{U}$ with the convention that the caller resamples r on RejSamp abort.

Proof. On input (x, w) with $(x, w) \in \mathcal{R}$, the prover obtains $\mathbf{u} = \mathbf{A}_{\text{com}} r_{\text{com}} + \mathbf{B}'_{\text{com}} \mathbf{s}$ for $\mathbf{s} = \text{Encode}(w)$, a deterministic challenge $c = H_{\text{chal}}(\text{crs}, x, \mathbf{u})$, and a masking vector $\mathbf{y}$ from $\text{RejSamp}(r_{\text{samp}}; c \cdot \mathbf{s})$. The verifier recomputes c from $(\text{crs}, x, \mathbf{u})$, so the challenge agrees. The response $\mathbf{z} = r_{\text{com}} + \mathbf{y}$ satisfies $\|\mathbf{z}\|_2 \leq \|r_{\text{com}}\|_2 + \|\mathbf{y}\|_2 \leq \beta_z$ by the output-bound guarantee of RejSamp . The relation sub-proof π_{rel} is accepted by $\text{Verify}_{\text{rel}}$ by completeness of the underlying commit-and-prove relation argument [39]. The only failure mode is the rejection-sampling abort, which occurs with probability $1 - 1/M$ per attempt and is absorbed by resampling. $\square$

Theorem 11. *Assume $\text{MSIS}_{n, m, q, \beta_{\text{SIS}}}$ is hard. There exists a PPT extractor $\mathcal{E}$ such that for every PPT prover $\mathcal{P}^*$ (possibly given tdkey) producing an accepting (x, π) with probability ε , $\mathcal{E}^{\mathcal{P}^*}(\text{crs}, x)$ outputs w with $(x, w) \in \mathcal{R}$ with probability at least $\varepsilon - \text{Adv}_{\text{B}_{\text{SIS}}}^{\text{MSIS}}(\lambda) - \text{Adv}_{\text{B}_{\text{rel}}, \text{rel}}^{\text{KS}}(\lambda) - \text{negl}(\lambda)$.*

Proof. The extractor runs $\mathcal{P}^*$ to obtain $(x, \pi = (\mathbf{u}, \mathbf{z}, \pi_{\text{rel}}))$. It then invokes the relation-proof extractor $\mathcal{E}_{\text{rel}}$ of [39] on π_{rel} , which, by knowledge soundness of the commit-and-prove argument, returns $(\hat{\mathbf{s}}, \hat{r})$ such that $\mathbf{u} = \mathbf{A}_{\text{com}} \hat{r} + \mathbf{B}'_{\text{com}} \hat{\mathbf{s}}$ and the quadratic relation holds for $\hat{\mathbf{s}}$. The extractor outputs $w = \text{Decode}(\hat{\mathbf{s}})$. Possession of tdkey by $\mathcal{P}^*$ does not aid the adversary: the trapdoor enables equivocation between *valid* witnesses but cannot produce a π_{rel} for an $\hat{\mathbf{s}}$ violating $\mathcal{R}$, since this would directly break the soundness of $\text{Prove}_{\text{rel}}$. If $\mathcal{P}^*$ produces π_{rel} for which extraction fails or yields an inconsistent $(\hat{\mathbf{s}}, \hat{r})$, this either contradicts knowledge soundness of the relation argument (advantage $\text{Adv}_{\text{B}_{\text{rel}}, \text{rel}}^{\text{KS}}$) or yields a short solution to the MSIS instance defined by $\mathbf{A}_{\text{com}}$ (advantage $\text{Adv}_{\text{B}_{\text{SIS}}}^{\text{MSIS}}$). $\square$

Theorem 12. *Assume $\text{MLWE}_{n, m, q, \chi}$ is hard. There exists a PPT simulator $\mathcal{S}$ such that for every PPT distinguisher $\mathcal{D}$ and every $(x, w) \in \mathcal{R}$,*

$$|\Pr[\mathcal{D}(\text{crs}, x, \text{Prove}(\text{crs}, \text{tdkey}, x, w; r)) = 1] - \Pr[\mathcal{D}(\text{crs}, x, \mathcal{S}(\text{crs}, x)) = 1]| \leq \text{Adv}_{\text{B}_{\text{LWE}}}^{\text{MLWE}}(\lambda) + \text{Adv}_{\text{B}_{\text{rel}}, \text{rel}}^{\text{ZK}}(\lambda) + \text{negl}(\lambda).$$

Proof. The simulator $\mathcal{S}$ samples $\mathbf{u} \leftarrow \mathcal{R}_q^n$ uniformly, programs the quantum random oracle so that $H_{\text{chal}}(\text{crs}, x, \mathbf{u})$ outputs a fresh $c \leftarrow \mathcal{R}_q$, samples $\mathbf{z}$ from the discrete-Gaussian distribution of width σ_z truncated to $\|\mathbf{z}\|_2 \leq \beta_z$ (the output distribution of RejSamp on input $c \cdot \mathbf{s}$), and invokes the zero-knowledge simulator $\mathcal{S}_{\text{rel}}$ of [39] on $(\text{crs}, x, \mathbf{u}, c, \mathbf{z})$ to obtain π_{rel} . The output $\pi = (\mathbf{u}, \mathbf{z}, \pi_{\text{rel}})$ verifies by construction. We argue indistinguishability via two hybrids: in H_1 the genuine proof is produced but π_{rel} is replaced by $\mathcal{S}_{\text{rel}}$'s output, introducing a gap of $\text{Adv}_{\mathcal{B}_{\text{rel}}, \text{rel}}^{\text{ZK}}(\lambda)$ by zero-knowledge of the relation argument; in H_2 the uniform $\mathbf{u}$ is sampled independently of $\mathbf{s}$, which is indistinguishable from $\mathbf{u} = \mathbf{A}_{\text{com}} r_{\text{com}} + \mathbf{B}'_{\text{com}} \mathbf{s}$ because $\mathbf{B}'_{\text{com}}$ is MLWE-indistinguishable from uniform and r_{com} is uniform, giving a gap of $\text{Adv}_{\mathcal{B}_{\text{LWE}}}^{\text{MLWE}}(\lambda)$. The QROM programming is sound by standard [39] adaptive-programmability arguments. $\square$

Theorem 13. *For every PPT adversary $\mathcal{A}$, every (x, w, w^*) with $(x, w), (x, w^*) \in \mathcal{R}$, and $(\text{crs}, \text{tdkey}) \leftarrow \text{Setup}(1^\lambda)$,*

$\text{Adv}_{\mathcal{A}, \text{NIZK}}^{\text{RD}}(\lambda) := |\Pr[\mathcal{A}(\text{crs}, x, w, r, \pi) = 1] - \Pr[\mathcal{A}(\text{crs}, x, w^, r^*, \pi) = 1]| \leq \Delta_{\text{smudge}}(\lambda) + \Delta_{\text{RS}}(\lambda)$, where $r \leftarrow \mathcal{U}$, $\pi = \text{Prove}(\text{crs}, \text{tdkey}, x, w; r)$, $r^* = \text{Fake}(\text{crs}, \text{tdkey}, x, w, r, w^*)$, Δ_{smudge} is the smudging statistical distance of the trapdoor shift $\mathbf{T}^\top(\mathbf{s}^* - \mathbf{s})$ against the commitment-randomness distribution, and Δ_{RS} is the statistical distance of RejSamp^{-1} on inputs of norm gap $\leq 2\beta_w$.*

Proof. By construction of Fake , the proof object π is identical in both distributions: $\mathbf{A}_{\text{com}} r_{\text{com}}^* + \mathbf{B}'_{\text{com}} \mathbf{s}^* = \mathbf{u}$ since $r_{\text{com}}^* = r_{\text{com}} + \mathbf{T}^\top(\mathbf{s}^* - \mathbf{s})$, c is deterministic in $(\text{crs}, x, \mathbf{u})$, and $\mathbf{z} = r_{\text{com}} + \mathbf{y} = r_{\text{com}}^* + \mathbf{y}^*$. The marginal distribution of r_{com}^* given π differs from a fresh uniform tape by $\Delta_{\text{smudge}}(\lambda)$: r_{com} is uniform and the added term $\mathbf{T}^\top(\mathbf{s}^* - \mathbf{s})$ has norm bounded by $\|\mathbf{T}\| \cdot 2\beta_w$, which is absorbed into r_{com} 's distribution up to negligible statistical distance by the smudging lemma. The marginal of r_{samp}^* given π differs from uniform by $\Delta_{\text{RS}}(\lambda)$: RejSamp^{-1} is well-defined and statistically faithful on bounded shifts, with $\Delta_{\text{RS}} \leq 2^{-\lambda}$ for the LNP [39] instantiation. The joint distribution is then statistically close, which yields the stated bound. $\square$

A.12 Concrete Parameters and Instantiations

Our Onyx system targets $\lambda = 128$ bits of post-quantum security, with parameters derived from the Lattice Estimator [117], Homomorphic Encryption Security Standard [147], and the following instantiations.

In our voting system, the dFHE [93] operates over the cyclotomic ring $R_q = \mathbb{Z}_q[X]/(X^{N_D} + 1)$ with ring degree $N_D = 2^{15} = 32768$ and modulus chain $\log q \leq 880$, satisfying [147]'s 128-bit PQ standard under RLWE [118]; ternary secret keys and discrete-Gaussian error of width $\sigma = 3.19$ [147] are used throughout, with a decryption-correctness norm bound $B = 2^{60}$. We set the deniability slack $\delta = 2^{14}$, which gives per-ciphertext detection probability $1/\delta = 2^{-14}$ and selector count $n_\delta = \delta^2 = 2^{28}$, with the online-offline decomposition of [93] relegating all δ -dependent work to a message-independent offline phase so that online voter encryption time is independent of δ .

The Universal Thresholdizer [94] has $n_T = 7$ trustees and threshold $t = 4$ (strict-minority corruption tolerance), circuit depth bound $d_{\text{UT}} = 2^{16}$ accommodating the $\Theta(\log^2(n_B + n_V) \cdot (\log n_B + \lambda)) \approx 37,000$ depth of the OddEvenMergeSort-based [148] cleansing-hiding pipeline for elections of up to $n_V = 10^4$ voters and $n_B = 5 \cdot 10^4$ cast

ballots, and share size $|s_i| \leq \text{poly}(\lambda, d_{\text{UT}}, n_T)$ tracked against $\text{LWE}_{n,m,q,\chi}$ [118]; the circuit class $\mathbb{C}$ encompasses the Eq, And, and CGate gadgets evaluated under threshold BGV.

VSS uses Π_{LA} [110] with $(n_R, t_R) = (5, 3)$, SHA-256 [149] as both the QROM [150] hash and the hash-commitment, secret field $\mathbb{Z}_N$ with $|N| = 256$ bits, and $\ell = \lceil \lambda / \log k \rceil = 3$ parallel repetitions over the exceptional set of $\mathbb{Z}_N$ to drive soundness error below 2^{-128} .

Our deniable lattice NIZK extends LNP [39] over a distinct cyclotomic $R_{q'} = \mathbb{Z}_{q'}[X]/(X^{d'} + 1)$ with ring degree $d' = 128$, $\log q' = 50$, MSIS rank $n_{\text{SIS}} = 14$, and MLWE rank $m_{\text{LWE}} = 21$ (LNP 128-bit [39]), masking width $\sigma_{\text{mask}} = 11 \cdot \beta_w$ with rejection-sampling repetition factor $M = 13$ [39], and equivocation trapdoor $\mathbf{T} \in R_{q'}^{\ell \times m_{\text{LWE}}}$ sampled via the Micciancio–Peikert procedure [40] with sub-Gaussian parameter $s_{\mathbf{T}} = \omega(\sqrt{\log N_D})$ chosen such that the smudging term $\mathbf{T}^\top (\mathbf{s}^* - \mathbf{s})$ has statistical distance $\leq 2^{-128}$ from the honest commitment randomness for any pair of valid witnesses of bounded norm; the witness embedding uses $\ell = 32$ ring elements covering (v, c_{id}, r_v, r_c) , and the SIS-keyed challenge derivation H_{chal} is the LNP [39] collision-resistant hash bound to crs .

The verifiable mix-net inherits the parameter set of Aranha et al. [111]: ring degree $N_M = 4096$, $\log q_M = 78$, plaintext modulus $p = 2$, ternary $B_{\text{Key}} = B_{\text{Err}} = 1$, BDLOP [112] commitment ranks $n_{\text{BDLOP}} = \ell_c + 2$ and $k_{\text{BDLOP}} = \ell_c + 3$, $\nu = 36$, $\hat{n} \geq 130$, statistical parameter $\text{sec} = 40$, $\xi_1 = \xi_2 = 4$ servers, which yields ≈ 168 bits of DKS^∞ [121] and ≈ 262 bits of SKS^2 [112] security. The three cyclotomic rings ($N_D = 32768$ for dFHE and cleansing-hiding, $d' = 128$ for NIZK, $N_M = 4096$ for MixNet) are intentionally distinct, each chosen for its workload; cross-ring relations are connected by encoding the dFHE ring statement as a system of $R_{q'}$ -equations per [39] mentioned.

With these parameters Onyx achieves $\lambda = 128$ bits of post-quantum security under the combined hardness of RLWE [118] at $N_D = 32768$ and $N_M = 4096$, MLWE and MSIS [39] at $d' = 128$, and the quantum collision resistance of SHA-256 [151], with the dominant PCR deniability gap $\text{Adv}_{\text{PCR}}^{\text{Onyx}} \leq 2/\delta + \text{negl}(\lambda) = 2^{-13} + \text{negl}(\lambda)$ per disclosed ballot.

A.13 More on Onyx Instantiations and Parameters

Deniable FHE. We instantiated DFHE with the BGV scheme [152] modified by [93], which targets $\lambda = 128$ bits of post-quantum security under the $\text{LWE}_{n,q,\chi}$ assumption [118]. Following [152], the ring modulus q is set to be super-polynomial in λ to ensure the biased decryption property on random inputs (Property 4 of Special FHE in [93]), the secret key $\mathbf{s} \in R^n$ is sampled from a narrow error distribution χ , and the public key matrix $\mathbf{A} \in R^{N \times n}$ is sampled uniformly at random, with error vector $\mathbf{e} \leftarrow \chi^N$ satisfying $\|\mathbf{e}\| \leq B = \text{poly}(\lambda)$; the decryption algorithm is modified to output 0 whenever the intermediate inner product $[\langle ct, \mathbf{s} \rangle]_q > B$, biasing decryption on random ciphertext-space elements to 0 with probability $1 - \text{negl}(\lambda)$. The bootstrapping procedure *boot* [153] is instantiated over the same BGV parameters under the circular security assumption [152], which can be removed for the leveled variant via a two-key chain $(pk_1, sk_1), (pk_2, sk_2)$ as noted in [93]. The fully deniable scheme sets $n_\delta = \delta^2$ components per ciphertext to achieve $1/\delta$ -deniability with compact ciphertext size $|dct| \leq \text{poly}(\lambda)$ independent of δ , while the encryption runtime scales as $O(\delta)$ and is amenable to an online–offline decomposition in which all but one component are computed message-independently in an offline phase as mentioned in [93] work.

Universal Thresholdizer. We adopt the UT from Boneh et al. [94], built upon Linear Secret Sharing Scheme $\{0,1\}$ -LSSS [94], Fully Homomorphic Encryption FHE [154]

(together with $\{0,1\}$ -LSSS they build TFHE), Pre-processing Zero-knowledge Proof PZK [155], Commitment scheme Com [156], which is merged by lattice-based UT [157] and $\forall(\mathbb{S}, N) : \text{TFHE}_{\mathbb{A}} \mid \mathbb{A} \subseteq 2^{\mathbb{S}}$ from Jain et al. [158]. We have formalized in Figure 28 how this UT operates, instantiated with $n_S, n_R \leq N$, depth bound d sufficient for the FDE circuits $\mathbb{C}_1, \mathbb{C}_2, \mathbb{C}_3, \mathbb{C}_{\text{dec}}$, and share sizes $|s_i| \leq \text{poly}(\lambda, d, N)$ which achieves λ -bit security under $\text{LWE}_{n,m,q,\chi}$ [118] assumption.

UT.Setup($1^\lambda, 1^d, \mathbb{A}, x$)	UT.Eval(pp, $s_i, \mathbb{C}$)
1: $(\text{tfhepk}, \text{tfhesk}_1, \dots, \text{tfhesk}_N) \leftarrow \text{TFHE.Setup}(1^\lambda, 1^d, \mathbb{A})$ 2: $\forall i \in [k]. \text{ct}_i \leftarrow \text{TFHE.Encrypt}(\text{tfhepk}, x_i)$ 3: $\forall i \in [N]. (\sigma_{V,i}, \sigma_{P,i}) \leftarrow \text{PZK.Pre}(1^\lambda)$ 4: $\forall i \in [N]. r_i \xleftarrow{R} \{0,1\}^\lambda$ 5: $\forall i \in [N]. \text{com}_i \leftarrow \text{Com}(\text{tfhesk}_i; r_i)$ 6: $\text{pp} \leftarrow (\text{tfhepk}, \{\text{ct}_i\}_{i \in [k]}, \{\sigma_{V,i}\}_{i \in [N]}, \{\text{com}_i\}_{i \in [N]})$ 7: $\forall i \in [N]. s_i \leftarrow (\text{tfhesk}_i, \sigma_{P,i}, r_i)$ 8: return $(\text{pp}, s_1, \dots, s_N)$	1: $\hat{\text{ct}} \leftarrow \text{TFHE.Eval}(\text{tfhepk}, \mathbb{C}, \{\text{ct}_i\}_{i=1}^k)$ 2: $p_i \leftarrow \text{TFHE.PartDec}(\text{tfhepk}, \hat{\text{ct}}, \text{tfhesk}_i)$ 3: $\Psi_i : \exists(\text{tfhesk}_i, r_i) \mid \text{com}_i \leftarrow \text{Com}(\text{tfhesk}_i; r_i)$ $\quad \wedge p_i \leftarrow \text{TFHE.PartDec}(\text{tfhepk}, \hat{\text{ct}}, \text{tfhesk}_i)$ 4: $\pi_i \leftarrow \text{PZK.Prove}(\sigma_{P,i}, \Psi_i, (\text{tfhesk}_i, r_i))$ 5: $y_i \leftarrow (p_i, \pi_i)$ 6: return y_i
UT.Combine(pp, B)	UT.Verify(pp, $y_i, \mathbb{C}$)
1: $B \leftarrow \{y_i\}_{i \in S} \mid (S \subseteq \{1, \dots, N\}) \vee (S \in \mathbb{A})$ 2: $\forall i \in S: y_i \leftarrow (p_i, \pi_i)$ 3: $y \leftarrow \text{TFHE.FinDec}(\text{tfhepk}, \{p_i\}_{i \in S})$ 4: return y	1: $\hat{\text{ct}} \leftarrow \text{TFHE.Eval}(\text{tfhepk}, \mathbb{C}, \{\text{ct}_i\}_{i=1}^k)$ 2: $\Psi_i : \exists(\text{tfhesk}_i, r_i) \mid \text{com}_i \leftarrow \text{Com}(\text{tfhesk}_i; r_i)$ $\quad \wedge p_i \leftarrow \text{TFHE.PartDec}(\text{tfhepk}, \hat{\text{ct}}, \text{tfhesk}_i)$ 3: $y_i \leftarrow (p_i, \pi_i)$ 4: return $\text{PZK.Verify}(\sigma_{V,i}, \Psi_i, \pi_i)$

Fig. 28: Universal Thresholdizer from TFHE = (Setup, Encrypt, Eval, PartDec, FinDec) [154], PZK = (PZK.Pre, PZK.Prove, PZK.Verify) [155], and Com = (Com) [156]

Verifiable Secret Sharing. We use Π with the hash-based scheme Π_{LA} of Bagheri [110], which can be viewed as an amplified variant of the simple VSS of Gennaro et al. [159] and outperforms the recent PQ-secure construction of Atapoor et al. [160] by a constant factor across all metrics. Π_{LA} targets $\lambda = 128$ bits of post-quantum security. The hash function H is instantiated with SHA-256 [149], providing 256-bit output and ≥ 128 -bit quantum collision resistance (Grover's algorithm [161] reduces preimage security by at most a factor of two, which leaves SHA-256 with 128-bit post-quantum preimage resistance [151]); H serves simultaneously as the quantum random oracle and the hash-based commitment scheme, with $c_i = H(s_i \parallel \gamma_i)$. The secret and shares are defined over $\mathbb{Z}_N$ with $|\mathbb{Z}_N| = 256$ bits, and the challenge $d = H(c_1 \parallel \dots \parallel c_n)$ is drawn from an exceptional set modulo N [110]; when $\mathbb{Z}_N$ is a ring whose largest exceptional set has size $k < 2^\lambda$, soundness is amplified by repeating the proof $\ell = \lceil \lambda / \log k \rceil$ times to achieve a negligible error below $2^{-\lambda}$. The scheme operates with parameters (n, t) satisfying $n \geq 2t + 1$, with private communication cost $|\mathbb{Z}_N| = 256$ bits per party and broadcast cost $n|H| + \frac{1}{2}n|\mathbb{Z}_N|$ bits, achieving λ -bit post-quantum security solely under the quantum collision resistance of SHA-256 in the quantum random oracle [150] model.

Mix-Net. The construction of Aranha et al. [111] variates the mixing-and-decryption e-voting paradigm from lattice-based assumptions, targeting $\kappa \approx 168$ bits of DKS^∞

security and $\kappa \approx 262$ bits of SKS^2 security as established by [112, 121]. The underlying encryption scheme is BGV [152] over the ring $R_q = \mathbb{Z}_q[X]/(X^N + 1)$ with $N = 4096$, plaintext modulus $p = 2$, and ciphertext modulus $q \approx 2^{78}$ (prime, $q \equiv 1 \pmod{2N}$), with secret-key and error bounds $B_{\text{Key}} = B_{\text{Err}} = 1$ (ternary distributions). Commitments use the BDLOP scheme [112] with $n = l_c + 2$, $k = l_c + 3$, $\nu = 36$, $B_{\text{Com}} = 1$, and $\hat{n} \geq 130$ parallel instances; the statistical security parameter is $\text{sec} = 40$ bits. The protocol is deployed with $\xi_1 = 4$ shuffle servers and $\xi_2 = 4$ decryption servers. The verifiable shuffle Π_{Mix} combines an extended BDLOP shuffle of commitments [162] for vectors in $R_q^{l_c}$ with an amortized exact zero-knowledge proof of shortness Π_{Small} [163] (instantiated with Reed-Solomon code parameters $g \approx 2^{20}$, $l \approx 2^{20.3}$, $\eta = 325$, run twice for negligible soundness), achieving a per-ballot shuffle proof of size 370τ KB and generation time 260.5τ ms. The verifiable distributed decryption Π_{Dec} applies noise drowning with per-share noise bounded by $2^{\text{sec}} \cdot B_{\text{Dec}}/p\xi_2$ and proves correctness via Π_{Lin} [112] and Π_{Bnd} [164], which achieves a per-ballot decryption proof of size 157τ KB and generation time 138.1τ ms.

A.14 Computational Cost of Onyx

The complexity of each procedure in Onyx is driven by three lattice rings: N_D (the dFHE ring degree), d' (the NIZK ring degree), and N_M (the MixNet ring degree); by the NIZK matrix dimensions (n, m, ℓ, k) over $\mathcal{R}_{q'}$; by the UT depth bound d_{UT} ; and by the working list length $M := n_B + n_V$ entering the cleansing pipeline. We write $T_{\text{NTT}}(N) := O(N \log N)$ for one ring multiplication.

Setup runs dFHE.Gen , UT.Setup , NIZK.Setup , and MixNet.Setup once. The dominant cost is UT.Setup , which secret-shares dsk over an (S_{t, n_T}) access structure and emits per-trustee material of size $\text{poly}(\lambda, d_{\text{UT}}, n_T)$; CRS sampling adds $O((nm + n\ell + k(m + \ell)) \cdot d')$ ring operations and MixNet.Setup adds $O(\xi_2 \cdot N_M)$. Total: time $O(n_T \cdot \text{poly}(\lambda, d_{\text{UT}}) + (nm + k(m + \ell))d' + \xi_2 N_M)$ and space $O(n_T \cdot \text{poly}(\lambda, d_{\text{UT}}) + |\text{crs}| + |pp_{\text{MN}}|)$.

Register executes $\Pi.\text{Share}$ over $\mathbb{Z}_N$ at n_R evaluation points with hash-based commitments, in time $O(n_R(t_R + \lambda))$ and per-voter share space $O(n_R \lambda)$, and one dFHE roster encryption in time $T_{\text{NTT}}(N_D)$. Aggregated over n_V voters, time is $O(n_V(n_R(t_R + \lambda) + T_{\text{NTT}}(N_D)))$ and the published roster R has size $O(n_V \cdot N_D \log q)$.

Vote produces two dFHE ciphertexts and one NIZK proof. Encryption costs $2 \cdot T_{\text{NTT}}(N_D)$. The proof computes a commitment $\mathbf{u} = \mathbf{A}_{\text{com}} r_{\text{com}} + \mathbf{B}'_{\text{com}} \mathbf{s}$ in time $O(nmd')$, derives the deterministic challenge in $O(\lambda)$, runs RejSamp with expected repetition factor M_{rs} and amortized cost $O(md')$, and emits the LNP relation sub-proof π_{rel} in time $O((nm + k(m + \ell))d' \log q')$. Per-voter time is therefore $O(T_{\text{NTT}}(N_D) + (nm + k(m + \ell))d' \log q')$, and the ballot $b = (C_v, C_c, \pi)$ together with the local state ls occupies $O(N_D \log q + (m + n)d' \log q')$.

Valid recomputes the challenge, verifies $\|\mathbf{z}\|_2 \leq \beta_z$, runs the relation verifier, and checks for exact replay against BB ; with hashed indexing this costs $O((nm + k(m + \ell))d' \log q' + \log n_B)$ time and $O(1)$ additional space.

Tally dominates the wall-clock budget. The Batcher odd-even merge sort [148] of the length- M keyed list issues $O(M \log^2 M)$ comparators, each instantiated as an $\text{Eq} + \text{CGate}$ circuit homomorphically evaluated under UT on dFHE ciphertexts at cost $O(\text{poly}(\lambda, d_{\text{UT}}) \cdot T_{\text{NTT}}(N_D))$ per comparator. The $M - 1$ post-sort equality and conditional-gate evaluations add $O(M \cdot \text{poly}(\lambda, d_{\text{UT}}) \cdot T_{\text{NTT}}(N_D))$. The verifiable shuffle

and the n_T partial decryptions contribute $O(n_T \cdot M \cdot T_{\text{NTT}}(N_M))$. Hence

$$T_{\text{Tally}} = O(M \log^2 M \cdot \text{poly}(\lambda, d_{\text{UT}}) \cdot T_{\text{NTT}}(N_D) + n_T M T_{\text{NTT}}(N_M)),$$

and the proof bundle $\Pi_{\text{tally}} = (\Pi_{\text{Sort}}, \pi_{\text{Mix}}, \{\pi_{\text{Dec}_j}\}_j)$ has size $O(M \log^2 M \cdot \text{poly}(\lambda, d_{\text{UT}}) + M(|\pi_{\text{Mix}}| + n_T |\pi_{\text{Dec}}|))$.

Verify replays the UT, MixNet.MixVerify, and MixNet.DDecVerify checks, which is linear in $|\Pi_{\text{tally}}|$ and runs in time $O(M \log^2 M \cdot \text{poly}(\lambda, d_{\text{UT}}) + n_T M T_{\text{NTT}}(N_M))$ with $O(|\Pi_{\text{tally}}|)$ read-only space.

Fake is a purely local equivocation. Each dFHE.Fake call rewrites one ciphertext's randomness in time $T_{\text{NTT}}(N_D)$, and NIZK.Fake computes $r_{\text{com}}^* = r_{\text{com}} + \mathbf{T}^\top(\mathbf{s}^* - \mathbf{s})$ and inverts RejSamp in $O((nm + \ell m)d')$. Per disclosed ballot, time is $O(T_{\text{NTT}}(N_D) + \ell m d')$ and the fabricated view occupies $O(N_D \log q + (m + |r_{\text{samp}}|)d' \log q')$; across k_{vot} ballots costs scale linearly in k_{vot} , which matches the linear amplification factor predicted by Theorem 6.

A.15 Proof of Theorem 1

Proof. We prove the contrapositive $(\neg \text{CR} \implies \neg \text{PCR}) \equiv (\text{PCR} \implies \text{CR})$: if Π does not satisfy CR, then Π does not satisfy PCR. Suppose for contradiction that there exists a PPT adversary $\mathcal{A}_{\text{CR}}$ and a non-negligible $\varepsilon(\lambda)$ such that for every PPT simulator $\mathcal{S}_{\text{CR}}$,

$$\text{Adv}_{\mathcal{A}_{\text{CR}}, \Pi}^{\text{CR}}(\lambda) := \left| \Pr[\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{CR-Real}} = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{S}_{\text{CR}}}^{\text{CR-Ideal}} = 1] \right| > \varepsilon(\lambda).$$

We construct a PPT adversary $\mathcal{A}_{\text{PCR}}$ such that for every PPT simulator $\mathcal{S}_{\text{PCR}}^*$,

$$\left| \Pr[\text{Exp}_{\Pi, \mathcal{A}_{\text{PCR}}}^{\text{PCR-Real}} = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{S}_{\text{PCR}}^*}^{\text{PCR-Ideal}} = 1] \right| \geq \varepsilon(\lambda) - \mu(\lambda)$$

for a negligible μ , contradicting Definition 2.

$\mathcal{A}_{\text{PCR}}$ runs $\mathcal{A}_{\text{CR}}$ as a black-box subroutine and forwards every CR-experiment interaction to the corresponding PCR-experiment interface, with a single deferred step at credential disclosure.

(i) *Setup, registration, corruption.* $\mathcal{A}_{\text{PCR}}$ forwards pk , the registration record R , and the corrupted credentials $\{\text{cred}_i\}_{i \in \mathcal{D}}$ output by the PCR experiment to $\mathcal{A}_{\text{CR}}$ exactly. When $\mathcal{A}_{\text{CR}}$ selects its corrupted set $\mathcal{V}$, $\mathcal{A}_{\text{PCR}}$ sets the PCR corruption set $\mathcal{D} := \mathcal{V}$. By design, these phases are identical in the two experiments.

(ii) *Coerced voter and behavioral distribution.* When $\mathcal{A}_{\text{CR}}$ specifies a target (j, α) , $\mathcal{A}_{\text{PCR}}$ instantiates the PCR coerced set as $\mathcal{H}_{\text{coerce}} := \{j\}$ with claim $\alpha_j := \alpha$. $\mathcal{A}_{\text{PCR}}$ then specifies the PCR behavioral distribution $\mathcal{B}$ to coincide with CR's $\mathcal{D}_{n_C}$ on every non-coerced voter and to contain the single entry (j, α) for the coerced voter. Under this $\mathcal{B}$, the bit-conditional adjustment of $\text{Exp}^{\text{PCR-Real}}$ produces a board containing exactly one honest ballot $\text{Vote}(\text{pk}, \text{cred}_j, \alpha)$ in either branch: in $\beta = 0$ the entry (j, α) is kept and voter j casts α ; in $\beta = 1$ the entry is replaced by $(j, \alpha_j) = (j, \alpha)$ and voter j casts α . This functionally matches CR-Real, in which voter j contributes a single ballot for α in $b = 0$ (cast by the voter) and the coercer issues an equivalent ballot in $b = 1$ using the disclosed real credential.

(iii) *Voting phase, deferred credential.* $\mathcal{A}_{\text{PCR}}$ forwards every $\mathcal{O}_{\text{board}}$ query of $\mathcal{A}_{\text{CR}}$ to PCR's board oracle and routes responses back unmodified. When $\mathcal{A}_{\text{CR}}$'s control flow reaches the moment at which the CR experiment delivers the credential artifact cred , $\mathcal{A}_{\text{PCR}}$ suspends $\mathcal{A}_{\text{CR}}$, recording its full internal state and the timestamped board view it last observed.

- (iv) *Tally and post-tally disclosure.* $\mathcal{A}_{\text{PCR}}$ allows the PCR experiment to proceed through $(r^*, \Pi^*) \leftarrow \mathcal{P}_{\text{Tally}}^{\mathcal{A}_{\text{PCR}}}(BB, R, \text{pk}, \{(s_i, h_i)\}, t)$ and observes the published tally. It then issues a single $\mathcal{O}_{\text{reveal}}(j)$ query, the only one required by the gating clause $\mathcal{H}_{\text{coerce}} \not\subseteq \text{Disc}$, which returns $(\text{cred}_j^\dagger, \{(\tilde{v}^k, \tilde{\rho}^k, b^k)\}_k)$, and extracts the credential component $\widetilde{\text{cred}} := \text{cred}_j^\dagger$.
- (v) *Resumption and output.* $\mathcal{A}_{\text{PCR}}$ resumes $\mathcal{A}_{\text{CR}}$ from its stashed state, which supplies $\widetilde{\text{cred}}$ as the CR-disclosed credential and forwarding the public board BB and tally r^* for any subsequent observations $\mathcal{A}_{\text{CR}}$ makes. $\mathcal{A}_{\text{CR}}$ outputs a final bit b' ; $\mathcal{A}_{\text{PCR}}$ outputs $d := b'$.

Distribution alignment between $\mathcal{A}_{\text{CR}}$'s simulated and genuine views. We argue that $\mathcal{A}_{\text{CR}}$'s view inside $\mathcal{A}_{\text{PCR}}$'s simulation is statistically close to its view in the genuine CR experiment, both on the Real and the Ideal axes.

Real side. On Real, the joint distribution of $(\text{pk}, R, \{\text{cred}_i\}_{i \in V}, BB, r^*, \widetilde{\text{cred}})$ that $\mathcal{A}_{\text{CR}}$ observes inside the simulation is generated by exactly the same protocol algorithms as in $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{CR-Real}}$ on the same inputs, with two exceptions:

- *Order of disclosure.* In CR-Real, $\widetilde{\text{cred}}$ is exposed during voting; in $\mathcal{A}_{\text{PCR}}$'s simulation, it is exposed post-tally. Any $\mathcal{O}_{\text{board}}$ insertions that $\mathcal{A}_{\text{CR}}$ would have made using $\widetilde{\text{cred}}$ during voting are absent from BB . Under Remark 1's anonymous-channel and append-only-board assumptions, $\mathcal{A}_{\text{CR}}$'s $\mathcal{O}_{\text{board}}$ insertions before $\widetilde{\text{cred}}$ are independent of $\widetilde{\text{cred}}$, so only the post- $\widetilde{\text{cred}}$ insertions are affected. For a ballot $\text{Vote}(\text{pk}, c, v)$ inserted with a credential c outside the registered roster, the cleansing-hiding pipeline of Section 5 removes the ballot homomorphically, so its contribution to r^* is masked by the same noise that the protocol applies to every fake-credential injection; for $c = \widetilde{\text{cred}}$, the contribution differs between $\beta = 0$ and $\beta = 1$ by a single additional counted vote, which under the post-quantum hiding of the cleansing pipeline is statistically indistinguishable from the random noise contributed by other injected ballots. Hence the missing post- $\widetilde{\text{cred}}$ insertions perturb the distribution of r^* by at most a negligible statistical distance $\mu_1(\lambda)$ relative to a genuine CR-Real run.
- *Credential distribution.* In CR-Real with $b = 1$, $\widetilde{\text{cred}}$ equals the registered cred_j ; in $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{PCR-Real}}$ with $\beta = 1$, the $\mathcal{O}_{\text{reveal}}(j)$ -disclosed credential is exactly cred_j . In CR-Real with $b = 0$, $\widetilde{\text{cred}} = \text{fakecred}(\text{cred}_j)$; in $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{PCR-Real}}$ with $\beta = 0$, the disclosed credential is the output of $\text{Fake}(\text{pk}, \text{cred}_j, \cdot, \cdot, \cdot)$, which by Definition 2 produces a credential by the same algorithmic procedure fakecred on the same protocol state. The two distributions therefore agree up to a statistical gap $\mu_2(\lambda) \leq \text{negl}(\lambda)$, which is inherited from Fake .

Now, if we map the CR bit b to the PCR bit β , the joint distribution of $\mathcal{A}_{\text{CR}}$'s view inside the simulation when the outer experiment is $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{PCR-Real}}$ with bit β is within $\mu_1 + \mu_2 = \text{negl}(\lambda)$ of its joint distribution in $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{CR-Real}}$ with $b = \beta$.

When the outer experiment is $\text{Exp}_{\Pi, \mathcal{S}_{\text{PCR}}}^{\text{PCR-Ideal}}$, the same reduction template applies: $\mathcal{A}_{\text{PCR}}$ receives the ideal leakage $(|D|, r^*, A)$ produced by $\mathcal{S}_{\text{PCR}}^*$ and uses it to drive a CR-Ideal-style simulation for $\mathcal{A}_{\text{CR}}$. The board length, the tally, the corrupted voters' assignments, and the claim list are all available from the PCR-Ideal interface. The credential and the local view are not exposed in PCR-Ideal, which mirrors CR-Ideal's absence of any credential or per-ballot disclosure. Consequently, $\mathcal{A}_{\text{CR}}$'s simulated view in this case is distributed identically to its view in $\text{Exp}_{\Pi, \mathcal{A}_{\text{CR}}}^{\text{CR-Ideal}}$ driven by the natural projection $\mathcal{S}_{\text{CR}}$ of $\mathcal{S}_{\text{PCR}}^*$ that emits only $(|D|, r^*, A)$. Denote the corresponding ideal-side slack $\mu_3(\lambda) \leq \text{negl}(\lambda)$.

Finally, if we combine the alignments, for every PPT $\mathcal{S}_{\text{PCR}}^*$ the projection $\mathcal{S}_{\text{CR}}(\mathcal{S}_{\text{PCR}}^*)$ is also PPT, and

$$\left| \Pr[\text{Exp}_{\Pi, \mathcal{A}_{\text{PCR}}}^{\text{PCR-Real}} = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{S}_{\text{PCR}}^*}^{\text{PCR-Ideal}} = 1] \right| \geq \text{Adv}_{\mathcal{A}_{\text{CR}}, \Pi}^{\text{CR}}(\lambda) - \mu_1(\lambda) - \mu_2(\lambda) - \mu_3(\lambda) > \varepsilon(\lambda) - \text{negl}(\lambda).$$

Since ε is non-negligible and each μ_k is negligible, the right-hand side is non-negligible. This contradicts assumed Π 's PCR-security, which shows our contrapositive proof. $\square$

A.16 Proof of Theorem 2

Proof. We proceed by a sequence of $G_0, G_1, \dots, G_8$ where $G_0 = \text{Exp}_{\Pi, \mathcal{A}}^{\text{PCR-Real}}$ and $G_8 = \text{Exp}_{\Pi, \mathcal{B}^*}^{\text{PCR-Ideal}}$. Let $\Pr[G_i \Rightarrow 1]$ denote the probability that experiment G_i outputs 1. The overall advantage comes via the triangle inequality $|\Pr[G_0 \Rightarrow 1] - \Pr[G_8 \Rightarrow 1]| \leq \sum_{i=0}^7 |\Pr[G_i \Rightarrow 1] - \Pr[G_{i+1} \Rightarrow 1]|$. Throughout, the experiment maintains an internal *shadow record* of every honest vote actually cast; this record is used only to compute the result r^* via $\text{Result} \circ \text{Policy} \circ \text{cleanse}$ and is never exposed to $\mathcal{A}$ except through transcripts produced by the building blocks.

$G_0 \rightarrow G_1$. We replace the trustee setup line 6 of **Setup** and every **UT.Eval** invocation inside **Tally** by the simulator $(\text{Sim}_S, \text{Sim}_E)$ guaranteed by Definition 19. Sim_S produces pp_{UT} and the corrupted shares $\{s_i\}_{i \in \mathcal{C}}$ without ever instantiating dk ; Sim_E produces every partial evaluation transcript Π_{Sort} together with the equality, conjunction, and conditional-gate transcripts $\{D_k, F_k, V'_k\}$ from the public circuit $\mathcal{C}$ and the shadow output $\mathbb{C}(dk)$ alone. Any PPT distinguisher between G_0 and G_1 yields a UT simulation-security distinguisher of identical advantage, so $|\Pr[G_0 \Rightarrow 1] - \Pr[G_1 \Rightarrow 1]| \leq \text{Adv}_{\mathcal{B}_{\text{U}}, \text{UT}}^{\text{SimSec}}(\lambda)$.

$G_1 \rightarrow G_2$. We replace $\text{MixNet.Mix}(pp_{\text{MN}}, \{V'_k\})$ with its shuffle simulator from Definition 26, which produces $(\{\hat{V}_k\}, \tilde{\pi}_{\text{Mix}})$ from $(pp_{\text{MN}}, \{V'_k\}, \{\hat{V}_k\})$ without knowing the secret permutation. By the shuffle zero-knowledge property of **MixNet**, $|\Pr[G_1 \Rightarrow 1] - \Pr[G_2 \Rightarrow 1]| \leq \text{Adv}_{\mathcal{B}_{\text{M}}, \text{MixNet}}^{\text{Shuf}}(\lambda)$.

$G_2 \rightarrow G_3$. For each honest trustee $j \notin \mathcal{C}$, replace $(\{t_{k,j}\}, \pi_{\text{Dec}_j}) \leftarrow \text{MixNet.DDec}(s_j, \{\hat{V}_k\})$ with the simulated output $(\{\tilde{t}_{k,j}\}, \tilde{\pi}_{\text{Dec}_j})$ produced by the decryption simulator of Definition 28 on input $(pp_{\text{MN}}, \{s_j\}_{j \in \mathcal{C}}, \{(\hat{V}_k, m_k)\})$, where each m_k is read from the shadow tally. A hybrid over the at most n_T honest trustees gives $|\Pr[G_2 \Rightarrow 1] - \Pr[G_3 \Rightarrow 1]| \leq n_T \cdot \text{Adv}_{\mathcal{B}_{\text{DD}}, \text{MixNet}}^{\text{DDec}}(\lambda)$. After hops G_1 – G_3 the entire tally bundle $\Pi^* = (\Pi_{\text{Sort}}, \tilde{\pi}_{\text{Mix}}, \{\tilde{\pi}_{\text{Dec}_j}\}_j)$ is a function of $(ep, \{\hat{V}_k\}, r^*)$ only, independent of any honest secret material.

$G_3 \rightarrow G_4$. We add an abort whenever two distinct ballot statements $(stmt, \mathbf{u}) \neq (stmt', \mathbf{u}')$ come during the experiment collide under the SIS-keyed challenge hash, i.e., $H_{\text{chal}}(\text{crs}, stmt, \mathbf{u}) = H_{\text{chal}}(\text{crs}, stmt', \mathbf{u}')$. Any such abort directly yields a collision against the quantum-collision-resistant H_{chal} , so $|\Pr[G_3 \Rightarrow 1] - \Pr[G_4 \Rightarrow 1]| \leq \text{Adv}_{\mathcal{B}_{\text{H}}, H}^{\text{CR}}(\lambda)$. After this hop, every challenge derived during the experiment is unique, which excludes a class of **Valid** acceptances of replayed-but-distinct ciphertext components.

$G_4 \rightarrow G_5$. For each of the q_b ballots produced by an honest casting, we replace the deniable lattice NIZK proof $\pi \leftarrow \text{NIZK.Prove}(\text{crs}, \text{tdkey}, \text{stmt}, \text{wit}; r_\pi)$ with the simulator output $\tilde{\pi} \leftarrow \mathcal{S}_{\text{NIZK}}(\text{crs}, \text{stmt})$, which depends only on the public statement (C_v, C_c, ep) . By Theorem 12 (computational zero-knowledge of NIZK under MLWE), a q_b -fold hybrid yields $|\Pr[G_4 \Rightarrow 1] - \Pr[G_5 \Rightarrow 1]| \leq q_b \cdot \text{Adv}_{\mathcal{B}_Z, \text{NIZK}}^{\text{ZK}}(\lambda)$. The witness (v, c_{id}, r_v, r_c) is preserved in the shadow record but no longer flows into the public ballot beyond the two ciphertexts.

$G_5 \rightarrow G_6$. For each honest ballot, we replace both $C_v = \text{dFHE.Enc}(dpk, v; r_v)$ and $C_c = \text{dFHE.Enc}(dpk, c_{id}; r_c)$ with encryptions of canonical messages $v^\circ \in [1, n_C]$ and $c^\circ \in \mathcal{C}$ under fresh randomness; the local state $\text{State}[i]$ retains the original randomness (r_v, r_c) alongside the new ciphertexts (C_v°, C_c°) . A standard $2q_b$ -step hybrid, in which each step replaces one ciphertext and reduces to dFHE IND-CPA (Definition 14), yields $|\Pr[G_5 \Rightarrow 1] - \Pr[G_6 \Rightarrow 1]| \leq 2q_b \cdot \text{Adv}_{\mathcal{B}_C, \text{dFHE}}^{\text{IND-CPA}}(\lambda)$. The published result r^* is unaffected because, after hops G_1 – G_3 , the tally bundle is generated from the shadow record, not from the on-board ciphertexts. Consistency between the on-board canonical ciphertext and the stored randomness will be re-established by Fake at disclosure.

$G_6 \rightarrow G_7$. For each $\mathcal{O}_{\text{reveal}}(j)$ query with $j \in \mathcal{H}_{\text{coerce}}$, we modify the oracle as follows. In G_6 , on $\beta = 0$ the oracle returns $(\text{cred}_j^*, \{(\alpha_j, \rho^{*,k}, b^k)\}_k)$ where each $\rho^{*,k}$ is produced via dFHE.Fake (applied twice, once for the vote and once for the credential ciphertext) and NIZK.Fake; on $\beta = 1$ it returns the genuine local view, which after G_6 contains randomness inconsistent with the canonical ciphertexts and is therefore also routed through dFHE.Fake and NIZK.Fake to remain consistent with b^k . In G_7 , we replace *both* branches by freshly sampled tuples $(\tilde{\text{cred}}, \tilde{r}_v, \tilde{r}_c, \tilde{r}_\pi, \tilde{b})$ where $\tilde{b} = (\text{dFHE.Enc}(dpk, \alpha_j; \tilde{r}_v), \text{dFHE.Enc}(dpk, \tilde{\text{cred}}; \tilde{r}_c), \tilde{\pi})$ with $\tilde{\pi}$ a fresh deniable-NIZK proof under witness $(\alpha_j, \tilde{\text{cred}}, \tilde{r}_v, \tilde{r}_c)$. By dFHE δ -deniability (Definition 16) applied to the vote and the credential ciphertext separately, and by NIZK randomness deniability (Theorem 13), a $3q_d$ -step hybrid yields

$$|\Pr[G_6 \Rightarrow 1] - \Pr[G_7 \Rightarrow 1]| \leq q_d \cdot (2 \cdot \text{Adv}_{\mathcal{B}_D, \text{dFHE}}^{\text{Den}}(\lambda) + \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{RD}}(\lambda)).$$

In G_7 , every disclosure is distributed identically in $\beta = 0$ and $\beta = 1$: a fresh credential drawn uniformly from $\mathcal{C}$, a fresh ballot encoding the claim α_j , and fresh randomness of the prescribed length.

$G_7 \rightarrow G_8 = \text{Exp}^{\text{PCR-Ideal}}$. We construct $\mathcal{B}^*$ that runs $\mathcal{A}$ internally and reproduces every component of G_7 from the ideal-game leakage $(|D|, r^*, A)$:

- *Public election parameters* ep , R , and $\{c_i\}_{i \in \mathcal{D}}$ are generated internally by $\mathcal{B}^*$ from scratch, since after G_1 – G_3 no honest secret material is needed.
- *Bulletin board* BB is reconstructed: $\mathcal{B}^*$ produces $|D|$ honest ballots, each being a fresh canonical-ciphertext pair with a simulated NIZK; these are interleaved with the adversary's own posts.
- *Tally bundle* Π^* is generated by the UT and MixNet simulators from the ideal r^* together with $\{\hat{V}_k\}$, exactly as in hops G_1 – G_3 .
- *Disclosures* are sampled identically to G_7 : for each $(j, \alpha_j) \in A$, $\mathcal{B}^*$ outputs a fresh credential and a fresh ballot encoding α_j with fresh randomness.

Every distribution coincides with G_7 except for negligible statistical value inherited from Fake (rejection-sampling residue of NIZK.Fake and smudging gap of dFHE.Fake).

Therefore $|\Pr[G_7 \Rightarrow 1] - \Pr[G_8 \Rightarrow 1]| \leq \text{negl}(\lambda)$. Now, combining the hops via the triangle inequality, for every PPT $\mathcal{A}$ there exists a PPT $\mathcal{B}^*$ such that

$$\begin{aligned} \text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{PCR}}(\lambda) &\leq \text{Adv}_{\mathcal{B}_U, \text{UT}}^{\text{SimSec}}(\lambda) + \text{Adv}_{\mathcal{B}_M, \text{MixNet}}^{\text{Shuf}}(\lambda) + n_T \cdot \text{Adv}_{\mathcal{B}_{DD}, \text{MixNet}}^{\text{DDec}}(\lambda) + \text{Adv}_{\mathcal{B}_H, H}^{\text{CR}}(\lambda) \\ &\quad + q_b \cdot \text{Adv}_{\mathcal{B}_Z, \text{NIZK}}^{\text{ZK}}(\lambda) + 2q_b \cdot \text{Adv}_{\mathcal{B}_C, \text{dFHE}}^{\text{IND-CPA}}(\lambda) \\ &\quad + q_d \cdot (2 \cdot \text{Adv}_{\mathcal{B}_D, \text{dFHE}}^{\text{Den}}(\lambda) + \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{RD}}(\lambda)) + \text{negl}(\lambda), \end{aligned}$$

matching the bound of the theorem. Under the assumptions of LWEE, MLWE, MSIS, DKS[∞], SKS², and quantum collision resistance of H , each individual advantage is negligible in λ , hence $\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{PCR}}(\lambda)$ is negligible. Therefore Onyx satisfies Persistent Coercion Resistance. $\square$

A.17 Proof of Theorem 3

Proof. We bound the adversary's verifiability advantage through $G_0, G_1, \dots, G_6$, where G_0 is the real verifiability experiment and G_6 aborts on every inconsistency between the published result r^* and the honestly valid tally. Each hop removes one class of adversary winning conditions and incurs a black-box reduction to one of the soundness assumptions of an underlying building block. Let Win_i denote the event that in G_i the adversary produces $(r^*, \Pi_{\text{tally}})$ with $\text{Verify}(ep, BB, R, r^*, \Pi_{\text{tally}}) = 1$ yet $r^* \neq \text{Result}(\text{Policy}(\text{cleanse}(\{(id, \nu_{id})\}_{id \notin \mathcal{D}} \cup \{(\text{cor}_\ell, \alpha_\ell)\}_\ell)))$ for any choice of corrupted-voter assignments $\{(\text{cor}_\ell, \alpha_\ell)\}_\ell$, with all honest voters who voted being happy. By definition, $\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{Ver}}(\lambda) = \Pr[\text{Win}_0]$.

$G_0 \rightarrow G_1$. For each accepted ballot $b_k = (C_v^k, C_c^k, \pi^k)$ on BB , we invoke the knowledge extractor $\mathcal{E}$ guaranteed by Theorem 11 on π^k to obtain a witness $w^k = (v^k, c^k, r_v^k, r_c^k)$ satisfying $(\text{stmt}^k, w^k) \in \mathcal{R}_{\text{bal}}$; we abort if extraction fails for any of the at most q_b accepted ballots. A q_b -fold hybrid over ballots reduces this event to the knowledge-soundness advantage of NIZK: $|\Pr[\text{Win}_0] - \Pr[\text{Win}_1]| \leq q_b \cdot \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{KS}}(\lambda)$. After G_1 , every ballot on the board has an extractable plaintext pair $(v^k, c^k) \in [1, n_C] \times \mathcal{C}$, well-formed encryption randomness, and is uniquely tied to the encrypted-domain entry $(V_k, K_k) = (C_v^k, (\text{Enc}(k-1; \mathbf{0}), C_c^k))$ in the cleansing-hiding pipeline.

$G_1 \rightarrow G_2$. We abort if any two distinct (statement, commitment) pairs arising during the experiment hash under H_{chal} to the same challenge, i.e., $H_{\text{chal}}(\text{crs}, \text{stmt}, \mathbf{u}) = H_{\text{chal}}(\text{crs}, \text{stmt}', \mathbf{u}')$ with $(\text{stmt}, \mathbf{u}) \neq (\text{stmt}', \mathbf{u}')$. Any such event directly yields a collision against the SIS-keyed hash, giving $|\Pr[\text{Win}_1] - \Pr[\text{Win}_2]| \leq \text{Adv}_{\mathcal{B}_H, H}^{\text{CR}}(\lambda)$. After G_2 , each extracted witness uniquely determines the verifier's challenge c and hence the response $\mathbf{z}$, which rules out any two distinct ballots that fold into the same verified statement under different witnesses.

$G_2 \rightarrow G_3$. We abort if the adversary submits an accepted ballot whose extracted credential c^k equals the credential c_{id} of some honest voter $id \notin \mathcal{D}$, a credential the adversary did not legitimately receive at registration. Under Remark 2, the honest registrar distributes credentials via $\Pi.\text{Share}$ over an untappable channel and publishes only the roster $R = (\text{dFHE}.\text{Enc}(dpk, c_{id}; \rho_{\text{reg}, id}))_{id}$; the underlying credentials remain hidden by VSS unpredictability (Definition 23) together with the indistinguishability of the roster ciphertexts. A hybrid over the at most n_V roster entries, each step reducing to dFHE IND-CPA

on a single encrypted credential, yields $|\Pr[\text{Win}_2] - \Pr[\text{Win}_3]| \leq n_V \cdot \text{Adv}_{\mathcal{B}_{C,\text{dFHE}}}^{\text{IND-CPA}}(\lambda)$. In G_3 , every adversary-submitted ballot either uses a corrupted-voter credential c_{id} with $id \in \mathcal{D}$ (legitimate from the verifiability standpoint and counted as the corresponding voter's choice) or a credential lying outside the registered roster, which will be filtered out by the cleansing-hiding pipeline.

$G_3 \rightarrow G_4$. We abort if any **UT.Verify** check during the tally accepts a partial evaluation $y_i^* \neq \text{UT.Eval}(pp_{\text{UT}}, s_i, \mathbb{C})$ produced by a corrupted trustee for some cleansing circuit $\mathbb{C} \in \{\text{Sort}_{K^\top, K^\perp}, \text{Eq}, \text{And}, \text{CGate}\}$. By the robustness of the universal thresholdizer (Definition 20), $|\Pr[\text{Win}_3] - \Pr[\text{Win}_4]| \leq \text{Adv}_{\mathcal{B}_{U,\text{UT}}}^{\text{Sound}}(\lambda)$. After G_4 , the sequence $\{(V'_k, K'_k)\}_k$ output by the cleansing-hiding pipeline is the homomorphic encryption of the correct sort-and-cleanse operation on the extracted plaintext pairs $\{(v^k, c^k)\}_k$ joined with the registered roster, and every encrypted equality and conditional-gate bit is the homomorphic encryption of the correct boolean outcome. Equivalently, the cleansed ciphertexts $\{V'_k\}$ encrypt the multiset $\{v^k : c^k \in \{c_{id}\}_{id \in [n_V]}\} \cup \{0\}^*$, with duplicates resolved by the re-voting policy encoded in the sort order.

$G_4 \rightarrow G_5$. We abort if **MixNet.MixVerify** accepts a shuffled output $\{\hat{V}_k\}$ whose plaintext multiset under ds_k differs from that of $\{V'_k\}$. By the verifiability property of **MixNet** (Definition 25), $|\Pr[\text{Win}_4] - \Pr[\text{Win}_5]| \leq \text{Adv}_{\mathcal{B}_{M,\text{MixNet}}}^{\text{ShufSound}}(\lambda)$. After G_5 , the mixed ciphertexts $\{\hat{V}_k\}$ encrypt precisely the same multiset of cleansed plaintexts as $\{V'_k\}$, permuted into an order computationally unlinked from the casting order.

$G_5 \rightarrow G_6$. We abort if **MixNet.DDecVerify** accepts a partial-decryption tuple $(\{t_{k,j}\}, \pi_{\text{Dec}_j})$ that does not correspond to the honest share under $\hat{V}_k$ and trustee secret s_j . By threshold verifiability of **MixNet** (Definition 27) applied per honest trustee,

$$|\Pr[\text{Win}_5] - \Pr[\text{Win}_6]| \leq n_T \cdot \text{Adv}_{\mathcal{B}_{\text{DD},\text{MixNet}}}^{\text{DDecSound}}(\lambda).$$

After G_6 , the combined plaintexts $\{m_k\} \leftarrow \text{MixNet.Comb}(\{\hat{V}_k\}, \{\{t_{k,j}\}_{j \in [n_T]}\})$ are exactly the correct decryptions of the mixed cleansing-hiding output under ds_k obtained by Lagrange reconstruction over any authorised set of $t+1$ honest shares guaranteed by the corruption bound $n_T - t - 1$ of Remark 2.

In G_6 , the published result $r^* = \text{Result}(\{m_k \mid m_k \neq 0\})$ is a deterministic function of the extracted plaintext pairs $\{(v^k, c^k)\}_k$ for accepted ballots and the registered roster R : hops G_3 and G_4 guarantee that the cleansing pipeline discards every ballot whose credential lies outside the roster and resolves duplicates by retaining the unique sorted entry that immediately precedes the corresponding roster entry; hops G_5 and G_6 guarantee that the mix-and-decrypt phase delivers exactly that multiset in cleartext. By hop G_3 , every adversary-submitted ballot that survives cleansing uses a corrupted-voter credential c_{id} with $id \in \mathcal{D}$, so r^* matches $\text{Result}(\text{Policy}(\text{cleanse}(\{(id, \nu_{id})\}_{id \in \mathcal{D}} \cup \{(\text{cor}_\ell, \alpha_\ell)\}_\ell)))$ for the assignment $\{(\text{cor}_\ell, \alpha_\ell)\}_\ell$ implicitly chosen by the adversary's ballots. This contradicts Win_6 , so $\Pr[\text{Win}_6] \leq \text{negl}(\lambda)$, the residual statistical error coming from the correctness of **dFHE** decryption and of **UT.Combine** reconstruction under the noise budget of the cleansing-hiding circuit depth. Now, summing the hops via the triangle inequality we will have:

$$\begin{aligned} \text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{Ver}}(\lambda) &\leq q_b \cdot \text{Adv}_{\mathcal{B}_N, \text{NIZK}}^{\text{KS}}(\lambda) + \text{Adv}_{\mathcal{B}_U, \text{UT}}^{\text{Sound}}(\lambda) + \text{Adv}_{\mathcal{B}_M, \text{MixNet}}^{\text{ShufSound}}(\lambda) + n_T \cdot \text{Adv}_{\mathcal{B}_{\text{DD}, \text{MixNet}}}^{\text{DDecSound}}(\lambda) \\ &\quad + n_V \cdot \text{Adv}_{\mathcal{B}_{C, \text{dFHE}}}^{\text{IND-CPA}}(\lambda) + \text{Adv}_{\mathcal{B}_H, H}^{\text{CR}}(\lambda) + \text{negl}(\lambda). \end{aligned}$$

Under the lattice assumptions (LWE, MLWE, MSIS, DKS[∞], SKS²) and the quantum collision resistance of H_{chal} , every term on the right-hand side is negligible in λ , so $\text{Adv}_{\mathcal{A}, \text{Onyx}}^{\text{Ver}}(\lambda)$ is negligible. Hence Onyx is verifiable. $\square$

A.18 Proof of Theorem 4

Proof Sketch. We show that no deterministic decision rule $\mathcal{A}: V \rightarrow \{0,1\}$ achieves greater advantage than $\mathcal{A}^*$; randomized rules follow by Jensen's inequality, since the advantage is linear in the rule's decision probabilities. Let $S_{\mathcal{A}} = \{V : \mathcal{A}(V) = 1\}$. The advantage of $\mathcal{A}$ is

$$\begin{aligned} \text{Adv}(\mathcal{A}) &= \Pr[\mathcal{A} = 1 \mid \beta = 1] - \Pr[\mathcal{A} = 1 \mid \beta = 0] \\ &= \sum_{V \in S_{\mathcal{A}}} (\Pr[V \mid \beta = 1] - \Pr[V \mid \beta = 0]). \end{aligned}$$

Each summand is positive iff $T_g(V) \geq 1$. Thus $\text{Adv}(\mathcal{A})$ is maximized exactly when $S_{\mathcal{A}} = M_{\alpha}^g$ (with ties broken arbitrarily, contributing 0). The maximum value is $\sum_{V \in M_{\alpha}^g} (\Pr[V \mid \beta = 1] - \Pr[V \mid \beta = 0])$. By the standard identity $\sum_V |a_V - b_V| = 2 \sum_{V: a_V \geq b_V} (a_V - b_V)$ for probability measures a, b on the same space, this equals the right-hand side of (3), which is exactly δ_{α}^g . $\square$

A.19 Proof of Theorem 5

Proof. We reduce to a univariate normal calculation. Let $X_{\alpha} := \sum_{i=1}^{n_V} \mathbf{1}[v_i = \alpha]$ be the count of votes for α among the n_V uncoerced voters; $X_{\alpha} \sim \text{Bin}(n_V, P_{\alpha}^{\circ})$. The coordinate ν_{α} of ν equals $X_{\alpha} + B_{\alpha}$ where $B_{\alpha} = 1$ under obey ($\beta = 1$) and $B_{\alpha} \sim \text{Bern}(P_{\alpha}^{\circ})$ under evade ($\beta = 0$); the remaining coordinates are governed by the same conditional distribution under both branches. By the sufficient-statistic argument of Theorem 4 together with the fact that ν_{α} is a sufficient statistic for the binary test (it is the only coordinate whose mean shifts), the optimal advantage equals

$$\delta_{\alpha}^{\text{ideal}} = \frac{1}{2} \text{TV}(\text{Bin}(n_V, P_{\alpha}^{\circ}) + 1, \text{Bin}(n_V, P_{\alpha}^{\circ}) + \text{Bern}(P_{\alpha}^{\circ})).$$

Let $\sigma^2 := n_V P_{\alpha}^{\circ} (1 - P_{\alpha}^{\circ})$. The two distributions both lie on $\{0, 1, \dots, n_V + 1\}$; their means differ by $1 - P_{\alpha}^{\circ}$ and their variances by $P_{\alpha}^{\circ} (1 - P_{\alpha}^{\circ})$, both $O(1)$ in n_V . The local CLT for the binomial [144, 145] states that, uniformly in k ,

$$\Pr[\text{Bin}(n_V, p) = k] = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(k - np)^2}{2\sigma^2}\right) (1 + O(n_V^{-1/2})).$$

Applying this to the two binomials and integrating the point-wise difference,

$$\begin{aligned} \frac{1}{2} \sum_k |\Pr[\text{Bin} + 1 = k] - \Pr[\text{Bin} + \text{Bern} = k]| &= \frac{1}{2} (1 - P_{\alpha}^{\circ}) \sum_k |\Pr[\text{Bin} = k - 1] - \Pr[\text{Bin} = k]| \\ &= \frac{1 - P_{\alpha}^{\circ}}{\sqrt{2\pi\sigma}} \int |\phi'(z)| \sigma dz \cdot (1 + O(n_V^{-1/2})) \\ &= \frac{1 - P_{\alpha}^{\circ}}{\sqrt{2\pi\sigma}} \cdot \sqrt{\frac{2}{\pi}} \cdot (1 + O(n_V^{-1/2})), \end{aligned}$$

using the standard identity $\int |\phi'(z)| dz = \sqrt{2/\pi}$ for the standard normal density ϕ . Maximizing over claims α (equivalently, minimizing over P_{α}° inside the worst-case adversary strategy) yields the stated rate, where the constant collapses to $1/\sqrt{2\pi P_{\alpha}^{\circ} (1 - P_{\alpha}^{\circ})}$ after gathering the $(1 - P_{\alpha}^{\circ})$ factor with the variance under the worst-case α (most-likely claim collides with most asymmetric marginal). $\square$

A.20 Proof of Theorem 6

Proof. We construct sequence H_0, H_1, H_2, H_3, H_4 with $H_0 = \text{Exp}^{\text{PCR-Real}}$ and $H_4 = \text{Exp}^{\text{PCR-Ideal}}$, and bound each adjacent gap.

$H_0 \rightarrow H_1$. We replace the genuine tally output (r^*, II^*) by the simulator's output $(r^*, \tilde{II}^*)$, where r^* is computed from the internal shadow record (the multiset of actually-cast votes) using $\text{Result} \circ \text{Policy} \circ \text{cleanse}$ (tally simulation). By the simulation-security property of the tally with advantage δ^{tally} ,

$$|\Pr[H_0=1] - \Pr[H_1=1]| \leq \delta^{\text{tally}}(II, \lambda).$$

$H_1 \rightarrow H_2$. For each honest ballot $b = (C_v, C_c, \pi)$ posted to the board, we replace C_v and C_c by canonical encryptions of fixed messages, keeping the internal shadow record unchanged. By a $2q_b$ -step hybrid over the honest ballots (where q_b is the total number of honest casts) and CPA-security of the encryption scheme,

$$|\Pr[H_1=1] - \Pr[H_2=1]| \leq \delta^{\text{board}}(II, \lambda).$$

After this hop the public board contains canonical ciphertexts unrelated to the actual votes; the tally is still correct because it is computed from the shadow record (board IND-CPA).

$H_2 \rightarrow H_3$. For each disclosure $\mathcal{O}_{\text{reveal}}(j)$ and for each of the k_{vot} ballots cast by voter j , we replace the disclosed (cred^*, ρ^*) pair (whether computed by Fake or the honest local state) with a fresh independent sample of $(\tilde{\text{cred}}, \tilde{\rho})$ drawn as a genuine honest casting of α_j . By Definition 11 each such substitution contributes at most $\varepsilon_{\text{ball}}$ to the gap; over $q_c \cdot k_{\text{vot}}$ substitutions a standard hybrid yields

$$|\Pr[H_2=1] - \Pr[H_3=1]| \leq q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}(II, \lambda).$$

After this hop every disclosure is drawn from the same distribution under $\beta=0$ and $\beta=1$: a fresh credential and fresh randomness consistent with the claim α_j (per-voter Fake replacement).

$H_3 \rightarrow H_4$. In H_3 , the only β -dependent quantity remaining in the adversary's view is the tally r^* , which depends on β through the shadow record's ballot count for α_j (versus a draw from $\mathcal{B}$) (ideal reduction). This is precisely the dependence captured by $\text{Exp}^{\text{PCR-Ideal}}$, so an information-theoretic argument shows

$$|\Pr[H_3=1] - \Pr[H_4=1]| \leq \delta_{\alpha}^{\text{ideal}}(II).$$

Finally, combining the four bounds by the triangle inequality and maximizing over $\mathcal{A}$ yields our claim. $\square$

A.21 Proof of Theorem 7

Proof Sketch. We consider four adversaries $\mathcal{A}^{(I)}, \mathcal{A}^{(T)}, \mathcal{A}^{(B)}, \mathcal{A}^{(D)}$, each focusing one term of the decomposition, and then take their best-of-four combination.

$\mathcal{A}^{(I)}$ runs the ideal-game distinguisher achieving $\delta_{\alpha^*}^{\text{ideal}}$. Since the Real experiment extends the Ideal view, $\mathcal{A}^{(I)}$ achieves at least this advantage in the Real experiment.

$\mathcal{A}^{(T)}$ is the adversary witnessing the tally simulator's gap: by definition of δ^{tally} there is a distinguisher between the genuine and simulated tally transcripts with this advantage, and this distinguisher transfers directly to the Real-vs-(Ideal-with-simulated-tally) test, contributing δ^{tally} to the gap.

$\mathcal{A}^{(B)}$ is the adversary witnessing δ^{board} via a CPA challenger on pk.

$\mathcal{A}^{(D)}$ is the per-disclosure distinguisher witnessing $\varepsilon_{\text{ball}}$ on *some* disclosed ballot. By the union bound, choosing a uniformly random target disclosure among the $q_c \cdot k_{\text{vot}}$

candidates retains an advantage of at least $\varepsilon_{\text{ball}}$, and summing over disclosures via the hybrid converse (a $q_c \cdot k_{\text{vot}}$ -step hybrid achieves $q_c \cdot k_{\text{vot}} \cdot \varepsilon_{\text{ball}}$ in the worst case, witnessed by the per-step distinguisher). The best of these four adversaries achieves advantage at least $\frac{1}{4}$ of the sum of the four terms, by averaging. $\square$

A.22 Proof of Theorem 8

Proof Sketch. We construct an explicit adversary against $\text{Exp}^{\text{PCR-Real}}(\Pi)$. Fix any $\mathcal{B}^*$ with support on $[1, n_C]$ and any claim α^* . The adversary $\mathcal{A}^*$ proceeds as follows: it coerces a single honest voter j ; after the tally is published it issues $\mathcal{O}_{\text{reveal}}(j)$ and obtains $(\text{cred}_j, \mathbf{v}_j, \mathbf{p}_j, \mathbf{b}_j)$. Since Π has no Fake, the disclosed $\mathbf{v}_j$ equals the genuine voted sequence, regardless of β . The adversary outputs $\beta' = 1$ if and only if $\mathbf{v}_j = (\alpha^*)$ as a sequence (i.e., every cast is for α^*). Under $\beta = 1$ (obey), the voter casts α^* at every step by definition, so $\Pr[\beta' = 1 | \beta = 1] = 1$. Under $\beta = 0$ (evade), the voter casts honestly per the behavioral tree; the probability that all $k_{\text{vot}} \geq 1$ casts are for α^* is at most $P_{\alpha^*}^\circ$ (taking the first cast only). Thus $\Pr[\beta' = 1 | \beta = 0] \leq P_{\alpha^*}^\circ$.

The advantage in the Real experiment is therefore at least $1 - P_{\alpha^*}^\circ$. By Theorem 4 this is a lower bound on $\delta_{\alpha^*}^{\text{Real}}(\Pi)$. Choosing $\alpha^* = \arg\min_{\alpha} P_{\alpha}^\circ$ gives

$$\delta^{\text{Real}}(\Pi) \geq 1 - \min_{\alpha} P_{\alpha}^\circ.$$

The persistent gap is $\delta^{\text{Real}} - \delta^{\text{Ideal}}$, which yields the first claim. The second follows by choosing $\mathcal{B}^*$ such that $P_{\alpha}^\circ \leq 1/n_C$ for some α (e.g., the uniform distribution over $[1, n_C]$ gives equality), and applying Theorem 5 to bound $\delta^{\text{Ideal}} = O(1/\sqrt{n_V})$. $\square$

A.23 Proof of Theorem 9

Proof. The Ideal view under independent coercion is the joint distribution $\bigotimes_j \mathcal{V}_{\alpha_j, \beta}^{\text{Ideal}}$ with shared β . By the standard TV sub-additivity for product measures sharing one coordinate [146], the TV distance between $\bigotimes_j \mathcal{V}_{\alpha_j, 1}$ and $\bigotimes_j \mathcal{V}_{\alpha_j, 0}$ is bounded by $\sum_j \text{TV}(\mathcal{V}_{\alpha_j, 1}, \mathcal{V}_{\alpha_j, 0})$, which is the displayed sum. For tightness: when the per-voter advantages are small ($\delta_j \ll 1/q_c$), the linear bound is tight up to factor 2 by the binomial approximation to the joint TV distance. $\square$

Policy on AI Usage

Grammarly’s AI commercial correction system has been used in the writing process of this work, including text editing and rephrasing, grammar, and spell checking.

References

1. M. Bernhard et al. Public evidence from secret ballots. In *Electronic Voting*, LNCS, pages 84–109. Springer, 2017.
2. J. Budurushi et al. An investigation into the usability of electronic voting systems for complex elections. *Ann. Telecommun.*, 71(7–8):309–322, 2016.
3. D. Rausch, N. Huber, and R. Küsters. Verifiable e-voting with a trustless bulletin board. In *IEEE CSF*, pages 457–472, 2025.
4. G. Kaim et al. Post-quantum online voting scheme. In *FC Workshops*, pages 290–305. Springer, 2021.
5. J. Mascelli and M. Rodden. “harvest now decrypt later”: Examining post-quantum cryptography and the data privacy risks for distributed ledger networks. Technical Report FEDS 2025-093, Board of Governors of the Federal Reserve System, 2025.
6. D.J. Bernstein and T. Lange. Post-quantum cryptography. *Nature*, 549(7671):188–194, 2017.
7. D. Sikeridis, P. Kampanakis, and M. Devetsikiotis. Post-quantum authentication in TLS 1.3: A performance study. In *NDSS*, 2020.
8. A. Aydeger et al. Post-quantum public key infrastructures: Hybrid certificates, cryptographic combiners, and migration strategies. In *IEEE CCNC*, pages 1–4, 2026.
9. D. Bernhard et al. Sok: A comprehensive analysis of game-based ballot privacy definitions. In *IEEE S&P*, pages 499–516, 2015.
10. V. Cortier et al. Election verifiability for Helios under weaker trust assumptions. In *ESORICS*, pages 327–344. Springer, 2014.
11. V. Cortier et al. Election verifiability for Helios under weaker trust assumptions. In *ESORICS*, volume 8713 of *LNCS*, pages 327–344. Springer, 2014.
12. V. Cortier et al. Sok: Verifiability notions for e-voting protocols. In *IEEE S&P*, pages 779–798, 2016.
13. R. Küsters, T. Truderung, and A. Vogt. Accountability: Definition and relationship to verifiability. In *ACM CCS*, pages 526–535, 2010.
14. F. Hao et al. Experience from UNITA elections: Reconciling revote, E2E verifiability and low coercion. Cryptology ePrint Archive, Paper 2025/1669, 2025.
15. I. Chillotti et al. A homomorphic LWE based e-voting scheme. In *PQCrypto*, pages 245–265. Springer, 2016.
16. V. Cortier, J. Lallemand, and B. Warinschi. Fifty shades of ballot privacy: Privacy against a malicious board. In *IEEE CSF*, pages 17–32, 2020.
17. C.C. Drăgan et al. Machine-checked proofs of privacy against malicious boards for Selene & co. In *IEEE CSF*, pages 335–347, 2022.
18. S. Delaune, S. Kremer, and M. Ryan. Coercion-resistance and receipt-freeness in electronic voting. In *IEEE CSFW*, pages 28–42, 2006.
19. T. Haines and B. Smyth. Surveying definitions of coercion resistance. Cryptology ePrint Archive, Paper 2019/822, 2019.
20. Ari Juels, Dario Catalano, and Markus Jakobsson. Coercion-resistant electronic elections. In *Proceedings of the 2005 ACM Workshop on Privacy in the Electronic Society (WPES)*, pages 61–70. ACM, 2005.
21. Thomas Haines and Ben Smyth. SoK: Surveying definitions of coercion resistance. In *Proceedings of the 2020 ACM Workshop on Privacy in the Electronic Society (WPES)*, pages 65–76. ACM, 2020.

22. Véronique Cortier, Pierrick Gaudry, and Quentin Yang. Is the jcy voting system really coercion-resistant? In *2024 IEEE 37th Computer Security Foundations Symposium (CSF)*, pages 186–200, 2024.
23. X. Boyen, T. Haines, and J. Müller. Epoque: Practical end-to-end verifiable post-quantum-secure e-voting. In *IEEE EuroS&P*, pages 272–291, 2021.
24. P. Hough, C. Sandsbråten, and T. Silde. More efficient lattice-based electronic voting from NTRU. *IACR Commun. Cryptol.*, 1(4), 2025.
25. R. Mosaheb. *Quantum-Safe Electronic Voting Schemes*. PhD thesis, Université du Luxembourg, 2025.
26. R. del Pino et al. Practical quantum-safe voting from lattices. In *ACM CCS*, pages 1565–1581, 2017.
27. I. Black et al. Practical quantum-safe voting from lattices, extended. Cryptology ePrint Archive, Paper 2022/1686, 2022.
28. N. Abapour et al. Blockchain-based economic voting with posterior security from lattices. In *IWSEC*, pages 247–266. Springer, 2025.
29. P. Sekar et al. Quantum-resistant electronic voting system using lattice cryptography techniques. In *Internet and Modern Society*, pages 257–270. Springer, 2025.
30. S.P. Mahanayak, B. Nikhita, and S. Bilgaiyan. Enhancing e-voting security with quantum-resistant encryption. *SN Comput. Sci.*, 4(5), 2023.
31. A. Junior Gabriel et al. Post-quantum cryptography system for secure electronic voting. *Open Comput. Sci.*, 9(1):292–298, 2019.
32. Chen Wu, Shaohua Tang, and Xingfu Yan. A homomorphic lwe-based verifiable electronic voting system. In *2018 IEEE Conference on Dependable and Secure Computing (DSC)*, pages 1–8, 2018.
33. R. Küsters, T. Truderung, and A. Vogt. Verifiability, privacy, and coercion-resistance: New insights from a case study. In *IEEE S&P*, pages 538–553, 2011.
34. A. De Caro, V. Iovino, and A. O’Neill. Deniable functional encryption. In *PKC*, pages 196–222. Springer, 2016.
35. Markus Jakobsson, Kazuo Sako, and Russell Impagliazzo. *Designated Verifier Proofs and Their Applications*, page 143–154. Springer Berlin Heidelberg, 1996.
36. Chen Li and Fangguo Zhang. Designated-verifier zk-snarks made easy. *International Journal of Information Security*, 25(1), January 2026.
37. Yevgeniy Dodis, Jonathan Katz, Adam Smith, and Shabsi Walfish. *Composability and On-Line Deniability of Authentication*, page 146–162. Springer Berlin Heidelberg, 2009.
38. Léo Colisson, Garazi Muguruza, and Florian Speelman. *Oblivious Transfer from Zero-Knowledge Proofs: Or How to Achieve Round-Optimal Quantum Oblivious Transfer and Zero-Knowledge Proofs on Quantum States*, page 3–38. Springer Nature Singapore, 2023.
39. Vadim Lyubashevsky, Ngoc Khanh Nguyen, and Maxime Plançon. Lattice-based zero-knowledge proofs and applications: Shorter, simpler, and more general. In *Advances in Cryptology – CRYPTO 2022*, volume 13508 of *LNCS*, pages 71–101. Springer, 2022.
40. D. Micciancio and C. Peikert. Trapdoors for lattices: Simpler, tighter, faster, smaller. In *EUROCRYPT*, volume 7237 of *LNCS*, pages 700–718. Springer, 2012.
41. Rafael Pass. *On Deniability in the Common Reference String and Random Oracle Model*, page 316–337. Springer Berlin Heidelberg, 2003.
42. B. Adida. Helios: Web-based open-audit voting. In *USENIX Security*, pages 335–348, 2008.
43. V. Cortier, P. Gaudry, and S. Glondu. Belenios: A simple private and verifiable electronic voting system. In *Foundations of Security, Protocols, and Equational Reasoning*, volume 11565 of *LNCS*. Springer, 2019.
44. J. Benaloh et al. ElectionGuard: A cryptographic toolkit to enable verifiable elections. In *USENIX Security*, pages 5485–5502, 2024.
45. ElectionGuard specification v2.1. https://github.com/microsoft/electionguard/releases/download/v2.1/EG_Spec_2_1.pdf. Accessed: 2024-12-23.

46. ElectionGuard specification. <https://www.electionguard.vote/spec/>. Accessed: 2024-07-20.
47. Michael R Clarkson, Stephen Chong, and Andrew C Myers. Civitas: Toward a secure voting system. In *2008 IEEE Symposium on Security and Privacy (sp 2008)*, pages 354–368. IEEE, 2008.
48. X. Boyen, T. Haines, and J. Mueller. Epoque: Practical end-to-end verifiable post-quantum-secure e-voting. Cryptology ePrint Archive, Paper 2021/304, 2021.
49. Rafaël del Pino, Vadim Lyubashevsky, Gregory Neven, and Gregor Seiler. Practical quantum-safe voting from lattices. Cryptology ePrint Archive, Paper 2017/1235, 2017.
50. D.F. Aranha et al. Verifiable mix-nets and distributed decryption for voting from lattice-based assumptions. Cryptology ePrint Archive, Paper 2022/422, 2022.
51. J. Benaloh. Verifiable secret-ballot elections. In *Ph.D. Thesis, Yale University*, 1987.
52. D.F. Aranha et al. Verifiable mix-nets and distributed decryption for voting from lattice-based assumptions. In *ACM CCS*, pages 1467–1481, 2023.
53. S. Bouaziz-Ermann et al. Lattice-based (partially) blind signature without restart. Cryptology ePrint Archive, Report 2020/260, 2020.
54. D. Micciancio and C. Peikert. Trapdoors for lattices: Simpler, tighter, faster, smaller. In *EUROCRYPT*, volume 7237 of *LNCS*, pages 700–718. Springer, 2012.
55. R. Bendlin, S. Krehbiel, and C. Peikert. How to share a lattice trapdoor: Threshold protocols for signatures and (H)IBE. In *ACNS*, volume 7954 of *LNCS*, pages 218–236. Springer, 2013.
56. C. Gentry, C. Peikert, and V. Vaikuntanathan. Trapdoors for hard lattices and new cryptographic constructions. In *ACM STOC*, pages 197–206, 2008.
57. V. Lyubashevsky, C. Peikert, and O. Regev. A toolkit for ring-LWE cryptography. In *EUROCRYPT*, volume 7881 of *LNCS*, pages 35–54. Springer, 2013.
58. E. Hauck et al. Lattice-based blind signatures, revisited. In *CRYPTO*, volume 12171 of *LNCS*. Springer, 2020.
59. A. Esser, A. May, and F. Zwegdinger. McEliece needs a break – solving McEliece-1284 and quasi-cyclic-2918 with modern ISD. In *EUROCRYPT*, volume 13277 of *LNCS*, pages 433–457. Springer, 2022.
60. Richard Lindner and Chris Peikert. Better key sizes (and attacks) for LWE-based encryption. In *Cryptographers Track at the RSA Conference*, pages 319–339. Springer, 2011.
61. M.R. Albrecht, R. Player, and S. Scott. On the concrete hardness of learning with errors. *J. Math. Cryptol.*, 9(3):169–203, 2015.
62. S. Park et al. Going from bad to worse: From internet voting to blockchain voting. *J. Cybersecur.*, 7(1), 2021.
63. I. Hwang, J. Seo, and Y. Song. Concretely efficient lattice-based polynomial commitment from standard assumptions. In *CRYPTO*, pages 414–448. Springer, 2024.
64. T.H. Yuen et al. Posterior security: Anonymity and message hiding of standard signatures. In *ACM CCS*, pages 528–542, 2025.
65. F. Baldimtsi et al. Non-interactive blind signatures: Post-quantum and stronger security. In *ASIACRYPT*, volume 15485 of *LNCS*. Springer, 2025.
66. J. Bos et al. CRYSTALS-Kyber: A CCA-secure module-lattice-based KEM. Cryptology ePrint Archive, Report 2017/634, 2017.
67. National Institute of Standards and Technology. SHA-3 standard: Permutation-based hash and extendable-output functions. Technical Report FIPS PUB 202, NIST, 2015.
68. J. Huang et al. The application of the blockchain technology in voting systems. *ACM Comput. Surv.*, 54:1–28, 2021.
69. S. Vivek et al. E-voting systems using blockchain: An exploratory literature survey. In *IEEE ICIRCA*, 2020.
70. W. Diffie and M.E. Hellman. New directions in cryptography. *IEEE Trans. Inf. Theory*, 22(6):644–654, 1976.

71. B. Guidi, A. Michienzi, and L. Ricci. Data persistence in decentralized social applications: the IPFS approach. In *IEEE CCNC*, 2021.
72. Y. Lin and C. Zhang. A method for protecting private data in IPFS. In *IEEE CSCWD*, 2021.
73. A.J. Gabriel et al. Post-quantum cryptography: A combination of post-quantum cryptography and steganography. In *ICITST*, pages 454–457, 2013.
74. A. Petzoldt. *Selecting and Reducing Key Sizes for Multivariate Cryptography*. PhD thesis, Technical University Darmstadt, 2013.
75. S. Agrawal, D. Boneh, and X. Boyen. Efficient lattice (H)IBE in the standard model. In *EUROCRYPT*, pages 553–572, 2010.
76. R. del Pino et al. Practical quantum-safe voting from lattices. In *ACM CCS*, pages 1565–1581, 2017.
77. R. del Pino et al. Practical quantum-safe voting from lattices. In *ACM CCS*, pages 1565–1581, 2017.
78. D. Stehlé and R. Steinfeld. Making NTRU as secure as worst-case problems over ideal lattices. In *EUROCRYPT*, volume 6632 of *LNCS*, pages 27–47. Springer, 2011.
79. C. Baum et al. More efficient commitments from structured lattice assumptions. In *SCN*, volume 11035 of *LNCS*, pages 368–385. Springer, 2018.
80. D.F. Aranha et al. Lattice-based proof of shuffle and applications to electronic voting. In *CT-RSA*, volume 12704 of *LNCS*, pages 227–251. Springer, 2021.
81. D.F. Aranha et al. Verifiable mix-nets and distributed decryption for voting from lattice-based assumptions. In *ACM CCS*, pages 1467–1481, 2023.
82. J. Bootle et al. More efficient amortization of exact zero-knowledge proofs for LWE. In *ESORICS*, volume 12973 of *LNCS*, pages 608–627. Springer, 2021.
83. N. Aragon et al. BIKE: Bit-flipping key encapsulation. https://bikesuite.org/files/v5.0/BIKE_Spec.2022.10.10.1.pdf, 2022.
84. C. Aguilar-Melchor et al. Efficient encryption from random quasi-cyclic codes. *IEEE Trans. Inf. Theory*, 64(5):3927–3943, 2018.
85. E. Bellini et al. Enhancing code based zero-knowledge proofs using rank metric. In *CANS*, volume 12579, pages 570–592, 2020.
86. F. Benhamouda et al. Better zero-knowledge proofs for lattice encryption and their application to group signatures. In *ASIACRYPT*, pages 551–572, 2014.
87. V. Lyubashevsky. Lattice signatures without trapdoors. In *EUROCRYPT*, pages 738–755, 2012.
88. C. Baum et al. Efficient commitments and zero-knowledge protocols from ring-SIS with applications to lattice-based threshold cryptosystems. Cryptology ePrint Archive 2016/997, 2016.
89. R. del Pino et al. Practical quantum-safe voting from lattices. In *ACM CCS*, pages 1565–1581, 2017.
90. L. Ducas and D. Micciancio. FHEW: Bootstrapping homomorphic encryption in less than a second. In *EUROCRYPT*, volume 9056 of *LNCS*, pages 617–640. Springer, 2015.
91. L. Ducas et al. Lattice signatures and bimodal gaussians. In *CRYPTO*, volume 8042 of *LNCS*, pages 40–56. Springer, 2013.
92. F. Laguillaumie et al. Lattice-based group signatures with logarithmic signature size. In *ASIACRYPT*, volume 8270 of *LNCS*, pages 41–61. Springer, 2013.
93. Shweta Agrawal, Shafi Goldwasser, and Saleet Mossel. *Deniable Fully Homomorphic Encryption from Learning with Errors*, page 641–670. Springer International Publishing, 2021.
94. D. Boneh et al. Threshold cryptosystems from threshold fully homomorphic encryption. In *CRYPTO*, pages 565–596. Springer, 2018.
95. V. Lyubashevsky, N.K. Nguyen, and M. Plançon. Lattice-based zero-knowledge proofs and applications: Shorter, simpler, and more general. In *CRYPTO*, pages 71–101. Springer, 2022.

96. Gurchetan S. Grewal, Mark D. Ryan, Sergiu Bursuc, and Peter Y.A. Ryan. Caveat coercitor: Coercion-evidence in electronic voting. In *2013 IEEE Symposium on Security and Privacy*, pages 367–381, 2013.
97. Roberto Araujo, Sébastien Foulle, and Jacques Traoré. A practical and secure coercion-resistant scheme for remote elections. In *Frontiers of Electronic Voting*, number 07311 in Dagstuhl Seminar Proceedings. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2008.
98. Roberto Araújo, Sébastien Foulle, and Jacques Traoré. A practical and secure coercion-resistant scheme for internet voting. In *Towards Trustworthy Elections*, volume 6000 of *Lecture Notes in Computer Science*, pages 330–342. Springer, 2010.
99. Oliver Spycher, Reto E. Koenig, Rolf Haenni, and Michael Schläpfer. A new approach towards coercion-resistant remote E-voting in linear time. In *Financial Cryptography and Data Security – FC 2011*, volume 7035 of *Lecture Notes in Computer Science*, pages 182–189. Springer, 2012.
100. Jeremy Clark and Urs Hengartner. Selections: Internet voting with over-the-shoulder coercion-resistance. In *Financial Cryptography and Data Security – FC 2011*, volume 7035 of *Lecture Notes in Computer Science*, pages 47–61. Springer, 2012.
101. Wouter Lueks, Iñigo Querejeta-Azurmendi, and Carmela Troncoso. Voteagain: a scalable coercion-resistant voting system. In *Proceedings of the 29th USENIX Conference on Security Symposium*, SEC’20, USA, 2020. USENIX Association.
102. Philipp Locher, Rolf Haenni, and Reto E. Koenig. Coercion-resistant internet voting with everlasting privacy. In *Financial Cryptography and Data Security – FC 2016 International Workshops, BITCOIN, VOTING, and WAHC*, volume 9604 of *Lecture Notes in Computer Science*, pages 161–175. Springer, 2016.
103. Roberto Araújo, Amira Barki, Solenn Brunet, and Jacques Traoré. Remote electronic voting can be efficient, verifiable and coercion-resistant. In *Financial Cryptography and Data Security – FC 2016 International Workshops, BITCOIN, VOTING, and WAHC*, volume 9604 of *Lecture Notes in Computer Science*, pages 224–232. Springer, 2016.
104. Sven Heiberg and Jan Willemson. Verifiable internet voting in Estonia. In *6th International Conference on Electronic Voting (EVOTE 2014)*, pages 1–8. IEEE, 2014.
105. T. ElGamal. A public key cryptosystem and a signature scheme based on discrete logarithms. In *CRYPTO*, volume 196 of *LNCS*, pages 10–18. Springer, 1985.
106. Anonymous Github — anonymous.4open.science. <https://anonymous.4open.science/r/onyx-paper>. [Accessed 21-05-2026].
107. J. Neyman and E. S. Pearson. *On the Problem of the Most Efficient Tests of Statistical Hypotheses*, page 73–108. Springer New York, 1992.
108. Véronique Cortier, Pierrick Gaudry, and Quentin Yang. *A Toolbox for Verifiable Tally-Hiding E-Voting Systems*, page 631–652. Springer Nature Switzerland, 2022.
109. David Mestel, Johannes Müller, and Pascal Reisert. How efficient are replay attacks against vote privacy? a formal quantitative analysis. In *2022 IEEE 35th Computer Security Foundations Symposium (CSF)*, pages 179–194, 2022.
110. Karim Bagheri. *II: A Unified Framework for Computational Verifiable Secret Sharing*, page 110–142. Springer Nature Switzerland, 2025.
111. Diego F. Aranha, Carsten Baum, Kristian Gjøsteen, and Tjerand Silde. Verifiable mix-nets and distributed decryption for voting from lattice-based assumptions. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security, CCS ’23*, page 1467–1481, New York, NY, USA, 2023. Association for Computing Machinery.
112. Carsten Baum, Ivan Damgård, Vadim Lyubashevsky, Sabine Oechsner, and Chris Peikert. More efficient commitments from structured lattice assumptions. In *Security and Cryptography for Networks – SCN 2018*, volume 11035 of *Lecture Notes in Computer Science*, pages 368–385. Springer, 2018.
113. G. Barthe et al. Easycrypt: A tutorial. In *Foundations of Security Analysis and Design VII*, pages 146–166. Springer, 2014.

114. Anonymous Github — anonymous.4open.science. <https://anonymous.4open.science/r/onyx-proof>. [Accessed 18-05-2026].
115. V. Cortier et al. Machine-checked proofs for electronic voting: Privacy and verifiability for Belenios. In *IEEE CSF*, pages 298–312, 2018.
116. Anonymous Github — anonymous.4open.science. <https://anonymous.4open.science/r/onyx/>. [Accessed 17-05-2026].
117. Martin R. Albrecht, Rachel Player, and Sam Scott. On the concrete hardness of Learning with Errors. *Journal of Mathematical Cryptology*, 9(3):169–203, 2015.
118. O. Regev. On lattices, learning with errors, random linear codes, and cryptography. *J. ACM*, 56(6), 2009.
119. Vadim Lyubashevsky. Fewer keys and more efficiency from the ring-lwe cryptosystem. In *CRYPTO*, volume 7417 of *LNCS*, pages 172–190. Springer, Berlin, Heidelberg, 2012. Presented at EUROCRYPT 2012, but formally introduces the module setting and is the basis for its adoption in subsequent schemes.
120. A. Langlois and D. Stehle. Worst-case to average-case reductions for module lattices. *Des. Codes Cryptogr.*, 75(3):565–599, 2015.
121. Adeline Langlois and Damien Stehlé. Worst-case to average-case reductions for module lattices. *Designs, Codes and Cryptography*, 75(3):565–599, 2015.
122. R. Canetti et al. Deniable encryption. In *CRYPTO*, volume 1294 of *LNCS*. Springer, 1997.
123. Daniel Hugenroth, Alberto Sonnino, Sam Cutler, and Alastair R. Beresford. Sloth: Key stretching and deniable encryption using secure elements on smartphones. *IACR Cryptol. ePrint Arch.*, 2023:1792, 2024.
124. K. Cohn-Gordon et al. A formal security analysis of the signal messaging protocol. In *IEEE EuroS&P*, pages 451–466, 2017.
125. Niusen Chen, Bo Chen, and Weisong Shi. An application-layer plausibly deniable encryption system for wearable devices. *Discover Internet of Things*, 4(1), August 2024.
126. N. Chen, B. Chen, and W. Shi. Mobiwear: A plausibly deniable encryption system for wearable mobile devices. In *EAI AC3*, pages 138–154. Springer, 2021.
127. GitHub - zorggomat/FractalCryptGUI: Free cross-platform deniable encryption cryptoarchiver — github.com. <https://github.com/zorggomat/FractalCryptGUI>. [Accessed 13-04-2026].
128. VeraCrypt – free open source disk encryption. <https://veracrypt.io/en/Home.html>. Accessed: 2025-12-16.
129. E. Anzuoni and T. Gagliardoni. Shufflecake: Plausible deniability for multiple hidden filesystems on linux. In *ACM CCS*, pages 3033–3047, 2023.
130. R. Canetti, S. Park, and O. Poburinnaya. Fully deniable interactive encryption. In *CRYPTO*, volume 12170 of *LNCS*. Springer, 2020.
131. A. Sahai and B. Waters. How to use indistinguishability obfuscation: Deniable encryption, and more. In *ACM STOC*, pages 475–484, 2014.
132. R. Bendlin et al. Lower and upper bounds for deniable public-key encryption. In *ASIACRYPT*, pages 125–142. Springer, 2011.
133. D. Dachman-Soled. On minimal assumptions for sender-deniable public key encryption. In *PKC*, pages 574–591. Springer, 2014.
134. D. Apon, X. Fan, and F.-H. Liu. Deniable attribute based encryption for branching programs from LWE. In *TCC*, volume 9986 of *LNCS*. Springer, 2016.
135. S. Goldwasser, S. Klein, and D. Wichs. The edited truth. In *TCC*, pages 305–340. Springer, 2017.
136. Shaoquan Jiang, Yeow Meng Chee, San Ling, Huaxiong Wang, and Chaoping Xing. A new framework for deniable secure key exchange. *Information and Computation*, 285:104866, 2022.
137. Ran Canetti, Ivan Damgård, Sebastian Kolby, Divya Ravi, and Sophia Yakubov. Deniable secret sharing. In *Theory of Cryptography: 23rd International Conference, TCC 2025, Aarhus, Denmark, December 1–5, 2025, Proceedings, Part II*, page 399–427, Berlin, Heidelberg, 2025. Springer-Verlag.

138. I. Damgård and J.B. Nielsen. Improved non-committing encryption schemes based on a general complexity assumption. In *CRYPTO*, volume 1880 of *LNCS*. Springer, 2000.
139. A. O’Neill, C. Peikert, and B. Waters. Bi-deniable public-key encryption. In *CRYPTO*, pages 525–542. Springer, 2011.
140. A. Peet. The puzzle of plausible deniability. *Synthese*, 203(1):156, 2024.
141. D. Bernhard, O. Kulyk, and M. Volkamer. Security proofs for participation privacy, receipt-freeness and ballot privacy for the helios voting scheme. In *ARES*. ACM, 2017.
142. M. Graf et al. Accountable bulletin boards: Definition and provably secure implementation. In *IEEE CSF*, pages 201–216, 2024.
143. C.-C. Drăgan et al. Machine-checked proofs of accountability: How to select who is to blame. In *ESORICS*, volume 14346 of *LNCS*. Springer, 2024.
144. Valentin V. Petrov. Sums of independent random variables. 1975.
145. Graeme Auld and Kritsana Neammanee. Explicit constants in the nonuniform local limit theorem for poisson binomial random variables. *Journal of Inequalities and Applications*, 2024(1), May 2024.
146. Gilles Barthe. An introduction to relational program verification, 2020.
147. Martin Albrecht, Melissa Chase, Hao Chen, Jintai Ding, Shafi Goldwasser, Sergey Gorbunov, Shai Halevi, Jeffrey Hoffstein, Kim Laine, Kristin Lauter, Satya Lokam, Daniele Micciancio, Dustin Moody, Travis Morrison, Amit Sahai, and Vinod Vaikuntanathan. Homomorphic encryption standard. Technical report, HomomorphicEncryption.org, 2018.
148. Christine Rüb. On the average running time of odd–even merge sort. *Journal of Algorithms*, 22(2):329–346, 1997.
149. National Institute of Standards and Technology. Secure Hash Standard (SHS). Technical Report FIPS PUB 180-4, NIST, 2015.
150. Dan Boneh, Özgür Dagdelen, Marc Fischlin, Anja Lehmann, Christian Schaffner, and Mark Zhandry. Random oracles in a quantum world. In *Advances in Cryptology – ASIACRYPT 2011*, volume 7073 of *Lecture Notes in Computer Science*, pages 41–69. Springer, 2011.
151. D.J. Bernstein. Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? Workshop Record of SHARCS’09, 2009.
152. Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. (Leveled) Fully Homomorphic Encryption without Bootstrapping. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference (ITCS)*, pages 309–325. ACM, 2012.
153. Craig Gentry. *A Fully Homomorphic Encryption Scheme*. PhD thesis, Stanford University, 2009. <https://crypto.stanford.edu/craig>.
154. C. Gentry, A. Sahai, and B. Waters. Homomorphic encryption from learning with errors: Conceptually-simpler, asymptotically-faster, attribute-based. In *CRYPTO*, volume 8042 of *LNCS*. Springer, 2013.
155. A. De Santis, S. Micali, and G. Persiano. Non-interactive zero-knowledge with preprocessing. In *CRYPTO*, volume 403 of *LNCS*. Springer, 1990.
156. M. Blum. Coin flipping by telephone a protocol for solving impossible problems. *SIGACT News*, 15(1):23–27, 1983.
157. D. Boneh et al. A lattice-based universal thresholdizer for cryptographic systems. Cryptology ePrint Archive, Paper 2017/251, 2017.
158. A. Jain, P.M.R. Rasmussen, and A. Sahai. Threshold fully homomorphic encryption. Cryptology ePrint Archive, Paper 2017/257, 2017.
159. Rosario Gennaro, Michael O. Rabin, and Tal Rabin. Simplified VSS and fast-track multiparty computations with applications to threshold cryptography. In *17th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 101–111. ACM, 1998.
160. Shahla Atapoor, Karim Bagheri, Daniele Cozzo, and Robi Pedersen. VSS from distributed ZK proofs and applications. In *Advances in Cryptology – ASIACRYPT 2023*, Lecture Notes in Computer Science. Springer, 2023. <https://eprint.iacr.org/2023/992>.

161. Lov K. Grover. A fast quantum mechanical algorithm for database search. In *28th Annual ACM Symposium on Theory of Computing (STOC)*, pages 212–219. ACM, 1996.
162. Diego F. Aranha, Carsten Baum, Kristian Gjøsteen, Tjerand Silde, and Thor Tunge. Lattice-based proof of shuffle and applications to electronic voting. In *Topics in Cryptology – CT-RSA 2021*, volume 12704 of *Lecture Notes in Computer Science*, pages 227–251. Springer, 2021.
163. Jonathan Bootle, Vadim Lyubashevsky, Ngoc Khanh Nguyen, and Gregor Seiler. More efficient amortization of exact zero-knowledge proofs for LWE. In *ESORICS 2021: 26th European Symposium on Research in Computer Security, Part II*, volume 12973 of *Lecture Notes in Computer Science*, pages 608–627. Springer, 2021.
164. Carsten Baum, Jonathan Bootle, Andrea Cerulli, Rafaël del Pino, Jens Groth, and Vadim Lyubashevsky. Sub-linear lattice-based zero-knowledge arguments for arithmetic circuits. In *Advances in Cryptology – CRYPTO 2018, Part II*, volume 10992 of *Lecture Notes in Computer Science*, pages 669–699. Springer, 2018.